

DEPARTMENT OF FINANCE
BUREAU OF LOCAL GOVERNMENT FINANCE

2nd Edition, 2018

PHILIPPINE VALUATION STANDARDS

*Incorporating the International Valuation
Standards (IVS) 2017 and Providing
Guidance Notes under Philippine Setting*

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2nd Edition, 2018

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and Providing Guidance Notes under Philippine Setting

**Philippine Valuation Standards 2nd Edition, 2018
Incorporating the International Valuation Standards (IVS) 2017
and Providing Guidance Notes under Philippine Setting**

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DEPARTMENT ORDER NO. 037.2018
2 July 2018

**SUBJECT : Prescribing the Philippine Valuation Standards (PVS) 2nd Edition, 2018
Incorporating the International Valuation Standards (IVS) 2017
and Providing Guidance Notes under Philippine Setting**

- 1. LEGAL BASES.** This Order is issued under the authority of the following laws:
- a. Section 201 of the Local Government Code 1991 mandates the Department of Finance (DOF) to promulgate the necessary rules and regulations for the classification, appraisal, and assessment of real property;
 - b. Section 33 (1) and (2), Chapter 4, Title II, Book IV of Executive Order (EO) No. 292, series of 1987, otherwise known as the Administrative Code of 1987, directs the Bureau of Local Government Finance (BLGF) to assist in the formulation and implementation of policies on local government revenue administration and fund management, exercise administrative and technical supervision and coordination over treasury and assessment operation of local governments;
 - d. Section 2 of Republic Act No. 9646, otherwise known as the Real Estate Service Act, provides the policy of the State to develop and nurture through proper and effective regulation and supervision a corps of technically competent, responsible and respected professional real estate service practitioners whose standards of practice and service shall be globally competitive and will promote the growth of the real estate industry; and
 - f. Section 4 (E) of Republic Act No. 10963, otherwise known as the Tax Reform for Acceleration and Inclusion (TRAIN), authorizes the Commissioner of the Bureau of Internal Revenue (BIR) to prescribe the real property values based on the current PVS.
- 2. COVERAGE AND OBJECTIVES.** The DOF recognizes the need for an efficient, effective and transparent real property valuation system in the country to ensure a robust and progressive real estate sector that will benefit the government and the private sectors, and foster greater confidence and transparency in valuation practice at the national and local levels for taxation and other purposes.

The PVS 2nd Edition intends to guide all concerned valuation stakeholders in the Philippines, mainly the attached bureaus and agencies of the DOF performing or requiring valuation service, to consistently use and apply internationally accepted valuation practices. In particular, the PVS 2nd Edition aims to: (i) raise the quality of public and private sector valuations and reporting of valuations, (ii) provide consistency and understanding between providers and users of valuations, especially at the local and national levels; (iii) promote transparency and reliability of valuations for taxation and other purposes; and (iv) reduce financial risk for users of valuations.

The PVS 2nd Edition shall primarily be used to guide valuation of real properties for property taxation by local government units (LGUs), the BLGF, and the BIR, in conjunction with the pertinent laws, rules and regulations related to their respective mandates. Other agencies and users who intend to refer to and adopt the PVS 2nd Edition, 2018, particularly with respect to IVS 2017, are encouraged to do so, subject to their respective mandates and the applicable guidelines and regulations of competent authority/ies.

- 3. THE PVS 2nd EDITION, 2018.** The PVS 2nd Edition, 2018- Incorporating the IVS 2017 and Providing Guidance Notes under Philippine Setting, which succeeds the PVS 1st Edition issued in 2009, is composed of three (3) parts, namely: (i) IVS 2017 in full adoption; (ii) Philippine Context Focusing on Valuation for Taxation and Other Purposes; and (iii) Guidance Notes.

The publication of the PVS 2nd Edition shall complement all issuances of the DOF pertaining to rules and regulations for the classification, appraisal and assessment of real property. It shall form part of the continuing professional development programs of the BLGF, being a duly recognized Institutional Member of the IVSC, as well as other supplemental manuals and guidelines for the LGUs.

- 5. COMPLIANCE.** All Provincial, City and Municipal Assessors, and heads of the BLGF and the BIR are hereby enjoined to properly, efficiently and strictly implement and comply with the provisions of the PVS 2nd Edition in their respective jurisdictions, subject to the provisions of pertinent laws on the matter.
- 7. SEPARABILITY CLAUSE.** Any portion or provision of this Order that may be declared unconstitutional or invalid shall not nullify other portions and provisions hereof as long as such remaining portion or provision can still subsist and be given effect in its entirety.
- 9. REPEALING CLAUSE.** This Department Order supersedes Department Order No. 37-09 dated 19 October 2009, and all other issuances, or parts thereof that are inconsistent herewith, are likewise hereby repealed or modified and amended accordingly.
- 11. EFFECTIVITY.** This Department Order shall take effect fifteen (15) days after its publication in the Official Gazette and the University of the Philippines Office of the National Administrative Register (ONAR) or in a newspaper of general circulation in the Philippines.


CARLOS G. DOMINGUEZ
Secretary
JUL 12 2018





Republic of the Philippines
DEPARTMENT OF FINANCE
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Manila 1004

MESSAGE

I commend the Bureau of Local Government Finance (BLGF) for its continuing effort to adapt international valuation standards for the Philippine setting. The effort is reflected in this second edition of the Philippine Valuation Standards (PVS).

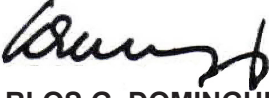
As a new institutional member of the International Valuation Standards Council (IVSC), the Philippines, through the BLGF, has come up with this publication, which reproduces the 2017 International Valuation Standards.

The publication will be disseminated to local government units (LGUs) as a guide to property valuations. This will surely be a worthwhile contribution to updating property valuation for taxation purposes.

Without common standards, property valuation by the local governments will be characterized by wide disparities. These disparities hamper our ability to collect the right taxes and improve our revenue intake.

This publication is extremely useful in building a modern taxation system that enables the state to make economic investments and ensure inclusive growth.




CARLOS G. DOMINGUEZ
Secretary
SEP 06 2018



Republic of the Philippines
DEPARTMENT OF FINANCE
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Manila 1004

MESSAGE

Promoting a region-wide move towards adopting the latest globally accepted valuation standards is vital to the country's drive to attain inclusive and sustainable growth. This is especially true given the huge growth in cross-border transactions and the ever-changing financial landscape.

One key area that requires reforms is the real estate sector, as it is not only close to personal and business decisions, but it also offers a progressive source of revenues for the government. Real property valuations need to be transparent, equitable, consistent and reliable as we build and sustain public confidence and investors' trust.

Over the past few years, our focus has been to achieve meaningful policy reforms that positively impact on the fiscal position of the country and promote greater equity for the people. It is in this context that we must strengthen the country's policy framework on valuation, improve the professional practice of real estate appraisal, and build trust in the process.

By issuing the Philippine Valuation Standards (PVS) 2nd Edition, we aim to step up the country's valuation practice, particularly at the local and national levels for taxation and other purposes, at par with latest global standards and best practices. It is for this reason that we must welcome and integrate international standards into our operations to institute benchmarks for excellence and better financial governance. Against this backdrop, we will be in a good position to raise more revenues, maintain sound fiscal management, and sustain economic growth.

ANTONETTE C. TIONKO
Undersecretary



Republic of the Philippines
DEPARTMENT OF FINANCE
BUREAU OF LOCAL GOVERNMENT FINANCE
8th Floor EDPC Building, BSP Complex, Roxas Boulevard, 1004 Manila

MESSAGE

The Bureau of Local Government Finance (BLGF) takes pride in publishing the Philippine Valuation Standards (PVS) 2nd Edition, 2018, embodying in full the IVS 2017 issued by the International Valuation Standards Council (IVSC), providing country context on taxation and other purposes, and issuing relevant guidance notes to support our mandate in assisting the local government units (LGUs).

This latest issuance aims to further help the LGUs value correctly real property as a vital local revenue source to improve public service delivery. Without a reliable property valuation, government's capacity to generate revenues is easily undermined. The PVS aims to foster trust and confidence in valuing properties not just for taxation but also for other purposes, such as rental, lease, financial reporting, regulatory compliance, etc. In all these, the key principles in valuation remain the same, and their methods must be consistent and uniformly aligned with the standards.

This publication is timely following the admission of the BLGF as an Institutional Member of the IVSC this year. Thus, the rollout of the PVS 2018 and continuous capacity building of local appraisers and assessment officers using internationally accepted best practices and tools are the next logical steps.

The publication of the PVS 2018 would not have been possible without the technical assistance grant of the Food and Agriculture Organization of the United Nations (UN FAO) and the Asian Development Bank. The BLGF is grateful for the guidance and support of the Department of Finance, and to all the stakeholders who provided invaluable inputs to the PVS, as well as the officials and personnel of the BLGF Central and Regional Offices whose work helped realize this publication.

NIÑO RAYMOND B. ALVINA

Executive Director

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Preface

The development and publication of the **Philippine Valuation Standards 2nd Edition, 2018 - Incorporating the International Valuation Standards (IVS) 2017 and Providing Guidance Notes under Philippine Setting** (hereinafter PVS 2nd Edition, 2018) is a recognition of the continued commitment of the Philippine Government to ensure that valuations in the country are compliant or aligned with international standards and best practices, and cognizant of the country context and legal requisites. This is in line with the government's concurrent efforts to raise more revenues, maintain sound fiscal management, and foster a conducive environment for investments towards inclusive economic growth.

The PVS 2nd Edition, 2018 acknowledges that consistency, objectivity and transparency are fundamental to building and sustaining public confidence and trust in valuation. It contains standards for undertaking valuation assignments using generally concepts and principles that promote transparency and consistency in valuation practice. For this purpose, the PVS 2nd Edition, 2018 adopted and reproduced the IVS 2017 in full. However, since there are departures or differences in the application of the Standards relative to existing Philippine laws and regulations, recognition and contextual discussion of such differences are necessary. Guidance Notes are also provided for areas of practice not specifically covered by the IVS or on issues in the application of the standards.

The development of the PVS was supported by the Technical Cooperation Project entitled "Mainstreaming Voluntary Guidelines on Governance of Tenure (VGGT)", implemented in partnership between the Government of the Philippines and Food and Agriculture Organization of the United Nations (UN FAO). The project aimed to improve the overall capacity of the Government of the Philippines in providing good land governance by updating the Land Sector Development Framework (LSDF), a 20-year development framework for land tenure governance, with VGGT principles and guidelines adopted, and covering the thematic areas of public land management, land administration, land information management, valuation, and capacity building.

International and national valuation experts were engaged, and consultative meetings and interactive exposure workshops of the PVS drafts with various stakeholders were undertaken in Metro Manila, Luzon, the Visayas and Mindanao in order to solicit comments and recommendations.

While Republic Act (RA) No. 9646 or the Real Estate Service Act of 2009 enabled the professionalization of real estate service practice, there remains key legislative reforms, such as the Valuation Reform Bill, that must be pursued. The Department of Finance (DOF), on the other hand, shall continue to promulgate the necessary rules in appraisal/valuation and assessment of real property as required under the Local Government Code of 1991.

Coverage and Objectives of the PVS 2nd Edition, 2018

The coverage of the PVS 2nd Edition, 2018 is provided under DOF Department Order No. 037.2018, entitled "*Prescribing the Philippine Valuation Standards (PVS) 2nd Edition, 2018 - Incorporating the International Valuation Standards (IVS) 2017 and Providing Guidance Notes under Philippine Setting.*"

Consistent with the IVS, and cognizant of the mandates under RA No. 7160 (Local Government Code of 1991) and RA No. 10963 (Tax Reform for Acceleration and Inclusion), the objectives of the PVS 2nd Edition, 2018 are as follows:

- To raise the quality of public and private sector valuations and reporting of valuations;
- To provide consistency and understanding between providers and users of valuations, especially at the local and national levels;
- To promote transparency and reliability of valuations for taxation and other purposes; and
- To reduce financial risk for users of valuations.

The PVS 2nd Edition, 2018 complements all issuances of the DOF pertaining to rules and regulations for the classification, appraisal and assessment of real property. It shall form part of the continuing professional development programs of the BLGF, as well as other supplemental manuals and guidelines for the LGUs.

Structure of the PVS 2nd Edition, 2018

The PVS 2nd Edition, 2018 is structured in three (3) parts, namely:

Part I: International Valuation Standards (IVS) 2017. This refers to the full, unabridged text reproduction of the IVS 2017. The main sections include:

- *IVS General Standards.* These set out requirements for the conduct of all valuation assignments including establishing the terms of a valuation engagement, bases of value, valuation approaches and methods, and reporting. They are designed to be applicable to valuations of all types of *assets* and for any valuation purpose.
- *IVS Asset Standards.* The Asset Standards include requirements related to specific types of *assets*. These requirements *must* be followed in conjunction with the General Standards when performing a valuation of a specific *asset* type. The Asset Standards include certain background information on the characteristics of each *asset* type that influence value and additional *asset*-specific requirements on common valuation approaches and methods used.

Part II: Philippine Context Focusing on Valuation for Taxation and Other Purposes. This contains the departures or differences in application of the IVS based on Philippine contextual setting or legal requisites, with particular focus on valuation for property taxation and other purposes under Philippine setting.

Part III: Guidance Notes. This includes Guidance Notes (GNs) deemed important within the context of the Philippines. The GNs consider the same guidance notes issued in PVS 2009 to provide guidance on specific valuation issues and how standards are to be applied. These GNs complement and expand on the Standards with which they have equal importance.

The BLGF as an Institutional Member of the IVSC

On 5 March 2018, the BLGF was officially admitted as an Institutional Member of the IVSC. In line with its mandate, the BLGF is expected to cooperate and collaborate with IVSC and its other members to establish a consistent and transparent framework for valuation practice worldwide; bridge the gap between local and international valuation standards (IVS); and provide trainings and capacity development programs, or issue locally applicable guidance and quality measures.

PART I

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Introduction

The International Valuation Standards Council (IVSC) is an independent, not-for-profit organisation committed to advancing quality in the valuation profession. Our primary objective is to build confidence and public trust in valuation by producing standards and securing their universal adoption and implementation for the valuation of *assets* across the world. We believe that International Valuation Standards (IVS) are a fundamental part of the financial system, along with high levels of professionalism in applying them.

Valuations are widely used and relied upon in financial and other markets, whether for inclusion in financial statements, for regulatory compliance or to support secured lending and transactional activity. The International Valuation Standards (IVS) are standards for undertaking valuation assignments using generally recognised concepts and principles that promote transparency and consistency in valuation practice. The IVSC also promotes leading practice approaches for the conduct and competency of professional valuers.

The IVSC Standards Board is the body responsible for setting the IVS. The Board has autonomy in the development of its agenda and approval of its publications. In developing the IVS, the Board:

- follows established due process in the development of any new standard, including consultation with stakeholders (valuers, users of valuation services, regulators, valuation professional organisations, etc) and public exposure of all new standards or material alterations to existing standards,
- liaises with other bodies that have a standard-setting function in the financial markets,
- conducts outreach activities including round-table discussions with invited constituents and targeted discussions with specific users or user groups.

The objective of the IVS is to increase the confidence and trust of users of valuation services by establishing transparent and consistent valuation practices. A standard will do one or more of the following:

- identify or develop globally accepted principles and definitions,
- identify and promulgate considerations for the undertaking of valuation assignments and the reporting of valuations,
- identify specific matters that require consideration and methods commonly used for valuing different types of *assets* or liabilities.

The IVS consist of mandatory requirements that *must* be followed in order to state that a valuation was performed in compliance with the IVS. Certain aspects of the standards do not direct or mandate any particular course of action, but provide fundamental principles and concepts that *must* be considered in undertaking a valuation.

The IVS are arranged as follows:

The IVS Framework

This serves as a preamble to the IVS. The IVS Framework consists of general principles for valuers following the IVS regarding objectivity, judgement, competence and acceptable departures from the IVS.

IVS General Standards

These set forth requirements for the conduct of all valuation assignments including establishing the terms of a valuation engagement, bases of value, valuation approaches and methods, and reporting. They are designed to be applicable to valuations of all types of *assets* and for any *valuation purpose*.

IVS Asset Standards

The *Asset Standards* include requirements related to specific types of *assets*. These requirements *must* be followed in conjunction with the General Standards when performing a valuation of a specific *asset* type. The *Asset Standards* include certain background information on the characteristics of each *asset* type that influence value and additional *asset*-specific requirements on common valuation approaches and methods used.

What is in this Book?

This book includes the IVS Framework, the IVS General Standards and the IVS *Asset Standards* approved by the IVSC Standards Board on 15 December 2016, with an effective date of **1 July 2017**. Early adoption of these standards is allowed.

Future Changes to these Standards

The IVSC Standards Board intends to continuously review the IVS and update or clarify the standards as needed to meet stakeholder and market needs. The Board has continuing projects that *may* result in additional standards being introduced or amendments being made to the standards in this publication at any time. News on current projects and any impending or approved changes can be found on the IVSC website at www.ivsc.org.

Glossary

10. Overview of Glossary

- 10.1. This glossary defines certain terms used in the International Valuation Standards.
- 10.2. This glossary does not attempt to define basic valuation, accounting or finance terms, as valuers are assumed to have an understanding of such terms (see definition of “*valuer*”).

20. Defined Terms

20.1. Asset or Assets

To assist in the readability of the standards and to avoid repetition, the words “*asset*” and “*assets*” refer generally to items that might be subject to a valuation engagement. Unless otherwise specified in the standard, these terms can be considered to mean “*asset*, group of *assets*, liability, group of liabilities, or group of *assets* and liabilities”.

20.2. Client

The word “*client*” refers to the person, persons, or entity for whom the valuation is performed. This *may* include external *clients* (ie, when a *valuer* is engaged by a third-party *client*) as well as internal *clients* (ie, valuations performed for an employer).

20.3. Jurisdiction

The word “*jurisdiction*” refers to the legal and regulatory environment in which a valuation engagement is performed. This generally includes laws and regulations set by governments (eg, country, state and municipal) and, depending on the *purpose*, rules set by certain regulators (eg, banking authorities and securities regulators).

20.4. May

The word “*may*” describes actions and procedures that *valuers* have a responsibility to consider. Matters described in this fashion require the *valuer’s* attention and understanding. How and whether the *valuer* implements these matters in the valuation engagement will depend on the exercise of professional judgement in the circumstances consistent with the objectives of the standards.

20.5. Must

The word “*must*” indicates an unconditional responsibility. The *valuer must* fulfill responsibilities of this type in all cases in which the circumstances exist to which the requirement applies.

20.6. Participant

The word “*participant*” refers to the relevant *participants* pursuant to the basis (or bases) of value used in a valuation engagement (see IVS 104 *Bases of Value*). Different bases of value require *valuers* to consider different perspectives, such as those of “market *participants*” (eg, Market Value, IFRS Fair Value) or a particular owner or prospective buyer (eg, Investment Value).

20.7. Purpose

The word “*purpose*” refers to the reason(s) a valuation is performed. Common *purposes* include (but are not limited to) financial reporting, tax reporting, litigation support, transaction support, and to support secured lending decisions.

20.8. Should

The word “*should*” indicates responsibilities that are presumptively mandatory. The *valuer must* comply with requirements of this type unless the *valuer* demonstrates that alternative actions which were followed under the circumstances were sufficient to achieve the objectives of the standards.

In the rare circumstances in which the *valuer* believes the objectives of the standard can be met by alternative means, the *valuer must* document why the indicated action was not deemed to be necessary and/or appropriate.

If a standard provides that the *valuer “should”* consider an action or procedure, consideration of the action or procedure is presumptively mandatory, while the action or procedure is not.

20.9. Significant and/or Material

Assessing significance and *materiality* require professional judgement. However, that judgement *should* be made in the following context:

- Aspects of a valuation (including inputs, assumptions, special assumptions, and methods and approaches applied) are considered to be *significant/material* if their application and/or impact on the valuation could reasonably be expected to influence the economic or other decisions of users of the valuation; and judgments about *materiality* are made in light of the overall valuation engagement and are affected by the size or nature of the *subject asset*.
- As used in these standards, “*material/materiality*” refers to *materiality* to the valuation engagement, which *may* be different from materiality considerations for other *purposes*, such as financial statements and their audits.

20.10. Subject or Subject Asset

These terms refer to the *asset(s)* valued in a particular valuation engagement.

20.11. Valuation Purpose or Purpose of Valuation

See “*Purpose*”.

20.12. Valuation Reviewer

A “*valuation reviewer*” is a professional *valuer* engaged to review the work of another *valuer*. As part of a valuation review, that professional *may* perform certain valuation procedures and/or provide an opinion of value.

20.13. Valuer

A “*valuer*” is an individual, group of individuals or a firm who possesses the necessary qualifications, ability and experience to execute a valuation in an objective, unbiased and competent manner. In some *jurisdictions*, licensing is required before one can act as a *valuer*.

20.14. Weight

The word “*weight*” refers to the amount of reliance placed on a particular indication of value in reaching a conclusion of value (eg, when a single method is used, it is afforded 100% *weight*).

20.15. Weighting

The word “*weighting*” refers to the process of analysing and reconciling differing indications of values, typically from different methods and/or approaches. This process does not include the averaging of valuations, which is not acceptable.

IVS Framework

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10. Compliance with Standards

- 10.1. When a statement is made that a valuation will be, or has been, undertaken in accordance with the IVS, it is implicit that the valuation has been prepared in compliance with all relevant standards issued by the IVSC.

20. Assets and Liabilities

- 20.1. The standards can be applied to the valuation of both *assets* and liabilities. To assist the legibility of these standards, the words *asset* or *assets* have been defined to include liability or liabilities and groups of *assets*, liabilities, or *assets* and liabilities, except where it is expressly stated otherwise, or is clear from the context that liabilities are excluded.

30. Valuer

- 30.1. *Valuer* has been defined as “an individual, group of individuals, or a firm possessing the necessary qualifications, ability and experience to undertake a valuation in an objective, unbiased and competent manner. In some *jurisdictions*, licensing is required before one can act as a *valuer*. Because a valuation reviewer *must* also be a *valuer*, to assist with the legibility of these standards, the term *valuer* includes *valuation reviewers* except where it is expressly stated otherwise, or is clear from the context that *valuation reviewers* are excluded.

40. Objectivity

- 40.1. The process of valuation requires the *valuer* to make impartial judgements as to the reliability of inputs and assumptions. For a valuation to be credible, it is important that those judgements are made in a way that promotes transparency and minimises the influence of any subjective factors on the process. Judgement used in a valuation *must* be applied objectively to avoid biased analyses, opinions and conclusions.

- 40.2. It is a fundamental expectation that, when applying these standards, appropriate controls and procedures are in place to ensure the necessary degree of objectivity in the valuation process so that the results are free from bias. The IVSC *Code of Ethical Principles for Professional Valuers* provides an example of an appropriate framework for professional conduct.

50. Competence

- 50.1. Valuations *must* be prepared by an individual or firm having the appropriate technical skills, experience and knowledge of the subject of the valuation, the market(s) in which it trades and the *purpose of the valuation*.
- 50.2. If a *valuer* does not possess all of the necessary technical skills, experience and knowledge to perform all aspects of a valuation, it is acceptable for the *valuer* to seek assistance from specialists in certain aspects of the overall assignment, providing this is disclosed in the scope of work (see IVS 101 *Scope of Work*) and the report (see IVS 103 *Reporting*).
- 50.3. The *valuer must* have the technical skills, experience and knowledge to understand, interpret and utilise the work of any specialists.

60. Departures

- 60.1. A “*departure*” is a circumstance where specific legislative, regulatory or other authoritative requirements *must* be followed that differ from some of the requirements within IVS. Departures are mandatory in that a *valuer must* comply with legislative, regulatory and other authoritative requirements appropriate to the *purpose* and *jurisdiction* of the *valuation* to be in compliance with IVS. A *valuer may* still state that the valuation was performed in accordance with IVS when there are departures in these circumstances.
- 60.2. The requirement to depart from IVS pursuant to legislative, regulatory or other authoritative requirements takes precedence over all other IVS requirements.
- 60.3. As required by IVS 101 *Scope of Work*, para 20.3 (n) and IVS 103 *Reporting*, para 10.2 the nature of any departures *must* be identified (for example, identifying that the valuation was performed in accordance with IVS and local tax regulations). If there are any departures that *significantly* affect the nature of the procedures performed, inputs and assumptions used, and/or valuation conclusion(s), a *valuer must* also disclose the specific legislative, regulatory or other authoritative requirements and the *significant* ways in which they differ from the requirements of IVS (for example, identifying that the relevant *jurisdiction* requires the use of only a market approach in a circumstance where IVS would indicate that the income approach *should* be used).
- 60.4. Departure deviations from IVS that are not the result of legislative, regulatory or other authoritative requirements are not permitted in valuations performed in accordance with IVS.

General Standards

IVS 101 Scope of Work

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10. Introduction

- 10.1. A scope of work (sometimes referred to as terms of engagement) describes the fundamental terms of a valuation engagement, such as the *asset(s)* being valued, the *purpose of the valuation* and the responsibilities of parties involved in the valuation.
- 10.2. This standard is intended to apply to a wide spectrum of valuation assignments, including:
- (a) valuations performed by *valuers* for their own employers (“in-house valuations”),
 - (b) valuations performed by *valuers* for *clients* other than their employers (“third-party valuations”), and
 - (c) valuation reviews where the reviewer *may* not be required to provide their own opinion of value.

20. General Requirements

- 20.1. All valuation advice and the work undertaken in its preparation *must* be appropriate for the intended *purpose*.
- 20.2. A *valuer must* ensure that the intended recipient(s) of the valuation advice understand(s) what is to be provided and any limitations on its use before it is finalised and reported.
- 20.3. A *valuer must* communicate the scope of work to its *client* prior to completion of the assignment, including the following:
- (a) Identity of the *valuer*: The *valuer may* be an individual, group of individuals or a firm. If the *valuer* has any material connection or

involvement with the subject *asset* or the other parties to the valuation assignment, or if there are any other factors that could limit the *valuer's* ability to provide an unbiased and objective valuation, such factors *must* be disclosed at the outset. If such disclosure does not take place, the valuation assignment is not in compliance with IVS. If the *valuer* needs to seek *material* assistance from others in relation to any aspect of the assignment, the nature of such assistance and the extent of reliance *must* be made clear.

- (b) Identity of the *client(s)* (if any): Confirmation of those for whom the valuation assignment is being produced is important when determining the form and content of the report to ensure that it contains information relevant to their needs.
- (c) Identity of other intended users (if any): It is important to understand whether there are any other intended users of the valuation report, their identity and their needs, to ensure that the report content and format meets those users' needs.
- (d) *Asset(s)* being valued: The *subject asset* in the valuation assignment *must* be clearly identified.
- (e) The valuation currency: The currency for the valuation and the final valuation report or conclusion *must* be established. For example, a valuation might be prepared in euros or US dollars. This requirement is particularly important for valuation assignments involving *assets* in multiple countries and/or cash flows in multiple currencies.
- (f) *Purpose of the valuation*: The *purpose* for which the valuation assignment is being prepared *must* be clearly identified as it is important that valuation advice is not used out of context or for *purposes* for which it is not intended. The *purpose of the valuation* will also typically influence or determine the basis/bases of value to be used.
- (g) Basis/bases of value used: As required by IVS 104 *Bases of Value*, the valuation basis *must* be appropriate for the *purpose of the valuation*. The source of the definition of any basis of value used *must* be cited or the basis explained. This requirement is not applicable to a valuation review where no opinion of value is to be provided and the reviewer is not required to comment on the basis of value used.
- (h) Valuation date: The valuation date *must* be stated. If the valuation date is different from the date on which the valuation report is issued or the date on which investigations are to be undertaken or completed then where appropriate, these dates *should* be clearly distinguished.
- (i) The nature and extent of the *valuer's* work and any limitations thereon: Any limitations or restrictions on the inspection, enquiry and/or analysis in the valuation assignment *must* be identified (see IVS *Framework*, paras 60.1-60.4) If relevant information is not available because the conditions of the assignment restrict the investigation, these restrictions and any necessary assumptions or special assumptions (see IVS 104 *Bases of Value*, paras 200.1-200.5) made as a result of the restriction *must* be identified.

- (j) The nature and sources of information upon which the *valuer* relies: The nature and source of any relevant information that is to be relied upon and the extent of any verification to be undertaken during the valuation process *must* be identified.
 - (k) *Significant* assumptions and/or special assumptions: All *significant* assumptions and special assumptions that are to be made in the conduct and reporting of the valuation assignment *must* be identified.
 - (l) The type of report being prepared: The format of the report, that is, how the valuation will be communicated, *must* be described.
 - (m) Restrictions on use, distribution and publication of the report: Where it is necessary or desirable to restrict the use of the valuation or those relying on it, the intended users and restrictions *must* be clearly communicated.
 - (n) That the valuation will be prepared in compliance with IVS and that the *valuer* will assess the appropriateness of all *significant* inputs: The nature of any departures *must* be explained, for example, identifying that the valuation was performed in accordance with IVS and local tax regulations. See *IVS Framework* paras 60.1-60.4 relating to departures.
- 20.4. Wherever possible, the scope of work *should* be established and agreed between parties to a valuation assignment prior to the *valuer* beginning work. However, in certain circumstances, the scope of a valuation engagement *may* not be clear at the start of that engagement. In such cases, as the scope becomes clear, *valuers must* communicate and agree the scope of work to their *client*.
- 20.5. A written scope of work *may* not be necessary. However, since *valuers* are responsible for communicating the scope of work to their *client*, a written scope of work *should* be prepared.
- 20.6. Some aspects of the scope of work *may* be addressed in documents such as standing engagement instructions, master services agreements or a company's internal policies and procedures.
- 30. Changes to Scope of Work**
- 30.1. Some of the items in para 20.3 *may* not be determinable until the valuation assignment is in progress, or changes to the scope *may* become necessary during the course of the assignment due to additional information becoming available or matters emerging that require further investigation. As such, whilst the scope of work *may* be established at the outset, it *may* also be established over time throughout the course of the assignment.
- 30.2. In valuation assignments where the scope of work changes over time, the items in para 20.3 and any changes made over time *must* be communicated to the *client* before the assignment is completed and the valuation report is issued.

IVS 102 Investigations and Compliance

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10. General Principle

- 10.1. To be compliant with IVS, valuation assignments, including valuation reviews, *must* be conducted in accordance with all of the principles set out in IVS that are appropriate for the *purpose* and the terms and conditions set out in the scope of work.

20. Investigations

- 20.1. Investigations made during the course of a valuation assignment *must* be appropriate for the *purpose of the valuation* assignment and the basis(es) of value. References to a valuation or valuation assignment in this standard include a valuation review.
- 20.2. Sufficient evidence *must* be assembled by means such as inspection, inquiry, computation and analysis to ensure that the valuation is properly supported. When determining the extent of evidence necessary, professional judgement is required to ensure the information to be obtained is adequate for the *purpose of the valuation*.
- 20.3. Limits *may* be agreed on the extent of the *valuer's* investigations. Any such limits *must* be noted in the scope of work. However, IVS 105 *Valuation Approaches and Methods*, para 10.7 requires *valuers* to perform sufficient analysis to evaluate all inputs and assumptions and their appropriateness for the *valuation purpose*. If limitations on investigations are so substantial that the *valuer* cannot sufficiently evaluate the inputs and assumptions, the valuation engagement *must* not state that it has been performed in compliance with IVS.
- 20.4. When a valuation assignment involves reliance on information supplied by a party other than the *valuer*, consideration *should* be given as to whether the information is credible or that the information *may* otherwise be relied upon without adversely affecting the credibility of the valuation opinion. *Significant* inputs provided to the *valuer* (eg, by management/owners), *may* require consideration, investigation and/or corroboration. In cases where credibility or reliability of information supplied cannot be supported, such information *should* not be used.
- 20.5. In considering the credibility and reliability of information provided, *valuers should* consider matters such as:
- (a) *the purpose of the valuation*,
 - (b) *the significance* of the information to the valuation conclusion,

- (c) the expertise of the source in relation to the subject matter, and
- (d) whether the source is independent of either the *subject asset* and/or the recipient of the valuation (see IVS 101 *Scope of Work*, paras 20.3 (a)).

20.6. The *purpose of the valuation*, the basis of value, the extent and limits on the investigations and any sources of information that *may* be relied upon are part of the valuation assignment's scope of work that *must* be communicated to all parties to the valuation assignment (see IVS 101 *Scope of Work*).

20.7. If, during the course of an assignment, it becomes clear that the investigations included in the scope of work will not result in a credible valuation, or information to be provided by third parties is either unavailable or inadequate, the valuation assignment will not comply with IVS.

30. Valuation Record

30.1. A record *must* be kept of the work performed during the valuation process and the basis for the work on which the conclusions were reached for a reasonable period after completion of the assignment, having regard to any relevant statutory, legal or regulatory requirements. Subject to any such requirements, this record *should* include the key inputs, all calculations, investigations and analyses relevant to the final conclusion, and a copy of any draft or final report(s) provided to the *client*.

40. Compliance with Other Standards

40.1. As noted in the IVS *Framework*, when statutory, legal, regulatory or other authoritative requirements *must* be followed that differ from some of the requirements within IVS, a *valuer must* follow the statutory, legal, regulatory or other authoritative requirements (called a "*departure*"). Such a valuation has still been performed in overall compliance with IVS.

40.2. Most other sets of requirements, such as those written by Valuation Professional Organisations, other professional bodies, or firms' internal policies and procedures, will not contradict IVS and, instead, typically impose additional requirements on *valuers*. Such standards *may* be followed in addition to IVS without being seen as *departures* as long as all of the requirements in IVS are fulfilled.

IVS 103 Reporting

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10. Introduction

- 10.1. It is essential that the valuation report communicates the information necessary for proper understanding of the valuation or valuation review. A report *must* provide the intended users with a clear understanding of the valuation.
- 10.2. To provide useful information, the report *must* set out a clear and accurate description of the scope of the assignment, its *purpose* and intended use (including any limitations on that use) and disclosure of any assumptions, special assumptions (IVS 104 *Bases of Value*, para 200.4), *significant* uncertainty or limiting conditions that directly affect the valuation.
- 10.3. This standard applies to all valuation reports or reports on the outcome of a valuation review which *may* range from comprehensive narrative reports to abbreviated summary reports.
- 10.4. For certain *asset* classes there *may* be variations from these standards or additional requirements to be reported upon. These are found in the relevant IVS *Asset Standards*.

20. General Requirements

- 20.1. The *purpose of the valuation*, the complexity of the *asset* being valued and the users' requirements will determine the level of detail appropriate to the valuation report. The format of the report *should* be agreed with all parties as part of establishing a scope of work (see IVS 101 *Scope of Work*).
- 20.2. Compliance with this standard does not require a particular form or format of report; however, the report *must* be sufficient to communicate to the intended users the scope of the valuation assignment, the work performed and the conclusions reached.
- 20.3. The report *should* also be sufficient for an appropriately experienced valuation professional with no prior involvement with the valuation engagement to review the report and understand the items in paras 30.1 and 40.1, as applicable.

30. Valuation Reports

- 30.1. Where the report is the result of an assignment involving the valuation of an *asset* or *assets*, the report *must* convey the following, at a minimum:
- (a) the scope of the work performed, including the elements noted in para 20.3 of IVS 101 *Scope of Work*, to the extent that each is applicable to the assignment,
 - (b) the approach or approaches adopted,
 - (c) the method or methods applied,
 - (d) the key inputs used,
 - (e) the assumptions made,
 - (f) the conclusion(s) of value and principal reasons for any conclusions reached, and
 - (g) the date of the report (which *may* differ from the valuation date).
- 30.2. Some of the above requirements *may* be explicitly included in a report or incorporated into a report through reference to other documents (engagement letters, scope of work documents, internal policies and procedures, etc).

40. Valuation Review Reports

- 40.1. Where the report is the result of a valuation review, the report *must* convey the following, at a minimum:
- (a) the scope of the review performed, including the elements noted in para 20.3 of IVS 101 *Scope of Work* to the extent each is applicable to the assignment,
 - (b) the valuation report being reviewed and the inputs and assumptions upon which that valuation was based,
 - (c) the reviewer's conclusions about the work under review, including supporting reasons, and
 - (d) the date of the report (which *may* differ from the valuation date).
- 40.2. Some of the above requirements *may* be explicitly included in a report or incorporated into a report through reference to other documents (eg, engagement letters, scope of work documents, internal policies and procedures, etc).

IVS 104 Bases of Value

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Compliance with this mandatory standard requires a valuer to select the appropriate basis (or bases) of value and follow all applicable requirements associated with that basis of value, whether those requirements are included as part of this standard (for IVS-defined bases of value) or not (for non-IVS-defined bases of value).

10. Introduction

- 10.1. Bases of value (sometimes called standards of value) describe the fundamental premises on which the reported values will be based. It is critical that the basis (or bases) of value be appropriate to the terms and *purpose of the valuation* assignment, as a basis of value *may* influence or dictate a *valuer's* selection of methods, inputs and assumptions, and the ultimate opinion of value.
- 10.2. A *valuer may* be required to use bases of value that are defined by statute, regulation, private contract or other document. Such bases have to be interpreted and applied accordingly.

- 10.3. While there are many different bases of value used in valuations, most have certain common elements: an assumed transaction, an assumed date of the transaction and the assumed parties to the transaction.
- 10.4. Depending on the basis of value, the assumed transaction could take a number of forms:
 - (a) a hypothetical transaction,
 - (b) an actual transaction,
 - (c) a purchase (or entry) transaction,
 - (d) a sale (or exit) transaction, and/or
 - (e) a transaction in a particular or hypothetical market with specified characteristics.
- 10.5. The assumed date of a transaction will influence what information and data a *valuer* consider in a valuation. Most bases of value prohibit the consideration of information or market sentiment that would not be known or knowable with reasonable due diligence on the measurement/valuation date by *participants*.
- 10.6. Most bases of value reflect assumptions concerning the parties to a transaction and provide a certain level of description of the parties. In respect to these parties, they could include one or more actual or assumed characteristics, such as:
 - (a) hypothetical,
 - (b) known or specific parties,
 - (c) members of an identified/described group of potential parties,
 - (d) whether the parties are subject to particular conditions or motivations at the assumed date (eg, duress), and/or
 - (e) an assumed knowledge level.

20. Bases of Value

- 20.1. In addition to the IVS-defined bases of value listed below, the IVS have also provided a non-exhaustive list of other non-IVS-defined bases of value prescribed by individual *jurisdictional* law or those recognised and adopted by international agreement:
 - (a) IVS-defined bases of value:
 1. Market Value (section 30),
 2. Market Rent (section 40),
 3. Equitable Value (section 50),
 4. Investment Value/Worth (section 60),
 5. Synergistic Value (section 70), and
 6. Liquidation Value (section 80).

(b) Other bases of value (non-exhaustive list):

1. Fair Value (International Financial Reporting Standards) (section 90),
2. Fair Market Value (Organisation for Economic Co-operation and Development) (section 100),
3. Fair Market Value (United States Internal Revenue Service) (section 110), and
4. Fair Value (Legal/Statutory) (section 120):
 - a. the Model Business Corporation Act, and
 - b. Canadian case law (Manning v Harris Steel Group Inc).

- 20.2. *Valuers must* choose the relevant basis (or bases) of value according to the terms and *purpose of the valuation* assignment. The *valuer's* choice of a basis (or bases) of value *should* consider instructions and input received from the *client* and/or its representatives. However, regardless of instructions and input provided to the *valuer*, the *valuer should* not use a basis (or bases) of value that is inappropriate for the intended *purpose of the valuation* (for example, if instructed to use an IVS-defined basis of value for financial reporting *purposes* under IFRS, compliance with IVS *may* require the *valuer* to use a basis of value that is not defined or mentioned in the IVS).
- 20.3. In accordance with IVS 101 *Scope of Work*, the basis of value *must* be appropriate for the *purpose* and the source of the definition of any basis of value used *must* be cited or the basis explained.
- 20.4. *Valuers* are responsible for understanding the regulation, case law and other interpretive guidance related to all bases of value used.
- 20.5. The bases of value illustrated in sections 90-120 of this standard are defined by organisations other than the IVSC and the onus is on the *valuer* to ensure they are using the relevant definition.

30. IVS-Defined Basis of Value – Market Value

- 30.1. Market Value is the estimated amount for which an *asset* or liability *should* exchange on the valuation date between a willing buyer and a willing seller in an arm's length transaction, after proper marketing and where the parties had each acted knowledgeably, prudently and without compulsion.
- 30.2. The definition of Market Value *must* be applied in accordance with the following conceptual framework:
- (a) "The estimated amount" refers to a price expressed in terms of money payable for the *asset* in an arm's length market transaction. *Market Value* is the most probable price reasonably obtainable in the market on the valuation date in keeping with the market value definition. It is the best price reasonably obtainable by the seller and the most advantageous price reasonably obtainable by the buyer. This estimate specifically excludes an estimated price inflated or deflated by special terms or circumstances such as atypical financing, sale and leaseback arrangements, special considerations or concessions granted by anyone

associated with the sale, or any element of value available only to a specific owner or purchaser.

- (b) “An *asset* or liability *should* exchange” refers to the fact that the value of an *asset* or liability is an estimated amount rather than a predetermined amount or actual sale price. It is the price in a transaction that meets all the elements of the Market Value definition at the valuation date.
- (c) “On the valuation date” requires that the value is time-specific as of a given date. Because markets and market conditions *may* change, the estimated value *may* be incorrect or inappropriate at another time. The valuation amount will reflect the market state and circumstances as at the valuation date, not those at any other date.
- (d) “Between a willing buyer” refers to one who is motivated, but not compelled to buy. This buyer is neither over eager nor determined to buy at any price. This buyer is also one who purchases in accordance with the realities of the current market and with current market expectations, rather than in relation to an imaginary or hypothetical market that cannot be demonstrated or anticipated to exist. The assumed buyer would not pay a higher price than the market requires. The present owner is included among those who constitute “the market”.
- (e) “And a willing seller” is neither an over eager nor a forced seller prepared to sell at any price, nor one prepared to hold out for a price not considered reasonable in the current market. The willing seller is motivated to sell the *asset* at market terms for the best price attainable in the open market after proper marketing, whatever that price *may* be. The factual circumstances of the actual owner are not a part of this consideration because the willing seller is a hypothetical owner.
- (f) “In an arm’s length transaction” is one between parties who do not have a particular or special relationship, eg, parent and subsidiary companies or landlord and tenant, that *may* make the price level uncharacteristic of the market or inflated. The Market Value transaction is presumed to be between unrelated parties, each acting independently.
- (g) “After proper marketing” means that the *asset* has been exposed to the market in the most appropriate manner to effect its disposal at the best price reasonably obtainable in accordance with the Market Value definition. The method of sale is deemed to be that most appropriate to obtain the best price in the market to which the seller has access. The length of exposure time is not a fixed period but will vary according to the type of *asset* and market conditions. The only criterion is that *there must* have been sufficient time to allow the *asset* to be brought to the attention of an adequate number of market *participants*. The exposure period occurs prior to the valuation date.
- (h) “Where the parties had each acted knowledgeably, prudently” presumes that both the willing buyer and the willing seller are reasonably informed about the nature and characteristics of the *asset*, its actual and potential uses, and the state of the market as of the valuation date. Each is further presumed to use that knowledge prudently to seek the price that is most favourable for their respective positions in the transaction. Prudence is assessed by referring to the state of the market at the

valuation date, not with the benefit of hindsight at some later date. For example, it is not necessarily imprudent for a seller to sell *assets* in a market with falling prices at a price that is lower than previous market levels. In such cases, as is true for other exchanges in markets with changing prices, the prudent buyer or seller will act in accordance with the best market information available at the time.

- (i) “And without compulsion” establishes that each party is motivated to undertake the transaction, but neither is forced or unduly coerced to complete it.

- 30.3. The concept of Market Value presumes a price negotiated in an open and competitive market where the *participants* are acting freely. The market for an *asset* could be an international market or a local market. The market could consist of numerous buyers and sellers, or could be one characterised by a limited number of market *participants*. The market in which the *asset* is presumed exposed for sale is the one in which the *asset* notionally being exchanged is normally exchanged.
- 30.4. The Market Value of an *asset* will reflect its highest and best use (see paras 140.1-140.5). The highest and best use is the use of an *asset* that maximises its potential and that is possible, legally permissible and financially feasible. The highest and best use *may* be for continuation of an *asset*’s existing use or for some alternative use. This is determined by the use that a market *participant* would have in mind for the *asset* when formulating the price that it would be willing to bid.
- 30.5. The nature and source of the valuation inputs *must* be consistent with the basis of value, which in turn *must* have regard to the *valuation purpose*. For example, various approaches and methods *may* be used to arrive at an opinion of value providing they use market-derived data. The market approach will, by definition, use market-derived inputs. To indicate Market Value, the income approach *should* be applied, using inputs and assumptions that would be adopted by *participants*. To indicate Market Value using the cost approach, the cost of an *asset* of equal utility and the appropriate depreciation *should* be determined by analysis of market-based costs and depreciation.
- 30.6. The data available and the circumstances relating to the market for the *asset* being valued *must* determine which valuation method or methods are most relevant and appropriate. If based on appropriately analysed market-derived data, each approach or method used *should* provide an indication of Market Value.
- 30.7. Market Value does not reflect attributes of an *asset* that are of value to a specific owner or purchaser that are not available to other buyers in the market. Such advantages *may* relate to the physical, geographic, economic or legal characteristics of an *asset*. Market Value requires the disregard of any such element of value because, at any given date, it is only assumed that there is a willing buyer, not a particular willing buyer.

40. IVS-Defined Basis of Value – Market Rent

- 40.1. Market Rent is the estimated amount for which an interest in real property *should* be leased on the valuation date between a willing lessor and a willing lessee on appropriate lease terms in an arm's length transaction, after proper marketing and where the parties had each acted knowledgeably, prudently and without compulsion.
- 40.2. Market Rent *may* be used as a basis of value when valuing a lease or an interest created by a lease. In such cases, it is necessary to consider the contract rent and, where it is different, the market rent.
- 40.3. The conceptual framework supporting the definition of Market Value shown above can be applied to assist in the interpretation of Market Rent. In particular, the estimated amount excludes a rent inflated or deflated by special terms, considerations or concessions. The "appropriate lease terms" are terms that would typically be agreed in the market for the type of property on the valuation date between market *participants*. An indication of Market Rent *should* only be provided in conjunction with an indication of the principal lease terms that have been assumed.
- 40.4. Contract Rent is the rent payable under the terms of an actual lease. It *may* be fixed for the duration of the lease, or variable. The frequency and basis of calculating variations in the rent will be set out in the lease and *must* be identified and understood in order to establish the total benefits accruing to the lessor and the liability of the lessee.
- 40.5. In some circumstances the Market Rent *may* have to be assessed based on terms of an existing lease (eg, for rental determination *purposes* where the lease terms are existing and therefore not to be assumed as part of a notional lease).
- 40.6. In calculating Market Rent, the *valuer must* consider the following:
- (a) in regard to a Market Rent subject to a lease, the terms and conditions of that lease are the appropriate lease terms unless those terms and conditions are illegal or contrary to overarching legislation, and
 - (b) in regard to a Market Rent that is not subject to a lease, the assumed terms and conditions are the terms of a notional lease that would typically be agreed in a market for the type of property on the valuation date between market *participants*.

50. IVS-Defined Basis of Value – Equitable Value

- 50.1. Equitable Value is the estimated price for the transfer of an *asset* or liability between identified knowledgeable and willing parties that reflects the respective interests of those parties.
- 50.2. Equitable Value requires the assessment of the price that is fair between two specific, identified parties considering the respective advantages or disadvantages that each will gain from the transaction. In contrast, Market Value requires any advantages or disadvantages that would not be available to, or incurred by, market *participants* generally to be disregarded.

50.3. Equitable Value is a broader concept than Market Value. Although in many cases the price that is fair between two parties will equate to that obtainable in the market, there will be cases where the assessment of Equitable Value will involve taking into account matters that have to be disregarded in the assessment of Market Value, such as certain elements of Synergistic Value arising because of the combination of the interests.

50.4. Examples of the use of Equitable Value include:

- (a) determination of a price that is equitable for a shareholding in a non-quoted business, where the holdings of two specific parties *may* mean that the price that is equitable between them is different from the price that might be obtainable in the market, and
- (b) determination of a price that would be equitable between a lessor and a lessee for either the permanent transfer of the leased *asset* or the cancellation of the lease liability.

60. IVS-Defined Basis of Value – Investment Value/Worth

- 60.1. Investment Value is the value of an *asset* to a particular owner or prospective owner for individual investment or operational objectives.
- 60.2. Investment Value is an entity-specific basis of value. Although the value of an *asset* to the owner *may* be the same as the amount that could be realised from its sale to another party, this basis of value reflects the benefits received by an entity from holding the *asset* and, therefore, does not involve a presumed exchange. Investment Value reflects the circumstances and financial objectives of the entity for which the valuation is being produced. It is often used for measuring investment performance.

70. IVS-Defined Basis of Value – Synergistic Value

- 70.1. Synergistic Value is the result of a combination of two or more *assets* or interests where the combined value is more than the sum of the separate values. If the synergies are only available to one specific buyer then Synergistic Value will differ from Market Value, as the Synergistic Value will reflect particular attributes of an *asset* that are only of value to a specific purchaser. The added value above the aggregate of the respective interests is often referred to as “marriage value.”

80. IVS-Defined Basis of Value – Liquidation Value

- 80.1. Liquidation Value is the amount that would be realised when an *asset* or group of *assets* are sold on a piecemeal basis. Liquidation Value *should* take into account the costs of getting the *assets* into saleable condition as well as those of the disposal activity. Liquidation Value can be determined under two different premises of value:
 - (a) an orderly transaction with a typical marketing period (see section 160), or
 - (b) a forced transaction with a shortened marketing period (see section 170).
- 80.2. A *valuer must* disclose which premise of value is assumed.

**90. Other Basis of Value – Fair Value
(International Financial Reporting Standards)**

- 90.1. IFRS 13 defines Fair Value as the price that would be received to sell an *asset* or paid to transfer a liability in an orderly transaction between market *participants* at the measurement date.
- 90.2. For financial reporting *purposes*, over 130 countries require or permit the use of International Accounting Standards published by the International Accounting Standards Board. In addition, the Financial Accounting Standards Board in the United States uses the same definition of Fair Value in Topic 820.

100. Other Basis of Value – Fair Market Value (Organisation for Economic Co-operation and Development (OECD))

- 100.1. The OECD defines Fair Market Value as the price a willing buyer would pay a willing seller in a transaction on the open market.
- 100.2. OECD guidance is used in many engagements for international tax *purposes*.

**110. Other Basis of Value – Fair Market Value
(United States Internal Revenue Service)**

- 110.1. For United States tax *purposes*, Regulation §20.2031-1 states: “The fair market value is the price at which the property would change hands between a willing buyer and a willing seller, neither being under any compulsion to buy or to sell and both having reasonable knowledge of relevant facts.”

**120. Other Basis of Value – Fair Value (Legal/Statutory)
in different jurisdictions**

- 120.1. Many national, state and local agencies use Fair Value as a basis of value in a legal context. The definitions can vary *significantly* and *may* be the result of legislative action or those established by courts in prior cases.
- 120.2. Examples of US and Canadian definitions of Fair Value are as follows:
- (a) The Model Business Corporation Act (MBCA) is a model set of law prepared by the Committee on Corporate Laws of the Section of Business Law of the American Bar Association and is followed by 24 States in the United States. The definition of Fair Value from the MBCA is the value of the corporation’s shares determined:
- (1) immediately before the effectuation of the corporate action to which the shareholder objects,
 - (2) using customary and current valuation concepts and techniques generally employed for similar businesses in the context of the transaction requiring appraisal, and
 - (3) without discounting for lack of marketability or minority status except, if appropriate, for amendments to the articles pursuant to section 13.02(a)(5).

- (b) In 1986, the Supreme Court of British Columbia in Canada issued a ruling in *Manning v Harris Steel Group Inc.* that stated: “Thus, a ‘fair’ value is one which is just and equitable. That terminology contains within itself the concept of adequate compensation (indemnity), consistent with the requirements of justice and equity.”

130. Premise of Value/Assumed Use

130.1. A Premise of Value or Assumed Use describes the circumstances of how an *asset* or liability is used. Different bases of value *may* require a particular Premise of Value or allow the consideration of multiple Premises of Value. Some common Premises of Value are:

- (a) highest and best use,
- (b) current use/existing use,
- (c) orderly liquidation, and
- (d) forced sale.

140. Premise of Value – Highest and Best Use

- 140.1. Highest and best use is the use, from a *participant* perspective, that would produce the highest value for an *asset*. Although the concept is most frequently applied to non-financial *assets* as many financial *assets* do not have alternative uses, there *may* be circumstances where the highest and best use of financial *assets* needs to be considered.
- 140.2. The highest and best use *must* be physically possible (where applicable), financially feasible, legally allowed and result in the highest value. If different from the current use, the costs to convert an *asset* to its highest and best use would impact the value.
- 140.3. The highest and best use for an *asset* *may* be its current or existing use when it is being used optimally. However, highest and best use *may* differ from current use or even be an orderly liquidation.
- 140.4. The highest and best use of an *asset* valued on a stand-alone basis *may* be different from its highest and best use as part of a group of *assets*, when its contribution to the overall value of the group *must* be considered.
- 140.5. The determination of the highest and best use involves consideration of the following:
- (a) To establish whether a use is physically possible, regard will be had to what would be considered reasonable by *participants*.
 - (b) To reflect the requirement to be legally permissible, any legal restrictions on the use of the *asset*, eg, town planning/zoning designations, need to be taken into account as well as the likelihood that these restrictions will change.
 - (c) The requirement that the use be financially feasible takes into account whether an alternative use that is physically possible and legally permissible will generate sufficient return to a typical *participant*, after taking into account the costs of conversion to that use, over and above the return on the existing use.

150. Premise of Value – Current Use/Existing Use

- 150.1. Current use/existing use is the current way an *asset*, liability, or group of *assets* and/or liabilities is used. The current use *may* be, but is not necessarily, also the highest and best use.

160. Premise of Value – Orderly Liquidation

- 160.1. An orderly liquidation describes the value of a group of *assets* that could be realised in a liquidation sale, given a reasonable period of time to find a purchaser (or purchasers), with the seller being compelled to sell on an as-is, where-is basis.
- 160.2. The reasonable period of time to find a purchaser (or purchasers) *may* vary by *asset* type and market conditions.

170. Premise of Value – Forced Sale

- 170.1. The term “forced sale” is often used in circumstances where a seller is under compulsion to sell and that, as a consequence, a proper marketing period is not possible and buyers *may* not be able to undertake adequate due diligence. The price that could be obtained in these circumstances will depend upon the nature of the pressure on the seller and the reasons why proper marketing cannot be undertaken. It *may* also reflect the consequences for the seller of failing to sell within the period available. Unless the nature of, and the reason for, the constraints on the seller are known, the price obtainable in a forced sale cannot be realistically estimated. The price that a seller will accept in a forced sale will reflect its particular circumstances, rather than those of the hypothetical willing seller in the Market Value definition. A “forced sale” is a description of the situation under which the exchange takes place, not a distinct basis of value.
- 170.2. If an indication of the price obtainable under forced sale circumstances is required, it will be necessary to clearly identify the reasons for the constraint on the seller, including the consequences of failing to sell in the specified period by setting out appropriate assumptions. If these circumstances do not exist at the valuation date, these *must* be clearly identified as special assumptions.
- 170.3. A forced sale typically reflects the most probable price that a specified property is likely to bring under all of the following conditions:
- (a) consummation of a sale within a short time period,
 - (b) the *asset* is subjected to market conditions prevailing as of the date of valuation or assumed timescale within which the transaction is to be completed,
 - (c) both the buyer and the seller are acting prudently and knowledgeably,
 - (d) the seller is under compulsion to sell,
 - (e) the buyer is typically motivated,
 - (f) both parties are acting in what they consider their best interests,
 - (g) a normal marketing effort is not possible due to the brief exposure time, and
 - (h) payment will be made in cash.

- 170.4. Sales in an inactive or falling market are not automatically “forced sales” simply because a seller might hope for a better price if conditions improved. Unless the seller is compelled to sell by a deadline that prevents proper marketing, the seller will be a willing seller within the definition of Market Value (see paras 30.1-30.7).
- 170.5. While confirmed “forced sale” transactions would generally be excluded from consideration in a valuation where the basis of value is Market Value, it can be difficult to verify that an arm’s length transaction in a market was a forced sale.

180. Entity-Specific Factors

- 180.1. For most bases of value, the factors that are specific to a particular buyer or seller and not available to *participants* generally are excluded from the inputs used in a market-based valuation. Examples of entity-specific factors that *may* not be available to *participants* include:
- (a) additional value or reduction in value derived from the creation of a portfolio of similar *assets*,
 - (b) unique synergies between the *asset* and other *assets* owned by the entity,
 - (c) legal rights or restrictions applicable only to the entity,
 - (d) tax benefits or tax burdens unique to the entity, and
 - (e) an ability to exploit an *asset* that is unique to that entity.
- 180.2. Whether such factors are specific to the entity, or would be available to others in the market generally, is determined on a case-by-case basis. For example, an *asset may* not normally be transacted as a stand-alone item but as part of a group of *assets*. Any synergies with related *assets* would transfer to *participants* along with the transfer of the group and therefore are not entity specific.
- 180.3. If the objective of the basis of value used in a valuation is to determine the value to a specific owner (such as Investment Value/Worth discussed in paras 60.1 and 60.2), entity-specific factors are reflected in the valuation of the *asset*. Situations in which the value to a specific owner *may* be required include the following examples:
- (a) supporting investment decisions, and
 - (b) reviewing the performance of an *asset*.

190. Synergies

- 190.1. “Synergies” refer to the benefits associated with combining *assets*. When synergies are present, the value of a group of *assets* and liabilities is greater than the sum of the values of the individual *assets* and liabilities on a stand-alone basis. Synergies typically relate to a reduction in costs, and/or an increase in revenue, and/or a reduction in risk.
- 190.2. Whether synergies *should* be considered in a valuation depends on the basis of value. For most bases of value, only those synergies available

to other *participants* generally will be considered (see discussion of Entity-Specific Factors in paras 180.1-180.3).

- 190.3. An assessment of whether synergies are available to other *participants* *may* be based on the amount of the synergies rather than a specific way to achieve that synergy.

200. Assumptions and Special Assumptions

- 200.1. In addition to stating the basis of value, it is often necessary to make an assumption or multiple assumptions to clarify either the state of the *asset* in the hypothetical exchange or the circumstances under which the *asset* is assumed to be exchanged. Such assumptions can have a *significant* impact on value.
- 200.2. These types of assumptions generally fall into one of two categories:
- (a) assumed facts that are consistent with, or could be consistent with, those existing at the date of valuation, and
 - (b) assumed facts that differ from those existing at the date of valuation.
- 200.3. Assumptions related to facts that are consistent with, or could be consistent with, those existing at the date of valuation *may* be the result of a limitation on the extent of the investigations or enquiries undertaken by the *valuer*. Examples of such assumptions include, without limitation:
- (a) an assumption that a business is transferred as a complete operational entity,
 - (b) an assumption that *assets* employed in a business are transferred without the business, either individually or as a group,
 - (c) an assumption that an individually valued *asset* is transferred together with other complementary *assets*, and
 - (d) an assumption that a holding of shares is transferred either as a block or individually.
- 200.4. Where assumed facts differ from those existing at the date of valuation, it is referred to as a “special assumption”. Special assumptions are often used to illustrate the effect of possible changes on the value of an *asset*. They are designated as “special” so as to highlight to a valuation user that the valuation conclusion is contingent upon a change in the current circumstances or that it reflects a view that would not be taken by *participants* generally on the valuation date. Examples of such assumptions include, without limitation:
- (a) an assumption that a property is freehold with vacant possession,
 - (b) an assumption that a proposed building had actually been completed on the valuation date,
 - (c) an assumption that a specific contract was in existence on the valuation date which had not actually been completed, and
 - (d) an assumption that a financial instrument is valued using a yield curve that is different from that which would be used by a *participant*.

200.5. All assumptions and special assumptions *must* be reasonable under the circumstances, be supported by evidence, and be relevant having regard to the *purpose* for which the valuation is required.

210. Transaction Costs

210.1. Most bases of value represent the estimated exchange price of an *asset* without regard to the seller's costs of sale or the buyer's costs of purchase and without adjustment for any taxes payable by either party as a direct result of the transaction.

IVS 105 Valuation Approaches and Methods

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10. Introduction

- 10.1. Consideration *must* be given to the relevant and appropriate valuation approaches. The three approaches described and defined below are the main approaches used in valuation. They are all based on the economic principles of price equilibrium, anticipation of benefits or substitution. The principal valuation approaches are:
- (a) market approach,
 - (b) income approach, and
 - (c) cost approach.
- 10.2. Each of these valuation approaches includes different, detailed methods of application.
- 10.3. The goal in selecting valuation approaches and methods for an *asset* is to find the most appropriate method under the particular circumstances. No one method is suitable in every possible situation. The selection process *should* consider, at a minimum:
- (a) the appropriate basis(es) of value and premise(s) of value, determined by the terms and *purpose of the valuation* assignment,
 - (b) the respective strengths and weaknesses of the possible valuation approaches and methods,
 - (c) the appropriateness of each method in view of the nature of the *asset*, and the approaches or methods used by *participants* in the relevant market, and
 - (d) the availability of reliable information needed to apply the method(s).
- 10.4. *Valuers* are not required to use more than one method for the valuation of an *asset*, particularly when the *valuer* has a high degree of confidence in the accuracy and reliability of a single method, given the facts and circumstances of the valuation engagement. However, *valuers should* consider the use of multiple approaches and methods and more than one

valuation approach or method *should* be considered and *may* be used to arrive at an indication of value, particularly when there are insufficient factual or observable inputs for a single method to produce a reliable conclusion. Where more than one approach and method is used, or even multiple methods within a single approach, the conclusion of value based on those multiple approaches and/or methods *should* be reasonable and the process of analysing and reconciling the differing values into a single conclusion, without averaging, *should* be described by the *valuer* in the report.

- 10.5. While this standard includes discussion of certain methods within the Cost, Market and Income approaches, it does not provide a comprehensive list of all possible methods that *may* be appropriate. Some of the many methods not addressed in this standard include option pricing methods (OPMs), simulation/Monte Carlo methods and probability-*weighted* expected-return methods (PWERM). It is the *valuer's* responsibility to choose the appropriate method(s) for each valuation engagement. Compliance with IVS *may* require the *valuer* to use a method not defined or mentioned in the IVS.
- 10.6. When different approaches and/or methods result in widely divergent indications of value, a *valuer should* perform procedures to understand why the value indications differ, as it is generally not appropriate to simply *weight* two or more divergent indications of value. In such cases, *valuers should* reconsider the guidance in para 10.3 to determine whether one of the approaches/methods provides a better or more reliable indication of value.
- 10.7. *Valuers should* maximise the use of relevant observable market information in all three approaches. Regardless of the source of the inputs and assumptions used in a valuation, a *valuer must* perform appropriate analysis to evaluate those inputs and assumptions and their appropriateness for the *valuation purpose*.
- 10.8. Although no one approach or method is applicable in all circumstances, price information from an active market is generally considered to be the strongest evidence of value. Some bases of value *may* prohibit a *valuer* from making subjective adjustments to price information from an active market. Price information from an inactive market *may* still be good evidence of value, but subjective adjustments *may* be needed.

20. Market Approach

- 20.1. The market approach provides an indication of value by comparing the *asset* with identical or comparable (that is similar) *assets* for which price information is available.
- 20.2. The market approach *should* be applied and afforded *significant weight* under the following circumstances:
 - (a) the subject *asset* has recently been sold in a transaction appropriate for consideration under the basis of value,
 - (b) the subject *asset* or substantially similar *assets* are actively publicly traded, and/or
 - (c) there are frequent and/or recent observable transactions in substantially similar *assets*.

20.3. Although the above circumstances would indicate that the market approach *should* be applied and afforded *significant weight*, when the above criteria are not met, the following are additional circumstances where the market approach *may* be applied and afforded *significant weight*. When using the market approach under the following circumstances, a *valuer should* consider whether any other approaches can be applied and *weighted* to corroborate the value indication from the market approach:

- (a) Transactions involving the subject *asset* or substantially similar *assets* are not recent enough considering the levels of volatility and activity in the market.
- (b) The *asset* or substantially similar *assets* are publicly traded, but not actively.
- (c) Information on market transactions is available, but the comparable *assets* have *significant* differences to the subject *asset*, potentially requiring subjective adjustments.
- (d) Information on recent transactions is not reliable (ie, hearsay, missing information, synergistic purchaser, not arm's-length, distressed sale, etc).
- (e) The critical element affecting the value of the *asset* is the price it would achieve in the market rather than the cost of reproduction or its income-producing ability.

20.4. The heterogeneous nature of many *assets* means that it is often not possible to find market evidence of transactions involving identical or similar *assets*. Even in circumstances where the market approach is not used, the use of market-based inputs *should* be maximised in the application of other approaches (eg, market-based valuation metrics such as effective yields and rates of return).

20.5. When comparable market information does not relate to the exact or substantially the same *asset*, the *valuer must* perform a comparative analysis of qualitative and quantitative similarities and differences between the comparable *assets* and the subject *asset*. It will often be necessary to make adjustments based on this comparative analysis. Those adjustments *must* be reasonable and *valuers must* document the reasons for the adjustments and how they were quantified.

20.6. The market approach often uses market multiples derived from a set of comparables, each with different multiples. The selection of the appropriate multiple within the range requires judgement, considering qualitative and quantitative factors.

30. Market Approach Methods

Comparable Transactions Method

30.1. The comparable transactions method, also known as the guideline transactions method, utilises information on transactions involving *assets* that are the same or similar to the subject *asset* to arrive at an indication of value.

30.2. When the comparable transactions considered involve the subject *asset*, this method is sometimes referred to as the prior transactions method.

- 30.3. If few recent transactions have occurred, the *valuer may* consider the prices of identical or similar *assets* that are listed or offered for sale, provided the relevance of this information is clearly established, critically analysed and documented. This is sometimes referred to as the comparable listings method and *should* not be used as the sole indication of value but can be appropriate for consideration together with other methods. When considering listings or offers to buy or sell, the *weight* afforded to the listings/offer price *should* consider the level of commitment inherent in the price and how long the listing/offer has been on the market. For example, an offer that represents a binding commitment to purchase or sell an *asset* at a given price *may* be given more *weight* than a quoted price without such a binding commitment.
- 30.4. The comparable transaction method can use a variety of different comparable evidence, also known as units of comparison, which form the basis of the comparison. For example, a few of the many common units of comparison used for real property interests include price per square foot (or per square metre), rent per square foot (or per square metre) and capitalisation rates. A few of the many common units of comparison used in business valuation include EBITDA (Earnings Before Interest, Tax, Depreciation and Amortisation) multiples, earnings multiples, revenue multiples and book value multiples. A few of the many common units of comparison used in financial instrument valuation include metrics such as yields and interest rate spreads. The units of comparison used by *participants* can differ between *asset* classes and across industries and geographies.
- 30.5. A subset of the comparable transactions method is matrix pricing, which is principally used to value some types of financial instruments, such as debt securities, without relying exclusively on quoted prices for the specific securities, but rather relying on the securities' relationship to other benchmark quoted securities and their attributes (ie, yield).
- 30.6. The key steps in the comparable transactions method are:
- (a) identify the units of comparison that are used by *participants* in the relevant market,
 - (b) identify the relevant comparable transactions and calculate the key valuation metrics for those transactions,
 - (c) perform a consistent comparative analysis of qualitative and quantitative similarities and differences between the comparable *assets* and the subject *asset*,
 - (d) make necessary adjustments, if any, to the valuation metrics to reflect differences between the subject *asset* and the comparable *assets* (see para 30.12(d)),
 - (e) apply the adjusted valuation metrics to the subject *asset*, and
 - (f) if multiple valuation metrics were used, reconcile the indications of value.
- 30.7. A *valuer should* choose comparable transactions within the following context:
- (a) evidence of several transactions is generally preferable to a single transaction or event,

- (b) evidence from transactions of very similar *assets* (ideally identical) provides a better indication of value than *assets* where the transaction prices require *significant* adjustments,
- (c) transactions that happen closer to the valuation date are more representative of the market at that date than older/dated transactions, particularly in volatile markets,
- (d) for most bases of value, the transactions *should* be “arm’s length” between unrelated parties,
- (e) sufficient information on the transaction *should* be available to allow the *valuer* to develop a reasonable understanding of the comparable *asset* and assess the valuation metrics/comparable evidence,
- (f) information on the comparable transactions *should* be from a reliable and trusted source, and
- (g) actual transactions provide better valuation evidence than intended transactions.

30.8. A *valuer* *should* analyse and make adjustments for any material differences between the comparable transactions and the subject *asset*. Examples of common differences that could warrant adjustments *may* include, but are not limited to:

- (a) material characteristics (age, size, specifications, etc),
- (b) relevant restrictions on either the subject *asset* or the comparable *assets*,
- (c) geographical location (location of the *asset* and/or location of where the *asset* is likely to be transacted/used) and the related economic and regulatory environments,
- (d) profitability or profit-making capability of the *assets*,
- (e) historical and expected growth,
- (f) yields/coupon rates,
- (g) types of collateral,
- (h) unusual terms in the comparable transactions,
- (i) differences related to marketability and control characteristics of the comparable and the subject *asset*, and
- (j) ownership characteristics (eg, legal form of ownership, amount percentage held).

Guideline publicly-traded comparable method

30.9. The guideline publicly-traded method utilises information on publicly-traded comparables that are the same or similar to the subject *asset* to arrive at an indication of value.

- 30.10. This method is similar to the comparable transactions method. However, there are several differences due to the comparables being publicly traded, as follows:
- (a) the valuation metrics/comparable evidence are available as of the valuation date,
 - (b) detailed information on the comparables are readily available in public filings, and
 - (c) the information contained in public filings is prepared under well understood accounting standards.
- 30.11. The method *should* be used only when the subject *asset* is sufficiently similar to the publicly-traded comparables to allow for meaningful comparison.
- 30.12. The key steps in the guideline publicly-traded comparable method are to:
- (a) identify the valuation metrics/comparable evidence that are used by *participants* in the relevant market,
 - (b) identify the relevant guideline publicly-traded comparables and calculate the key valuation metrics for those transactions,
 - (c) perform a consistent comparative analysis of qualitative and quantitative similarities and differences between the publicly-traded comparables and the subject *asset*,
 - (d) make necessary adjustments, if any, to the valuation metrics to reflect differences between the subject *asset* and the publicly-traded comparables,
 - (e) apply the adjusted valuation metrics to the subject *asset*, and
 - (f) if multiple valuation metrics were used, *weight* the indications of value.
- 30.13. A *valuer should* choose publicly-traded comparables within the following context:
- (a) consideration of multiple publicly-traded comparables is preferred to the use of a single comparable,
 - (b) evidence from similar publicly-traded comparables (for example, with similar market segment, geographic area, size in revenue and/or *assets*, growth rates, profit margins, leverage, liquidity and diversification) provides a better indication of value than comparables that require *significant* adjustments, and
 - (c) securities that are actively traded provide more meaningful evidence than thinly-traded securities.
- 30.14. A *valuer should* analyse and make adjustments for any material differences between the guideline publicly-traded comparables and the subject *asset*. Examples of common differences that could warrant adjustments *may* include, but are not limited to:

- (a) material characteristics (age, size, specifications, etc),
- (b) relevant discounts and premiums (see para 30.17),
- (c) relevant restrictions on either the subject *asset* or the comparable *assets*,
- (d) geographical location of the underlying company and the related economic and regulatory environments,
- (e) profitability or profit-making capability of the *assets*,
- (f) historical and expected growth,
- (g) differences related to marketability and control characteristics of the comparable and the subject *asset*, and
- (h) type of ownership.

Other Market Approach Considerations

- 30.15. The following paragraphs address a non-exhaustive list of certain special considerations that *may* form part of a market approach valuation.
- 30.16. Anecdotal or “rule-of-thumb” valuation benchmarks are sometimes considered to be a market approach. However, value indications derived from the use of such rules *should* not be given substantial *weight* unless it can be shown that buyers and sellers place *significant* reliance on them.
- 30.17. In the market approach, the fundamental basis for making adjustments is to adjust for differences between the subject *asset* and the guideline transactions or publicly-traded securities. Some of the most common adjustments made in the market approach are known as discounts and premiums.
- (a) Discounts for Lack of Marketability (DLOM) *should* be applied when the comparables are deemed to have superior marketability to the subject *asset*. A DLOM reflects the concept that when comparing otherwise identical *assets*, a readily marketable *asset* would have a higher value than an *asset* with a long marketing period or restrictions on the ability to sell the *asset*. For example, publicly-traded securities can be bought and sold nearly instantaneously while shares in a private company *may* require a *significant* amount of time to identify potential buyers and complete a transaction. Many bases of value allow the consideration of restrictions on marketability that are inherent in the subject *asset* but prohibit consideration of marketability restrictions that are specific to a particular owner. DLOMs *may* be quantified using any reasonable method, but are typically calculated using option pricing models, studies that compare the value of publicly-traded shares and restricted shares in the same company, or studies that compare the value of shares in a company before and after an initial public offering.
 - (b) Control Premiums (sometimes referred to as *Market Participant Acquisition Premiums* or *MPAPs*) and Discounts for Lack of Control (DLOC) are applied to reflect differences between the comparables and the subject *asset* with regard to the ability to make decisions and the changes that can be made as a result of exercising control. All

else being equal, *participants* would generally prefer to have control over a subject *asset* than not. However, *participants'* willingness to pay a Control Premium or DLOC will generally be a factor of whether the ability to exercise control enhances the economic benefits available to the owner of the subject *asset*. Control Premiums and DLOCs *may* be quantified using any reasonable method, but are typically calculated based on either an analysis of the specific cash flow enhancements or reductions in risk associated with control or by comparing observed prices paid for controlling interests in publicly-traded securities to the publicly-traded price before such a transaction is announced. Examples of circumstances where Control Premiums and DLOC *should* be considered include where:

1. shares of public companies generally do not have the ability to make decisions related to the operations of the company (they lack control). As such, when applying the guideline public comparable method to value a subject *asset* that reflects a controlling interest, a control premium *may* be appropriate, or
 2. the guideline transactions in the guideline transaction method often reflect transactions of controlling interests. When using that method to value a subject *asset* that reflects a minority interest, a DLOC *may* be appropriate.
- (c) Blockage discounts are sometimes applied when the subject *asset* represents a large block of shares in a publicly-traded security such that an owner would not be able to quickly sell the block in the public market without negatively influencing the publicly-traded price. Blockage discounts *may* be quantified using any reasonable method but typically a model is used that considers the length of time over which a *participant* could sell the subject shares without negatively impacting the publicly-traded price (ie, selling a relatively small portion of the security's typical daily trading volume each day). Under certain bases of value, particularly fair value for financial reporting *purposes*, blockage discounts are prohibited.

40. Income Approach

- 40.1. The income approach provides an indication of value by converting future cash flow to a single current value. Under the income approach, the value of an *asset* is determined by reference to the value of income, cash flow or cost savings generated by the *asset*.
- 40.2. The income approach *should* be applied and afforded *significant weight* under the following circumstances:
- (a) the income-producing ability of the *asset* is the critical element affecting value from a *participant* perspective, and/or
 - (b) reasonable projections of the amount and timing of future income are available for the subject *asset*, but there are few, if any, relevant market comparables.

- 40.3. Although the above circumstances would indicate that the income approach *should* be applied and afforded *significant weight*, the following are additional circumstances where the income approach *may* be applied and afforded *significant weight*. When using the income approach under the following circumstances, a *valuer should* consider whether any other approaches can be applied and *weighted* to corroborate the value indication from the income approach:
- (a) the income-producing ability of the subject *asset* is only one of several factors affecting value from a *participant* perspective,
 - (b) there is *significant* uncertainty regarding the amount and timing of future income-related to the subject *asset*,
 - (c) there is a lack of access to information related to the subject *asset* (for example, a minority owner *may* have access to historical financial statements but not forecasts/budgets), and/or
 - (d) the subject *asset* has not yet begun generating income, but is projected to do so.
- 40.4. A fundamental basis for the income approach is that investors expect to receive a return on their investments and that such a return *should* reflect the perceived level of risk in the investment.
- 40.5. Generally, investors can only expect to be compensated for systematic risk (also known as “market risk” or “undiversifiable risk”).

50. Income Approach Methods

- 50.1. Although there are many ways to implement the income approach, methods under the income approach are effectively based on discounting future amounts of cash flow to present value. They are variations of the Discounted Cash Flow (DCF) method and the concepts below apply in part or in full to all income approach methods.

Discounted Cash Flow (DCF) Method

- 50.2. Under the DCF method the forecasted cash flow is discounted back to the valuation date, resulting in a present value of the *asset*.
- 50.3. In some circumstances for long-lived or indefinite-lived *assets*, DCF *may* include a terminal value which represents the value of the *asset* at the end of the explicit projection period. In other circumstances, the value of an *asset may* be calculated solely using a terminal value with no explicit projection period. This is sometimes referred to as an income capitalisation method.
- 50.4. The key steps in the DCF method are:
- (a) choose the most appropriate type of cash flow for the nature of the subject *asset* and the assignment (ie, pre-tax or post-tax, total cash flows or cash flows to equity, real or nominal, etc),
 - (b) determine the most appropriate explicit period, if any, over which the cash flow will be forecast,
 - (c) prepare cash flow forecasts for that period,

- (d) determine whether a terminal value is appropriate for the subject *asset* at the end of the explicit forecast period (if any) and then determine the appropriate terminal value for the nature of the *asset*,
- (e) determine the appropriate discount rate, and
- (f) apply the discount rate to the forecasted future cash flow, including the terminal value, if any.

Type of Cash Flow

- 50.5. When selecting the appropriate type of cash flow for the nature of *asset* or assignment, *valuers must* consider the factors below. In addition, the discount rate and other inputs *must* be consistent with the type of cash flow chosen.
- (a) Cash flow to whole *asset* or partial interest: Typically cash flow to the whole *asset* is used. However, occasionally other levels of income *may* be used as well, such as cash flow to equity (after payment of interest and principle on debt) or dividends (only the cash flow distributed to equity owners). Cash flow to the whole *asset* is most commonly used because an *asset should* theoretically have a single value that is independent of how it is financed or whether income is paid as dividends or reinvested.
 - (b) The cash flow can be pre-tax or post-tax: If a post-tax basis is used, the tax rate applied *should* be consistent with the basis of value and in many instances would be a *participant* tax rate rather than an owner-specific one.
 - (c) Nominal versus real: Real cash flow does not consider inflation whereas nominal cash flows include expectations regarding inflation. If expected cash flow incorporates an expected inflation rate, the discount rate has to include the same inflation rate.
 - (d) Currency: The choice of currency used *may* have an impact on assumptions related to inflation and risk. This is particularly true in emerging markets or in currencies with high inflation rates.
- 50.6. The type of cash flow chosen *should* be in accordance with *participant's* viewpoints. For example, cash flows and discount rates for real property are customarily developed on a pre-tax basis while cash flows and discount rates for businesses are normally developed on a post-tax basis. Adjusting between pre-tax and post-tax rates can be complex and prone to error and *should* be approached with caution.
- 50.7. When a valuation is being developed in a currency ("the valuation currency") that differs from the currency used in the cash flow projections ("the functional currency"), a *valuer should* use one of the following two currency translation methods:
- (a) Discount the cash flows in the functional currency using a discount rate appropriate for that functional currency. Convert the present value of the cash flows to the valuation currency at the spot rate on the valuation date.

- (b) Use a currency exchange forward curve to translate the functional currency projections into valuation currency projections and discount the projections using a discount rate appropriate for the valuation currency. When a reliable currency exchange forward curve is not available (for example, due to lack of liquidity in the relevant currency exchange markets), it *may* not be possible to use this method and only the method described in para 50.7(a) can be applied.

Explicit Forecast Period

- 50.8. The selection criteria will depend upon the *purpose of the valuation*, the nature of the *asset*, the information available and the required bases of value. For an *asset* with a short life, it is more likely to be both possible and relevant to project cash flow over its entire life.
- 50.9. *Valuers should* consider the following factors when selecting the explicit forecast period:
- (a) the life of the *asset*,
 - (b) a reasonable period for which reliable data is available on which to base the projections,
 - (c) the minimum explicit forecast period which *should* be sufficient for an *asset* to achieve a stabilised level of growth and profits, after which a terminal value can be used,
 - (d) in the valuation of cyclical *assets*, the explicit forecast period *should* generally include an entire cycle, when possible, and
 - (e) for finite-lived *assets* such as most financial instruments, the cash flows will typically be forecast over the full life of the *asset*.
- 50.10. In some instances, particularly when the *asset* is operating at a stabilised level of growth and profits at the valuation date, it *may* not be necessary to consider an explicit forecast period and a terminal value *may* form the only basis for value (sometimes referred to as an income capitalisation method).
- 50.11. The intended holding period for one investor *should* not be the only consideration in selecting an explicit forecast period and *should* not impact the value of an *asset*. However, the period over which an *asset* is intended to be held *may* be considered in determining the explicit forecast period if the objective of the valuation is to determine its investment value.

Cash Flow Forecasts

- 50.12. Cash flow for the explicit forecast period is constructed using prospective financial information (PFI) (projected income/inflows and expenditure/outflows).
- 50.13. As required by para 50.12, regardless of the source of the PFI (eg, management forecast), a *valuer must* perform analysis to evaluate the PFI, the assumptions underlying the PFI and their appropriateness for the *valuation purpose*. The suitability of the PFI and the underlying assumptions will depend upon the *purpose of the valuation* and the required bases of value. For example, cash flow used to determine market value *should* reflect

PFI that would be anticipated by *participants*; in contrast, investment value can be measured using cash flow that is based on the reasonable forecasts from the perspective of a particular investor.

- 50.14. The cash flow is divided into suitable periodic intervals (eg, weekly, monthly, quarterly or annually) with the choice of interval depending upon the nature of the *asset*, the pattern of the cash flow, the data available, and the length of the forecast period.
- 50.15. The projected cash flow *should* capture the amount and timing of all future cash inflows and outflows associated with the subject *asset* from the perspective appropriate to the basis of value.
- 50.16. Typically, the projected cash flow will reflect one of the following:
- (a) contractual or promised cash flow,
 - (b) the single most likely set of cash flow,
 - (c) the probability-*weighted* expected cash flow, or
 - (d) multiple scenarios of possible future cash flow.
- 50.17. Different types of cash flow often reflect different levels of risk and *may* require different discount rates. For example, probability-*weighted* expected cash flows incorporate expectations regarding all possible outcomes and are not dependent on any particular conditions or events (note that when a probability-*weighted* expected cash flow is used, it is not always necessary for *valuers* to take into account distributions of all possible cash flows using complex models and techniques. Rather, *valuers may* develop a limited number of discrete scenarios and probabilities that capture the array of possible cash flows). A single most likely set of cash flows *may* be conditional on certain future events and therefore could reflect different risks and warrant a different discount rate.
- 50.18. While *valuers* often receive PFI that reflects accounting income and expenses, it is generally preferable to use cash flow that would be anticipated by *participants* as the basis for valuations. For example, accounting non-cash expenses, such as depreciation and amortisation, *should* be added back, and expected cash outflows relating to capital expenditures or to changes in working capital *should* be deducted in calculating cash flow.
- 50.19. *Valuers must* ensure that seasonality and cyclicity in the subject has been appropriately considered in the cash flow forecasts.

Terminal Value

- 50.20. Where the *asset* is expected to continue beyond the explicit forecast period, *valuers must* estimate the value of the *asset* at the end of that period. The terminal value is then discounted back to the valuation date, normally using the same discount rate as applied to the forecast cash flow.
- 50.21. The terminal value *should* consider:
- (a) whether the *asset* is deteriorating/finite-lived in nature or indefinite-lived, as this will influence the method used to calculate a terminal value,

- (b) whether there is future growth potential for the *asset* beyond the explicit forecast period,
- (c) whether there is a pre-determined fixed capital amount expected to be received at the end of the explicit forecast period,
- (d) the expected risk level of the *asset* at the time the terminal value is calculated,
- (e) for cyclical *assets*, the terminal value *should* consider the cyclical nature of the *asset* and *should* not be performed in a way that assumes “peak” or “trough” levels of cash flows in perpetuity, and
- (f) the tax attributes inherent in the *asset* at the end of the explicit forecast period (if any) and whether those tax attributes would be expected to continue into perpetuity.

50.22. *Valuers may* apply any reasonable method for calculating a terminal value. While there are many different approaches to calculating a terminal value, the three most commonly used methods for calculating a terminal value are:

- (a) Gordon growth model/constant growth model (appropriate only for indefinite-lived *assets*),
- (b) market approach/exit value (appropriate for both deteriorating/finite-lived *assets* and indefinite-lived *assets*), and
- (c) salvage value/disposal cost (appropriate only for deteriorating/finite-lived *assets*).

Gordon Growth Model/Constant Growth Model

50.23. The constant growth model assumes that the *asset* grows (or declines) at a constant rate into perpetuity.

Market Approach/Exit Value

50.24. The market approach/exit value method can be performed in a number of ways, but the ultimate goal is to calculate the value of the *asset* at the end of the explicit cash flow forecast.

50.25. Common ways to calculate the terminal value under this method include application of a market-evidence based capitalisation factor or a market multiple.

50.26. When a market approach/exit value is used, *valuers should* comply with the requirements in the market approach and market approach methods section of this standard (sections 20 and 30). However, *valuers should* also consider the expected market conditions at the end of the explicit forecast period and make adjustments accordingly.

Salvage Value/Disposal Cost

50.27. The terminal value of some *assets may* have little or no relationship to the preceding cash flow. Examples of such *assets* include wasting *assets* such as a mine or an oil well.

- 50.28. In such cases, the terminal value is typically calculated as the salvage value of the *asset*, less costs to dispose of the *asset*. In circumstances where the costs exceed the salvage value, the terminal value is negative and referred to as a disposal cost or an *asset* retirement obligation.

Discount Rate

- 50.29. The rate at which the forecast cash flow is discounted *should* reflect not only the time value of money, but also the risks associated with the type of cash flow and the future operations of the *asset*.
- 50.30. *Valuers may* use any reasonable method for developing a discount rate. While there are many methods for developing or determining the reasonableness of a discount rate, a non-exhaustive list of common methods includes:
- (a) the capital *asset* pricing model (CAPM),
 - (b) the *weighted* average cost of capital (WACC),
 - (c) the observed or inferred rates/yields,
 - (d) the internal rate of return (IRR),
 - (e) the *weighted* average return on *assets* (WARA), and
 - (f) the build-up method (generally used only in the absence of market inputs).
- 50.31. In developing a discount rate, a *valuer should* consider:
- (a) the risk associated with the projections made in the cash flow used,
 - (b) the type of *asset* being valued. For example, discount rates used in valuing debt would be different to those used when valuing real property or a business,
 - (c) the rates implicit in transactions in the market,
 - (d) the geographic location of the *asset* and/or the location of the markets in which it would trade,
 - (e) the life/term of the *asset* and the consistency of inputs. For example, the risk-free rate considered would differ for an *asset* with a three-year life versus a 30-year life,
 - (f) the type of cash flow being used (see para 50.5), and
 - (g) the bases of value being applied. For most bases of value, the discount rate *should* be developed from the perspective of a *participant*.

60. Cost Approach

- 60.1. The cost approach provides an indication of value using the economic principle that a buyer will pay no more for an *asset* than the cost to obtain an *asset* of equal utility, whether by purchase or by construction, unless undue time, inconvenience, risk or other factors are involved. The approach

provides an indication of value by calculating the current replacement or reproduction cost of an *asset* and making deductions for physical deterioration and all other relevant forms of obsolescence.

60.2. The cost approach *should* be applied and afforded *significant weight* under the following circumstances:

- (a) *participants* would be able to recreate an *asset* with substantially the same utility as the subject *asset*, without regulatory or legal restrictions, and the *asset* could be recreated quickly enough that a *participant* would not be willing to pay a *significant* premium for the ability to use the subject *asset* immediately,
- (b) the *asset* is not directly income-generating and the unique nature of the *asset* makes using an income approach or market approach unfeasible, and/or
- (c) the basis of value being used is fundamentally based on replacement cost, such as replacement value.

60.3. Although the circumstances in para 60.2 would indicate that the cost approach *should* be applied and afforded *significant weight*, the following are additional circumstances where the cost approach *may* be applied and afforded *significant weight*. When using the cost approach under the following circumstances, a *valuer should* consider whether any other approaches can be applied and *weighted* to corroborate the value indication from the cost approach:

- (a) *participants* might consider recreating an *asset* of similar utility, but there are potential legal or regulatory hurdles or *significant* time involved in recreating the *asset*,
- (b) when the cost approach is being used as a reasonableness check to other approaches (for example, using the cost approach to confirm whether a business valued as a going-concern might be more valuable on a liquidation basis), and/or
- (c) the *asset* was recently created, such that there is a high degree of reliability in the assumptions used in the cost approach.

60.4. The value of a partially completed *asset* will generally reflect the costs incurred to date in the creation of the *asset* (and whether those costs contributed to value) and the expectations of *participants* regarding the value of the property when complete, but consider the costs and time required to complete the *asset* and appropriate adjustments for profit and risk.

70. Cost Approach Methods

70.1. Broadly, there are three cost approach methods:

- (a) replacement cost method: a method that indicates value by calculating the cost of a similar *asset* offering equivalent utility,
- (b) reproduction cost method: a method under the cost that indicates value by calculating the cost to recreating a replica of an *asset*, and
- (c) summation method: a method that calculates the value of an *asset* by the addition of the separate values of its component parts.

Replacement Cost Method

- 70.2. Generally, replacement cost is the cost that is relevant to determining the price that a *participant* would pay as it is based on replicating the utility of the *asset*, not the exact physical properties of the *asset*.
- 70.3. Usually replacement cost is adjusted for physical deterioration and all relevant forms of obsolescence. After such adjustments, this can be referred to as depreciated replacement cost.
- 70.4. The key steps in the replacement cost method are:
- (a) calculate all of the costs that would be incurred by a typical *participant* seeking to create or obtain an *asset* providing equivalent utility,
 - (b) determine whether there is any depreciation related to physical, functional and external obsolescence associated with the subject *asset*, and
 - (c) deduct total depreciation from the total costs to arrive at a value for the subject *asset*.
- 70.5. The replacement cost is generally that of a modern equivalent *asset*, which is one that provides similar function and equivalent utility to the *asset* being valued, but which is of a current design and constructed or made using current cost-effective materials and techniques.

Reproduction Cost Method

- 70.6. Reproduction cost is appropriate in circumstances such as the following:
- (a) the cost of a modern equivalent *asset* is greater than the cost of recreating a replica of the subject *asset*, or
 - (b) the utility offered by the subject *asset* could only be provided by a replica rather than a modern equivalent.
- 70.7. The key steps in the reproduction cost method are:
- (a) calculate all of the costs that would be incurred by a typical *participant* seeking to create an exact replica of the subject *asset*,
 - (b) determine whether there is any depreciation related to physical, functional and external obsolescence associated with the subject *asset*, and
 - (c) deduct total depreciation from the total costs to arrive at a value for the subject *asset*.

Summation Method

- 70.8. The summation method, also referred to as the underlying *asset* method, is typically used for investment companies or other types of *assets* or entities for which value is primarily a factor of the values of their holdings.
- 70.9. The key steps in the summation method are:
- (a) value each of the component *assets* that are part of the subject *asset* using the appropriate valuation approaches and methods, and

- (b) add the value of the component *assets* together to reach the value of the subject *asset*.

Cost Considerations

70.10. The cost approach *should* capture all of the costs that would be incurred by a typical *participant*.

70.11. The cost elements *may* differ depending on the type of the *asset* and *should* include the direct and indirect costs that would be required to replace/recreate the *asset* as of the valuation date. Some common items to consider include:

(a) direct costs:

1. materials, and
2. labour.

(b) indirect costs:

1. transport costs,
2. installation costs,
3. professional fees (design, permit, architectural, legal, etc),
4. other fees (commissions, etc),
5. overheads,
6. taxes,
7. finance costs (eg, interest on debt financing), and
8. profit margin/entrepreneurial profit to the creator of the *asset* (eg, return to investors).

70.12. An *asset* acquired from a third party would presumably reflect their costs associated with creating the *asset* as well as some form of profit margin to provide a return on their investment. As such, under bases of value that assume a hypothetical transaction, it *may* be appropriate to include an assumed profit margin on certain costs which can be expressed as a target profit, either a lump sum or a percentage return on cost or value. However, financing costs, if included, *may* already reflect *participants'* required return on capital deployed, so *valuers should* be cautious when including both financing costs and profit margins.

70.13. When costs are derived from actual, quoted or estimated prices by third party suppliers or contractors, these costs will already include a third parties' desired level of profit.

70.14. The actual costs incurred in creating the subject *asset* (or a comparable reference *asset*) *may* be available and provide a relevant indicator of the cost of the *asset*. However, adjustments *may* need to be made to reflect the following:

- (a) cost fluctuations between the date on which this cost was incurred and the valuation date, and
- (b) any atypical or exceptional costs, or savings, that are reflected in the cost data but that would not arise in creating an equivalent.

80. Depreciation/Obsolescence

- 80.1. In the context of the cost approach, “depreciation” refers to adjustments made to the estimated cost of creating an *asset* of equal utility to reflect the impact on value of any obsolescence affecting the subject *asset*. This meaning is different from the use of the word in financial reporting or tax law where it generally refers to a method for systematically expensing capital expenditure over time.
- 80.2. Depreciation adjustments are normally considered for the following types of obsolescence, which *may* be further divided into subcategories when making adjustments:
- (a) Physical obsolescence: Any loss of utility due to the physical deterioration of the *asset* or its components resulting from its age and usage.
 - (b) Functional obsolescence: Any loss of utility resulting from inefficiencies in the subject *asset* compared to its replacement such as its design, specification or technology being outdated.
 - (c) External or economic obsolescence: Any loss of utility caused by economic or locational factors external to the *asset*. This type of obsolescence can be temporary or permanent.
- 80.3. Depreciation/obsolescence *should* consider the physical and economic lives of the *asset*:
- (a) The physical life is how long the *asset* could be used before it would be worn out or beyond economic repair, assuming routine maintenance but disregarding any potential for refurbishment or reconstruction.
 - (b) The economic life is how long it is anticipated that the *asset* could generate financial returns or provide a non-financial benefit in its current use. It will be influenced by the degree of functional or economic obsolescence to which the *asset* is exposed.
- 80.4. Except for some types of economic or external obsolescence, most types of obsolescence are measured by making comparisons between the subject *asset* and the hypothetical *asset* on which the estimated replacement or reproduction cost is based. However, when market evidence of the effect of obsolescence on value is available, that evidence *should* be considered.
- 80.5. Physical obsolescence can be measured in two different ways:
- (a) curable physical obsolescence, ie, the cost to fix/cure the obsolescence, or

- (b) incurable physical obsolescence which considers the *asset's* age, expected total and remaining life where the adjustment for physical obsolescence is equivalent to the proportion of the expected total life consumed. Total expected life *may* be expressed in any reasonable way, including expected life in years, mileage, units produced, etc.

80.6. There are two forms of functional obsolescence:

- (a) excess capital cost, which can be caused by changes in design, materials of construction, technology or manufacturing techniques resulting in the availability of modern equivalent *assets* with lower capital costs than the subject *asset*, and
- (b) excess operating cost, which can be caused by improvements in design or excess capacity resulting in the availability of modern equivalent *assets* with lower operating costs than the subject *asset*.

80.7. Economic obsolescence *may* arise when external factors affect an individual *asset* or all the *assets* employed in a business and *should* be deducted after physical deterioration and functional obsolescence. For real estate, examples of economic obsolescence include:

- (a) adverse changes to demand for the products or services produced by the *asset*,
- (b) oversupply in the market for the *asset*,
- (c) a disruption or loss of a supply of labour or raw material, or
- (d) the *asset* being used by a business that cannot afford to pay a market rent for the *assets* and still generate a market rate of return.

80.8. Cash or cash equivalents do not suffer obsolescence and are not adjusted. Marketable *assets* are not adjusted below their market value determined using the market approach.

Asset Standards

IVS 200 Businesses and Business Interests

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10. Overview

- 10.1. The principles contained in the General Standards apply to valuations of businesses and business interests. This standard contains additional requirements that apply to valuations of businesses and business interests.

20. Introduction

- 20.1. The definition of what constitutes a business *may* differ depending on the *purpose of a valuation*. However, generally a business conducts a commercial, industrial, service or investment activity. Businesses can take many forms, such as corporations, partnerships, joint ventures and sole proprietorships. The value of a business *may* differ from the sum of the values of the individual *assets* or liabilities that make up that business. When a business value is greater than the sum of the recorded and unrecorded net tangible and identifiable intangible *assets* of the business, the excess value is often referred to as going concern value or goodwill.

- 20.2. When valuing individual *assets* or liabilities owned by a business, *valuers should* follow the applicable standard for that type of *asset* or liability (IVS 210 Intangible Assets, IVS 400 Real Property Interests, etc).
- 20.3. *Valuers must* establish whether the valuation is of the entire entity, shares or a shareholding in the entity (whether a controlling or non-controlling interest), or a specific business activity of the entity. The type of value being provided *must* be appropriate to the *purpose of the valuation* and communicated as part of the scope of the engagement (see IVS 101 *Scope of Work*). It is especially critical to clearly define the business or business interest being valued as, even when a valuation is performed on an entire entity, there *may* be different levels at which that value could be expressed. For example:
- (a) Enterprise value: Often described as the total value of the equity in a business plus the value of its debt or debt-related liabilities, minus any cash or cash equivalents available to meet those liabilities.
 - (b) Total invested capital value: The total amount of money currently invested in a business, regardless of the source, often reflected as the value of total *assets* less current liabilities and cash.
 - (c) Operating Value: The total value of the operations of the business, excluding the value of any non-operating *assets* and liabilities.
 - (d) Equity value: The value of a business to all of its equity shareholders.
- 20.4. Valuations of businesses are required for different *purposes* including acquisitions, mergers and sales of businesses, taxation, litigation, insolvency proceedings and financial reporting. Business valuations *may* also be needed as an input or step in other valuations such as the valuation of stock options, particular class(es) of stock, or debt.

30. Bases of Value

- 30.1. In accordance with IVS 104 Bases of Value, a *valuer must* select the appropriate basis(es) of value when valuing a business or business interest.
- 30.2. Often, business valuations are performed using bases of value defined by entities/organisations other than the IVSC (some examples of which are mentioned in IVS 104 Bases of Value) and it is the *valuer's* responsibility to understand and follow the regulation, case law and/or other interpretive guidance related to those bases of value as of the valuation date.

40. Valuation Approaches and Methods

- 40.1. The three principal valuation approaches described in IVS 105 *Valuation Approaches and Methods* *may* be applied to the valuation of businesses and business interests.
- 40.2. When selecting an approach and method, in addition to the requirements of this standard, a *valuer must* follow the requirements of IVS 105 *Valuation Approaches and Methods*, including para 10.3.

50. Market Approach

- 50.1. The market approach is frequently applied in the valuation of businesses and business interests as these *assets* often meet the criteria in IVS 105 *Valuation Approaches and Methods*, para 20.2 or 20.3. When valuing

businesses and business interests under the Market Approach, *valuers should* follow the requirements of IVS 105 *Valuation Approaches and Methods*, sections 20 and 30.

- 50.2. The three most common sources of data used to value businesses and business interests using the market approach are:
- (a) public stock markets in which ownership interests of similar businesses are traded,
 - (b) the acquisition market in which entire businesses or controlling interests in businesses are bought and sold, and
 - (c) prior transactions in shares or offers for the ownership of the subject business.
- 50.3. There *must* be a reasonable basis for comparison with, and reliance upon, similar businesses in the market approach. These similar businesses *should* be in the same industry as the subject business or in an industry that responds to the same economic variables. Factors that *should* be considered in assessing whether a reasonable basis for comparison exists include:
- (a) similarity to the subject business in terms of qualitative and quantitative business characteristics,
 - (b) amount and verifiability of data on the similar business, and
 - (c) whether the price of the similar business represents an arm's length and orderly transaction.
- 50.4. When applying a market multiple, adjustments such as those in para 60.8 *may* be appropriate to both the subject company and the comparable companies.
- 50.5. *Valuers should* follow the requirements of IVS 105 *Valuation Approaches and Methods*, paras 30.7 -30.8 when selecting and adjusting comparable transactions.
- 50.6. *Valuers should* follow the requirements of IVS 105 *Valuation Approaches and Methods*, paras 30.13 -30.14 when selecting and adjusting comparable public company information.

60. Income Approach

- 60.1. The income approach is frequently applied in the valuation of businesses and business interests as these *assets* often meet the criteria in IVS 105 *Valuation Approaches and Methods*, paras 40.2 or 40.3.
- 60.2. When the income approach is applied, *valuers should* follow the requirements of IVS 105 *Valuation Approaches and Methods*, sections 40 and 50.
- 60.3. Income and cash flow related to a business or business interest can be measured in a variety of ways and *may* be on a pre-tax or post-tax basis. The capitalisation or discount rate applied *must* be consistent with the type of income or cash flow used.

- 60.4. The type of income or cash flow used *should* be consistent with the type of interest being valued. For example:
- (a) enterprise value is typically derived using cash flows before debt servicing costs and an appropriate discount rate applicable to enterprise-level cash flows, such as a *weighted*-average cost of capital, and
 - (b) equity value *may* be derived using cash flows to equity, that is, after debt servicing costs and an appropriate discount rate applicable to equity-level cash flows, such as a cost of equity.
- 60.5. The income approach requires the estimation of a capitalisation rate when capitalising income or cash flow and a discount rate when discounting cash flow. In estimating the appropriate rate, factors such as the level of interest rates, rates of return expected by *participants* for similar investments and the risk inherent in the anticipated benefit stream are considered (see IVS 105 *Valuation Approaches and Methods*, paras 50.29-50.31).
- 60.6. In methods that employ discounting, expected growth *may* be explicitly considered in the forecasted income or cash flow. In capitalisation methods, expected growth is normally reflected in the capitalisation rate. If a forecasted cash flow is expressed in nominal terms, a discount rate that takes into account the expectation of future price changes due to inflation or deflation *should* be used. If a forecasted cash flow is expressed in real terms, a discount rate that takes no account of expected price changes due to inflation or deflation *should* be used.
- 60.7. Under the income approach, the historical financial statements of a business entity are often used as guide to estimate the future income or cash flow of the business. Determining the historical trends over time through ratio analysis *may* help provide the necessary information to assess the risks inherent in the business operations in the context of the industry and the prospects for future performance.
- 60.8. Adjustments *may* be appropriate to reflect differences between the actual historic cash flows and those that would be experienced by a buyer of the business interest on the valuation date. Examples include:
- (a) adjusting revenues and expenses to levels that are reasonably representative of expected continuing operations,
 - (b) presenting financial data of the subject business and comparison businesses on a consistent basis,
 - (c) adjusting non-arm's length transactions (such as contracts with customers or suppliers) to market rates,
 - (d) adjusting the cost of labour or of items leased or otherwise contracted from related parties to reflect market prices or rates,
 - (e) reflecting the impact of non-recurring events from historic revenue and expense items. Examples of non-recurring events include losses caused by strikes, new plant start-up and weather phenomena. However, the forecast cash flows *should* reflect any non-recurring revenues or expenses that can be reasonably anticipated and past occurrences *may* be indicative of similar events in the future, and

- (f) adjusting the inventory accounting to compare with similar businesses, whose accounts *may* be kept on a different basis from the subject business, or to more accurately reflect economic reality.

60.9. When using an income approach it *may* also be necessary to make adjustments to the valuation to reflect matters that are not captured in either the cash flow forecasts or the discount rate adopted. Examples *may* include adjustments for the marketability of the interest being valued or whether the interest being valued is a controlling or non-controlling interest in the business. However, *valuers should* ensure that adjustments to the valuation do not reflect factors that were already reflected in the cash flows or discount rate. For example, whether the interest being valued is a controlling or non-controlling interest is often already reflected in the forecasted cash flows.

60.10. While many businesses *may* be valued using a single cash flow scenario, *valuers may* also apply multi-scenario or simulation models, particularly when there is *significant* uncertainty as to the amount and/or timing of future cash flows.

70. Cost Approach

70.1. The cost approach cannot normally be applied in the valuation of businesses and business interests as these *assets* seldom meet the criteria in IVS 105 *Valuation Approaches and Methods*, paras 70.2 or 70.3. However, the cost approach is sometimes applied in the valuation of businesses, particularly when:

- (a) the business is an early stage or start-up business where profits and/or cash flow cannot be reliably determined and comparisons with other businesses under the market approach is impractical or unreliable,
- (b) the business is an investment or holding business, in which case the summation method is as described in IVS 105 *Valuation Approaches and Methods*, paras 70.8-70.9, and/or
- (c) the business does not represent a going concern and/or the value of its *assets* in a liquidation *may* exceed the business' value as a going concern.

70.2. In the circumstances where a business or business interest is valued using a cost approach, *valuers should* follow the requirements of IVS 105 *Valuation Approaches and Methods*, sections 70 and 80.

80. Special Considerations for Businesses and Business Interests

80.1. The following sections address a non-exhaustive list of topics relevant to the valuation of businesses and business interests:

- (a) Ownership Rights (section 90).
- (b) Business Information (section 100).
- (c) Economic and Industry Considerations (section 110).
- (d) Operating and Non-Operating Assets (section 120).
- (e) Capital Structure Considerations (section 130).

90. Ownership Rights

- 90.1. The rights, privileges or conditions that attach to the ownership interest, whether held in proprietorship, corporate or partnership form, require consideration in the valuation process. Ownership rights are usually defined within a *jurisdiction* by legal documents such as articles of association, clauses in the memorandum of the business, articles of incorporation, bylaws, partnership agreements and shareholder agreements (collectively “corporate documents”). In some situations, it *may* also be necessary to distinguish between legal and beneficial ownership.
- 90.2. Corporate documents *may* contain restrictions on the transfer of the interest or other provisions relevant to value. For example, corporate documents *may* stipulate that the interest *should* be valued as a pro rata fraction of the entire issued share capital regardless of whether it is a controlling or non-controlling interest. In each case, the rights of the interest being valued and the rights attaching to any other class of interest need to be considered at the outset.
- 90.3. Care *should* be taken to distinguish between rights and obligations inherent to the interest and those that *may* be applicable only to a particular shareholder (ie, those contained in an agreement between current shareholders which *may* not apply to a potential buyer of the ownership interest). Depending on the basis(es) of value used, the *valuer may* be required to consider only the rights and obligations inherent to the subject interest or both those rights and considerations inherent to the subject interest and those that apply to a particular owner.
- 90.4. All the rights and preferences associated with a subject business or business interest *should* be considered in a valuation, including:
- (a) if there are multiple classes of stock, the valuation *should* consider the rights of each different class, including, but not limited to:
 1. liquidation preferences,
 2. voting rights,
 3. redemption, conversion and participation provisions, and
 4. put and/or call rights.
 - (b) When a controlling interest in a business *may* have a higher value than a non-controlling interest. Control premiums or discounts for lack of control *may* be appropriate depending on the valuation method(s) applied (see IVS 105 *Valuation Approaches and Methods*, para 30.17.(b)). In respect of actual premiums paid in completed transactions, the *valuer should* consider whether the synergies and other factors that caused the acquirer to pay those premiums are applicable to the subject *asset* to a comparable degree.

100. Business Information

- 100.1. The valuation of a business entity or interest frequently requires reliance upon information received from management, representatives of the management or other experts. As required by IVS 105 *Valuation Approaches and Methods*, para 10.7, a *valuer must* assess the

reasonableness of information received from management, representatives of management or other experts and evaluate whether it is appropriate to rely on that information for the *valuation purpose*. For example, prospective financial information provided by management *may* reflect owner-specific synergies that *may* not be appropriate when using a basis of value that requires a *participant* perspective.

- 100.2. Although the value on a given date reflects the anticipated benefits of future ownership, the history of a business is useful in that it *may* give guidance as to the expectations for the future. *Valuers should* therefore consider the business' historical financial statements as part of a valuation engagement. To the extent the future performance of the business is expected to deviate *significantly* from historical experience, a *valuer must* understand why historical performance is not representative of the future expectations of the business.

110. Economic and Industry Considerations

- 110.1. Awareness of relevant economic developments and specific industry trends is essential for all valuations. Matters such as political outlook, government policy, exchange rates, inflation, interest rates and market activity *may* affect *assets* in different locations and/or sectors of the economy quite differently. These factors can be particularly important in the valuation of businesses and business interests, as businesses *may* have complex structures involving multiple locations and types of operations. For example, a business *may* be impacted by economic and industry factors specific related to:

- (a) the registered location of the business headquarters and legal form of the business,
- (b) the nature of the business operations and where each aspect of the business is conducted (ie, manufacturing *may* be done in a different location to where research and development is conducted),
- (c) where the business sells its goods and/or services,
- (d) the currency(ies) the business uses,
- (e) where the suppliers of the business are located, and
- (f) what tax and legal *jurisdictions* the business is subject to.

120. Operating and Non-Operating Assets

- 120.1. The valuation of an ownership interest in a business is only relevant in the context of the financial position of the business at a point in time. It is important to understand the nature of *assets* and liabilities of the business and to determine which items are required for use in the income-producing operations of the business and which ones are redundant or "excess" to the business at the valuation date.
- 120.2. Most valuation methods do not capture the value of *assets* that are not required for the operation of the business. For example, a business valued using a multiple of EBITDA would only capture the value the *assets* utilised in generating that level of EBITDA. If the business had non-operating *assets* or liabilities such as an idle manufacturing plant, the value of that

non-operating plant would not be captured in the value. Depending on the level of value appropriate for the valuation engagement (see para 20.3), the value of non-operating *assets* *may* need to be separately determined and added to the operating value of the business.

- 120.3. Businesses *may* have unrecorded *assets* and/or liabilities that are not reflected on the balance sheet. Such *assets* could include intangible *assets*, machinery and equipment that is fully depreciated and legal liabilities/lawsuits.
- 120.4. When separately considering non-operating *assets* and liabilities, a *valuer* *should* ensure that the income and expenses associated with non-operating *assets* are excluded from the cash flow measurements and projections used in the valuation. For example, if a business has a *significant* liability associated with an underfunded pension and that liability is valued separately, the cash flows used in the valuation of the business *should* exclude any “catch-up” payments related to that liability.
- 120.5. If the valuation considers information from publicly-traded businesses, the publicly-traded stock prices implicitly include the value of non-operating *assets*, if any. As such, *valuers* *must* consider adjusting information from publicly-traded businesses to exclude the value, income and expenses associated with non-operating *assets*.

130. Capital Structure Considerations

- 130.1. Businesses are often financed through a combination of debt and equity. However, in many cases, *valuers* *may* be asked to value only equity or a particular class of equity in a business. While equity or a particular class of equity can occasionally be valued directly, more often the enterprise value of the business is determined and then that value is allocated between debt and any types of equity.
- 130.2. When the value of debt is equal to its carrying value/book value, allocations of value *may* be straightforward. For example, in such cases it *may* be appropriate to deduct the book value of debt from enterprise value to calculate equity value (sometimes referred to as a “waterfall” method of value allocation). However, *valuers* *should* not necessarily assume that the value of debt and its book value are equal.
- 130.3. In circumstances where the value of debt *may* differ from its book value, *valuers* *should* either value the debt directly or use a method that appropriately allocates value to debt and any equity securities such as a probability-*weighted* expected return method or an option-pricing model.

IVS 210 Intangible Assets

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10. Overview

- 10.1. The principles contained in the General Standards apply to valuations of intangible *assets* and valuations with an intangible *assets* component. This standard contains additional requirements that apply to valuations of intangible *assets*.

20. Introduction

- 20.1. An intangible *asset* is a non-monetary *asset* that manifests itself by its economic properties. It does not have physical substance but grants rights and/or economic benefits to its owner.
- 20.2. Specific intangible *assets* are defined and described by characteristics such as their ownership, function, market position and image. These characteristics differentiate intangible *assets* from one another.
- 20.3. There are many types of intangible *assets*, but they are often considered to fall into one or more of the following categories (or goodwill):
- (a) Marketing-related: Marketing-related intangible *assets* are used primarily in the marketing or promotion of products or services. Examples include trademarks, trade names, unique trade design and internet domain names.
 - (b) Customer-related: Customer-related intangible *assets* include customer lists, backlog, customer contracts, and contractual and non-contractual customer relationships.
 - (c) Artistic-related: Artistic-related intangible *assets* arise from the right to benefits from artistic works such as plays, books, films and music, and from non-contractual copyright protection.

- (d) Contract-related: Contract-related intangible *assets* represent the value of rights that arise from contractual agreements. Examples include licensing and royalty agreements, service or supply contracts, lease agreements, permits, broadcast rights, servicing contracts, employment contracts and non-competition agreements and natural resource rights.
- (e) Technology-based: Technology-related intangible *assets* arise from contractual or non-contractual rights to use patented technology, unpatented technology, databases, formulae, designs, software, processes or recipes.

- 20.4. Although similar intangible *assets* within the same class will share some characteristics with one another, they will also have differentiating characteristics that will vary according to the type of intangible *asset*. In addition, certain intangible *assets*, such as brands, *may* represent a combination of categories in para 20.3,
- 20.5. Particularly in valuing an intangible *asset*, *valuers must* understand specifically what needs to be valued and the *purpose of the valuation*. For example, customer data (names, addresses, etc) typically has a very different value from customer contracts (those contracts in place on the valuation date) and customer relationships (the value of the ongoing customer relationship including existing and future contracts). What intangible *assets* need to be valued and how those intangible *assets* are defined *may* differ depending on the *purpose of the valuation*, and the differences in how intangible *assets* are defined can lead to *significant* differences in value.
- 20.6. Generally, goodwill is any future economic benefit arising from a business, an interest in a business or from the use of a group of *assets* which has not been separately recognised in another *asset*. The value of goodwill is typically measured as the residual amount remaining after the values of all identifiable tangible, intangible and monetary *assets*, adjusted for actual or potential liabilities, have been deducted from the value of a business. It is often represented as the excess of the price paid in a real or hypothetical acquisition of a company over the value of the company's other identified *assets* and liabilities. For some *purposes*, goodwill *may* need to be further divided into transferable goodwill (that which can be transferred to third parties) and non-transferable or "personal" goodwill.
- 20.7. As the amount of goodwill is dependent on which other tangible and intangible *assets* are recognised, its value can be different when calculated for different *purposes*. For example, in a business combination accounted for under IFRS or US GAAP, an intangible *asset* is only recognised to the extent that it:
- (a) is separable, ie, capable of being separated or divided from the entity and sold, transferred, licensed, rented or exchanged, either individually or together with a related contract, identifiable *asset* or liability, regardless of whether the entity intends to do so, or
 - (b) arises from contractual or other legal rights, regardless of whether those rights are transferable or separable from the entity or from other rights and obligations.

- 20.8. While the aspects of goodwill can vary depending on the *purpose of the valuation*, goodwill frequently includes elements such as:
- (a) company-specific synergies arising from a combination of two or more businesses (eg, reductions in operating costs, economies of scale or product mix dynamics),
 - (b) opportunities to expand the business into new and different markets,
 - (c) the benefit of an assembled workforce (but generally not any intellectual property developed by members of that workforce),
 - (d) the benefit to be derived from future *assets*, such as new customers and future technologies, and
 - (e) assemblage and going concern value.
- 20.9. *Valuers may* perform direct valuations of intangible *assets* where the value of the intangible *assets* is the *purpose* of the analysis or one part of the analysis. However, when valuing businesses, business interests, real property, and machinery and equipment, *valuers should* consider whether there are intangible *assets* associated with those *assets* and whether those directly or indirectly impact the *asset* being valued. For example, when valuing a hotel based on an income approach, the contribution to value of the hotel's brand *may* already be reflected in the profit generated by the hotel.
- 20.10. Intangible *asset* valuations are performed for a variety of *purposes*. It is the *valuer's* responsibility to understand the *purpose of a valuation* and whether intangible *assets should* be valued, whether separately or grouped with other *assets*. A non-exhaustive list of examples of circumstances that commonly include an intangible *asset* valuation component is provided below:
- (a) For financial reporting *purposes*, *valuations* of intangible *assets* are often required in connection with accounting for business combinations, *asset* acquisitions and sales, and impairment analysis.
 - (b) For tax reporting *purposes*, intangible *asset* valuations are frequently needed for transfer pricing analyses, estate and gift tax planning and reporting, and ad valorem taxation analyses.
 - (c) Intangible *assets may* be the subject of litigation, requiring valuation analysis in circumstances such as shareholder disputes, damage calculations and marital dissolutions (divorce).
 - (d) Other statutory or legal events *may* require the valuation of intangible *assets* such as compulsory purchases/ eminent domain proceedings.
 - (e) *Valuers* are often asked to value intangible *assets* as part of general consulting, collateral lending and transactional support engagements.

30. Bases of Value

- 30.1. In accordance with IVS 104 Bases of Value, a *valuer must* select the appropriate basis(es) of value when valuing intangible *assets*.

- 30.2. Often, intangible *asset* valuations are performed using bases of value defined by entities/organisations other than the IVSC (some examples of which are mentioned in IVS 104 Bases of Value) and the *valuer must* understand and follow the regulation, case law, and other interpretive guidance related to those bases of value as of the valuation date.

40. Valuation Approaches and Methods

- 40.1. The three valuation approaches described in IVS 105 Valuation Approaches can all be applied to the valuation of intangible *assets*.
- 40.2. When selecting an approach and method, in addition to the requirements of this standard, a *valuer must* follow the requirements of IVS 105 Valuation Approaches, including para 10.3.

50. Market Approach

- 50.1. Under the market approach, the value of an intangible *asset* is determined by reference to market activity (for example, transactions involving identical or similar *assets*).
- 50.2. Transactions involving intangible *assets* frequently also include other *assets*, such as a business combination that includes intangible *assets*.
- 50.3. *Valuers must* comply with paras 20.2 and 20.3 of IVS 105 when determining whether to apply the market approach to the valuation of intangible *assets*. In addition, *valuers should* only apply the market approach to value intangible *assets* if both of the following criteria are met:
- (a) information is available on arm's length transactions involving identical or similar intangible *assets* on or near the valuation date, and
 - (b) sufficient information is available to allow the *valuer* to adjust for all *significant* differences between the subject intangible *asset* and those involved in the transactions.
- 50.4. The heterogeneous nature of intangible *assets* and the fact that intangible *assets* seldom transact separately from other *assets* means that it is rarely possible to find market evidence of transactions involving identical *assets*. If there is market evidence at all, it is usually in respect of *assets* that are similar, but not identical.
- 50.5. Where evidence of either prices or valuation multiples is available, *valuers should* make adjustments to these to reflect differences between the subject *asset* and those involved in the transactions. These adjustments are necessary to reflect the differentiating characteristics of the subject intangible *asset* and the *assets* involved in the transactions. Such adjustments *may* only be determinable at a qualitative, rather than quantitative, level. However, the need for *significant* qualitative adjustments *may* indicate that another approach would be more appropriate for the valuation.
- 50.6. Consistent with the above, examples of intangible *assets* for which the market approach is sometimes used include:
- (a) broadcast spectrum,

(b) internet domain names, and

(c) taxi medallions.

50.7. The guideline transactions method is generally the only market approach method that can be applied to intangible *assets*.

50.8. In rare circumstances, a security sufficiently similar to a subject intangible *asset* *may* be publicly traded, allowing the use of the guideline public company method. One example of such securities is contingent value rights (CVRs) that are tied to the performance of a particular product or technology.

60. Income Approach

60.1. Under the income approach, the value of an intangible *asset* is determined by reference to the present value of income, cash flows or cost savings attributable to the intangible *asset* over its economic life.

60.2. *Valuers must* comply with paras 40.2 and 40.3 of IVS 105 *Valuation Approaches and Methods* when determining whether to apply the income approach to the valuation of intangible *assets*.

60.3. Income related to intangible *assets* is frequently included in the price paid for goods or a service. It *may* be challenging to separate the income related to the intangible *asset* from income related to other tangible and intangible *assets*. Many of the income approach methods are designed to separate the economic benefits associated with a subject intangible *asset*.

60.4. The income approach is the most common method applied to the valuation of intangible *assets* and is frequently used to value intangible *assets* including the following:

(a) technology,

(b) customer-related intangibles (eg, backlog, contracts, relationships),

(c) tradenames/trademarks/brands,

(d) operating licenses (eg, franchise agreements, gaming licenses, broadcast spectrum), and

(e) non-competition agreements.

Income Approach Methods

60.5. There are many income approach methods. The following methods are discussed in this standard in more detail:

(a) excess earnings method,

(b) relief-from-royalty method,

(c) premium profit method or with-and-without method,

(d) greenfield method, and

(e) distributor method.

Excess Earnings Method

- 60.6. The excess earnings method estimates the value of an intangible *asset* as the present value of the cash flows attributable to the subject intangible *asset* after excluding the proportion of the cash flows that are attributable to other *assets* required to generate the cash flows (“contributory *assets*”). It is often used for valuations where there is a requirement for the acquirer to allocate the overall price paid for a business between tangible *assets*, identifiable intangible *assets* and goodwill.
- 60.7. Contributory *assets* are *assets* that are used in conjunction with the subject intangible *asset* in the realisation of prospective cash flows associated with the subject intangible *asset*. *Assets* that do not contribute to the prospective cash flows associated with the subject intangible *asset* are not contributory *assets*.
- 60.8. The excess earnings method can be applied using several periods of forecasted cash flows (“multi-period excess earnings method” or “MPEEM”), a single period of forecasted cash flows (“single-period excess earnings method”) or by capitalising a single period of forecasted cash flows (“capitalised excess earnings method” or the “formula method”).
- 60.9. The capitalised excess earnings method or formula method is generally only appropriate if the intangible *asset* is operating in a steady state with stable growth/decay rates, constant profit margins and consistent contributory *asset* levels/charges.
- 60.10. As most intangible *assets* have economic lives exceeding one period, frequently follow non-linear growth/decay patterns and *may* require different levels of contributory *assets* over time, the MPEEM is the most commonly used excess earnings method as it offers the most flexibility and allows *valuers* to explicitly forecast changes in such inputs.
- 60.11. Whether applied in a single-period, multi-period or capitalised manner, the key steps in applying an excess earnings method are to:
- (a) forecast the amount and timing of future revenues driven by the subject intangible *asset* and related contributory *assets*,
 - (b) forecast the amount and timing of expenses that are required to generate the revenue from the subject intangible *asset* and related contributory *assets*,
 - (c) adjust the expenses to exclude those related to creation of new intangible *assets* that are not required to generate the forecasted revenue and expenses. Profit margins in the excess earnings method *may* be higher than profit margins for the overall business because the excess earnings method excludes investment in certain new intangible *assets*. For example:
 1. research and development expenditures related to development of new technology would not be required when valuing only existing technology, and
 2. marketing expenses related to obtaining new customers would not be required when valuing existing customer-related intangible *assets*.

- (d) identify the contributory *assets* that are needed to achieve the forecasted revenue and expenses. Contributory *assets* often include working capital, fixed *assets*, assembled workforce and identified intangible *assets* other than the subject intangible *asset*,
- (e) determine the appropriate rate of return on each contributory *asset* based on an assessment of the risk associated with that *asset*. For example, low-risk *assets* like working capital will typically have a relatively lower required return. Contributory intangible *assets* and highly specialised machinery and equipment often require relatively higher rates of return,
- (f) in each forecast period, deduct the required returns on contributory *assets* from the forecast profit to arrive at the excess earnings attributable to only the subject intangible *asset*,
- (g) determine the appropriate discount rate for the subject intangible *asset* and present value or capitalise the excess earnings, and
- (h) if appropriate for the *purpose of the valuation* (see paras 110.1-110.4), calculate and add the tax amortisation benefit (TAB) for the subject intangible *asset*.

60.12. Contributory *asset* charges (CACs) *should* be made for all the current and future tangible, intangible and financial *assets* that contribute to the generation of the cash flow, and if an *asset* for which a CAC is required is involved in more than one line of business, its CAC *should* be allocated to the different lines of business involved.

60.13. The determination of whether a CAC for elements of goodwill is appropriate *should* be based on an assessment of the relevant facts and circumstances of the situation, and the *valuer should* not mechanically apply CACs or alternative adjustments for elements of goodwill if the circumstances do not warrant such a charge. Assembled workforce, as it is quantifiable, is typically the only element of goodwill for which a CAC *should* be taken. Accordingly, *valuers must* ensure they have a strong basis for applying CACs for any elements of goodwill other than assembled workforce.

60.14. CACs are generally computed on an after-tax basis as a fair return on the value of the contributory *asset*, and in some cases a return of the contributory *asset* is also deducted. The appropriate return on a contributory *asset* is the investment return a typical *participant* would require on the *asset*. The return of a contributory *asset* is a recovery of the initial investment in the *asset*. There *should* be no difference in value regardless of whether CACs are computed on a pre-tax or after-tax basis.

60.15. If the contributory *asset* is not wasting in nature, like working capital, only a fair return on the *asset* is required.

60.16. For contributory intangible *assets* that were valued under a relief-from-royalty method, the CAC *should* be equal to the royalty (generally adjusted to an after-tax royalty rate).

- 60.17. The excess earnings method *should* be applied only to a single intangible *asset* for any given stream of revenue and income (generally the primary or most important intangible *asset*). For example, in valuing the intangible *assets* of a company utilising both technology and a tradename in delivering a product or service (ie, the revenue associated with the technology and the tradename is the same), the excess earnings method *should* only be used to value one of the intangible *assets* and an alternative method *should* be used for the other *asset*. However, if the company had multiple product lines, each using a different technology and each generating distinct revenue and profit, the excess earnings method *may* be applied in the valuation of the multiple different technologies.

Relief-from-Royalty Method

- 60.18. Under the relief-from-royalty method, the value of an intangible *asset* is determined by reference to the value of the hypothetical royalty payments that would be saved through owning the *asset*, as compared with licensing the intangible *asset* from a third party. Conceptually, the method *may* also be viewed as a discounted cash flow method applied to the cash flow that the owner of the intangible *asset* could receive through licensing the intangible *asset* to third parties.
- 60.19. The key steps in applying a relief-from-royalty method are to:
- (a) develop projections associated with the intangible *asset* being valued for the life of the subject intangible *asset*. The most common metric projected is revenue, as most royalties are paid as a percentage of revenue. However, other metrics such as a per-unit royalty *may* be appropriate in certain valuations,
 - (b) develop a royalty rate for the subject intangible *asset*. Two methods can be used to derive a hypothetical royalty rate. The first is based on market royalty rates for comparable or similar transactions. A prerequisite for this method is the existence of comparable intangible *assets* that are licensed at arm's length on a regular basis. The second method is based on a split of profits that would hypothetically be paid in an arm's length transaction by a willing licensee to a willing licensor for the rights to use the subject intangible *asset*,
 - (c) apply the selected royalty rate to the projections to calculate the royalty payments avoided by owning the intangible *asset*,
 - (d) estimate any additional expenses for which a licensee of the subject *asset* would be responsible. This can include upfront payments required by some licensors. A royalty rate *should* be analysed to determine whether it assumes expenses (such as maintenance, marketing and advertising) are the responsibility of the licensor or the licensee. A royalty rate that is "gross" would consider all responsibilities and expenses associated with ownership of a licensed *asset* to reside with the licensor, while a royalty that is "net" would consider some or all responsibilities and expenses associated with the licensed *asset* to reside with the licensee. Depending on whether the royalty is "gross" or "net", the valuation *should* exclude or include, respectively, a deduction for expenses such as maintenance, marketing or advertising expenses related to the hypothetically licensed *asset*.

- (e) if the hypothetical costs and royalty payments would be tax deductible, it *may* be appropriate to apply the appropriate tax rate to determine the after-tax savings associated with ownership of the intangible *asset*. However, for certain *purposes* (such as transfer pricing), the effects of taxes are generally not considered in the valuation and this step *should* be skipped,
- (f) determine the appropriate discount rate for the subject intangible *asset* and present value or capitalise the savings associated with ownership of the intangible *asset*, and
- (g) if appropriate for the *purpose of the valuation* (see paras 110.1-110.4), calculate and add the TAB for the subject intangible *asset*.

60.20. Whether a royalty rate is based on market transactions or a profit split method (or both), its selection *should* consider the characteristics of the subject intangible *asset* and the environment in which it is utilised. The consideration of those characteristics form the basis for selection of a royalty rate within a range of observed transactions and/or the range of profit available to the subject intangible *asset* in a profit split. Factors that *should* be considered include the following:

- (a) Competitive environment: The size of the market for the intangible *asset*, the availability of realistic alternatives, the number of competitors, barriers to entry and presence (or absence) of switching costs.
- (b) Importance of the subject intangible to the owner: Whether the subject *asset* is a key factor of differentiation from competitors, the importance it plays in the owner's marketing strategy, its relative importance compared with other tangible and intangible *assets*, and the amount the owner spends on creation, upkeep and improvement of the subject *asset*.
- (c) Life cycle of the subject intangible: The expected economic life of the subject *asset* and any risks of the subject intangible becoming obsolete.

60.21. When selecting a royalty rate, a *valuer should* also consider the following:

- (a) When entering a licence arrangement, the royalty rate *participants* would be willing to pay depends on their profit levels and the relative contribution of the licensed intangible *asset* to that profit. For example, a manufacturer of consumer products would not license a tradename at a royalty rate that leads to the manufacturer realising a lower profit selling branded products compared with selling generic products.
- (b) When considering observed royalty transactions, a *valuer should* understand the specific rights transferred to the licensee and any limitations. For example, royalty agreements *may* include *significant* restrictions on the use of a licensed intangible *asset* such as a restriction to a particular geographic area or for a product. In addition, the *valuer should* understand how the payments under the licensing agreement are structured, including whether there are upfront payments, milestone payments, puts/calls to acquire the licensed property outright, etc.

With-and-Without Method

- 60.22. The with-and-without method indicates the value of an intangible *asset* by comparing two scenarios: one in which the business uses the subject intangible *asset* and one in which the business does not use the subject intangible *asset* (but all other factors are kept constant).
- 60.23. The comparison of the two scenarios can be done in two ways:
- (a) calculating the value of the business under each scenario with the difference in the business values being the value of the subject intangible *asset*, and
 - (b) calculating, for each future period, the difference between the profits in the two scenarios. The present value of those amounts is then used to reach the value of the subject intangible *asset*.
- 60.24. In theory, either method *should* reach a similar value for the intangible *asset* provided the *valuer* considers not only the impact on the entity's profit, but additional factors such as differences between the two scenarios in working capital needs and capital expenditures.
- 60.25. The with-and-without method is frequently used in the valuation of non-competition agreements but *may* be appropriate in the valuation of other intangible *assets* in certain circumstances.
- 60.26. The key steps in applying the with-and-without method are to:
- (a) prepare projections of revenue, expenses, capital expenditures and working capital needs for the business assuming the use of all of the *assets* of the business including the subject intangible *asset*. These are the cash flows in the "with" scenario,
 - (b) use an appropriate discount rate to present value the future cash flows in the "with" scenario, and/or calculate the value of the business in the "with" scenario,
 - (c) prepare projections of revenue, expenses, capital expenditures and working capital needs for the business assuming the use of all of the *assets* of the business except the subject intangible *asset*. These are the cash flows in the "without" scenario,
 - (d) use an appropriate discount rate for the business, present value the future cash flows in the "with" scenario and/or calculate the value of the business in the "with" scenario,
 - (e) deduct the present value of cash flows or the value of the business in the "without" scenario from the present value of cash flows or value of the business in the "with" scenario, and
 - (f) if appropriate for the *purpose of the valuation* (see paras 110.1-110.4), calculate and add the TAB for the subject intangible *asset*.
- 60.27. As an additional step, the difference between the two scenarios *may* need to be probability-*weighted*. For example, when valuing a non-competition agreement, the individual or business subject to the agreement *may* choose not to compete, even if the agreement were not in place.

60.28. The differences in value between the two scenarios *should* be reflected solely in the cash flow projections rather than by using different discount rates in the two scenarios.

Greenfield Method

60.29. Under the greenfield method, the value of the subject intangible is determined using cash flow projections that assume the only *asset* of the business at the valuation date is the subject intangible. All other tangible and intangible *assets must* be bought, built or rented.

60.30. The greenfield method is conceptually similar to the excess earnings method. However, instead of subtracting contributory *asset* charges from the cash flow to reflect the contribution of contributory *assets*, the greenfield method assumes that the owner of the subject *asset* would have to build, buy or rent the contributory *assets*. When building or buying the contributory *assets*, the cost of a replacement *asset* of equivalent utility is used rather than a reproduction cost.

60.31. The greenfield method is often used to estimate the value of "enabling" intangible *assets* such as franchise agreements and broadcast spectrum.

60.32. The key steps in applying the greenfield method are to:

- (a) prepare projections of revenue, expenses, capital expenditures and working capital needs for the business assuming the subject intangible *asset* is the only *asset* owned by the subject business at the valuation date, including the time period needed to "ramp up" to stabilised levels,
- (b) estimate the timing and amount of expenditures related to the acquisition, creation or rental of all other *assets* needed to operate the subject business,
- (c) using an appropriate discount rate for the business, present value the future cash flows to determine the value of the subject business with only the subject intangible in place, and
- (d) if appropriate for the *purpose of the valuation* (see paras 110.1-110.4), calculate and add the TAB for the subject intangible *asset*.

Distributor Method

60.33. The distributor method, sometimes referred to as the disaggregated method, is a variation of the multi-period excess earnings method sometimes used to value customer-related intangible *assets*. The underlying theory of the distributor method is that businesses that are comprised of various functions are expected to generate profits associated with each function. As distributors generally only perform functions related to distribution of products to customers rather than development of intellectual property or manufacturing, information on profit margins earned by distributors is used to estimate the excess earnings attributable to customer-related intangible *assets*.

60.34. The distributor method is appropriate to value customer-related intangible *assets* when another intangible *asset* (for example, technology or a brand) is deemed to be the primary or most *significant* intangible *asset* and is valued under a multi-period excess earnings method.

60.35. The key steps in applying the distributor method are to:

- (a) prepare projections of revenue associated with existing customer relationships. This *should* reflect expected growth in revenue from existing customers as well as the effects of customer attrition,
- (b) identify comparable distributors that have customer relationships similar to the subject business and calculate the profit margins achieved by those distributors,
- (c) apply the distributor profit margin to the projected revenue,
- (d) identify the contributory *assets* related to performing a distribution function that are needed to achieve the forecast revenue and expenses. Generally distributor contributory *assets* include working capital, fixed *assets* and workforce. However, distributors seldom require other *assets* such as trademarks or technology. The level of required contributory *assets should* also be consistent with *participants* performing only a distribution function,
- (e) determine the appropriate rate of return on each contributory *asset* based on an assessment of the risk associated with that *asset*,
- (f) in each forecast period, deduct the required returns on contributory *assets* from the forecast distributor profit to arrive at the excess earnings attributable to only the subject intangible *asset*,
- (g) determine the appropriate discount rate for the subject intangible *asset* and present value the excess earnings, and
- (h) if appropriate for the *purpose of the valuation* (see paras 110.1-110.4), calculate and add the TAB for the subject intangible *asset*.

70. Cost Approach

- 70.1. Under the cost approach, the value of an intangible *asset* is determined based on the replacement cost of a similar *asset* or an *asset* providing similar service potential or utility.
- 70.2. *Valuers must* comply with paras 60.2 and 60.3 of IVS 105 *Valuation Approaches and Methods* when determining whether to apply the cost approach to the valuation of intangible *assets*.
- 70.3. Consistent with these criteria, the cost approach is commonly used for intangible *assets* such as the following:
 - (a) acquired third-party software,
 - (b) internally-developed and internally-used, non-marketable software, and
 - (c) assembled workforce.
- 70.4. The cost approach *may* be used when no other approach is able to be applied; however, a *valuer should* attempt to identify an alternative method before applying the cost approach in situations where the subject *asset* does not meet the criteria in paras 60.2 and 60.3 of IVS 105 *Valuation Approaches and Methods*.

- 70.5. There are broadly two main methods that fall under the cost approach: replacement cost and reproduction cost. However, many intangible *assets* do not have physical form that can be reproduced and *assets* such as software, which can be reproduced, generally derive value from their function/utility rather than their exact lines of code. As such, the replacement cost is most commonly applied to the valuation of intangible *assets*.
- 70.6. The replacement cost method assumes that a *participant* would pay no more for the *asset* than the cost that would be incurred to replace the *asset* with a substitute of comparable utility or functionality.
- 70.7. *Valuers should* consider the following when applying the replacement cost method:
- (a) the direct and indirect costs of replacing the utility of the *asset*, including labour, materials and overhead,
 - (b) whether the subject intangible *asset* is subject to obsolescence. While intangible *assets* do not become functionally or physically obsolete, they can be subject to economic obsolescence,
 - (c) whether it is appropriate to include a profit mark-up on the included costs. An *asset* acquired from a third party would presumably reflect their costs associated with creating the *asset* as well as some form of profit to provide a return on investment. As such, under bases of value (see IVS 104 Bases of Value) that assume a hypothetical transaction, it *may* be appropriate to include an assumed profit mark-up on costs. As noted in IVS 105 *Valuation Approaches and Methods*, costs developed based on estimates from third parties would be presumed to already reflect a profit mark-up, and
 - (d) opportunity costs *may* also be included, which reflect costs associated with not having the subject intangible *asset* in place for some period of time during its creation.

80. Special Considerations for Intangible Assets

- 80.1. The following sections address a non-exhaustive list of topics relevant to the valuation of intangible *assets*.
- (a) Discount Rates/Rates of Return for Intangible Assets (section 90).
 - (b) Intangible Asset Economic Lives (section 100).
 - (c) Tax Amortisation Benefit (section 110).

90. Discount Rates/Rates of Return for Intangible Assets

- 90.1. Selecting discount rates for intangible *assets* can be challenging as observable market evidence of discount rates for intangible *assets* is rare. The selection of a discount rate for an intangible *asset* generally requires *significant* professional judgment.
- 90.2. In selecting a discount rate for an intangible *asset*, *valuers should* perform an assessment of the risks associated with the subject intangible *asset* and consider observable discount rate benchmarks.

- 90.3. When assessing the risks associated with an intangible *asset*, a *valuer should* consider factors including the following:
- (a) intangible *assets* often have higher risk than tangible *assets*,
 - (b) if an intangible *asset* is highly specialised to its current use, it *may* have higher risk than *assets* with multiple potential uses,
 - (c) single intangible *assets may* have more risk than groups of *assets* (or businesses),
 - (d) intangible *assets* used in risky (sometimes referred to as non-routine) functions *may* have higher risk than intangible *assets* used in more low-risk or routine activities. For example, intangible *assets* used in research and development activities *may* be higher risk than those used in delivering existing products or services,
 - (e) the life of the *asset*. Similar to other investments, intangible *assets* with longer lives are often considered to have higher risk, all else being equal,
 - (f) intangible *assets* with more readily estimable cash flow streams, such as backlog, *may* have lower risk than similar intangible *assets* with less estimable cash flows, such as customer relationships.
- 90.4. Discount rate benchmarks are rates that are observable based on market evidence or observed transactions. The following are some of the benchmark rates that a *valuer should* consider:
- (a) risk-free rates with similar maturities to the life of the subject intangible *asset*,
 - (b) cost of debt or borrowing rates with maturities similar to the life of the subject intangible *asset*,
 - (c) cost of equity or equity rates or return for *participants* for the subject intangible *asset*,
 - (d) *weighted* average cost of capital (WACC) of *participants* for the subject intangible *asset* or of the company owning/using the subject intangible *asset*,
 - (e) in contexts involving a recent business acquisition including the subject intangible *asset*, the Internal Rate of Return (IRR) for the transaction *should* be considered, and
 - (f) in contexts involving a valuation of all *assets* of a business, the *valuer should* perform a *weighted* average return on *assets* (WARA) analysis to confirm reasonableness of selected discount rates.

100. Intangible Asset Economic Lives

- 100.1. An important consideration in the valuation of an intangible *asset*, particularly under the income approach, is the economic life of the *asset*. This *may* be a finite period limited by legal, technological, functional or economic factors; other *assets may* have an indefinite life. The economic life of an intangible *asset* is a different concept than the remaining useful life for accounting or tax *purposes*.

- 100.2. Legal, technological, functional and economic factors *must* be considered individually and together in making an assessment of the economic life. For example, a pharmaceutical technology protected by a patent *may* have a remaining legal life of five years before expiry of the patent, but a competitor drug with improved efficacy *may* be expected to reach the market in three years. This might cause the economic life of the patent to be assessed as only three years. In contrast, the expected economic life of the technology could extend beyond the life of the patent if the knowhow associated with the technology would have value in production of a generic drug beyond the expiration of the patent.
- 100.3. In estimating the economic life of an intangible *asset*, a *valuer should* also consider the pattern of use or replacement. Certain intangible *assets may* be abruptly replaced when a new, better or cheaper alternative becomes available, while others *may* be replaced slowly over time, such as when a software developer releases a new version of software every year but only replaces a portion of the existing code with each new release.
- 100.4. For customer-related intangibles, attrition is a key factor in estimating an economic life as well as the cash flows used to value the customer-related intangibles. Attrition applied in the valuation of intangible *assets* is a quantification of expectations regarding future losses of customers. While it is a forward-looking estimate, attrition is often based on historical observations of attrition.
- 100.5. There are a number of ways to measure and apply historical attrition:
- (a) a constant rate of loss (as a percentage of prior year balance) over the life of the customer relationships *may* be assumed if customer loss does not appear to be dependent on age of the customer relationship,
 - (b) a variable rate of loss *may* be used over the life of the customer relationships if customer loss is dependent on age of the customer relationship. In such circumstances, generally younger/new customers are lost at a higher rate than older, more established customer relationships,
 - (c) attrition *may* be measured based on either revenue or number of customers/customer count as appropriate, based on the characteristics of the customer group,
 - (d) customers *may* need to be segregated into different groups. For example, a company that sells products to distributors and retailers *may* experience different attrition rates for each group. Customers *may* also be segregated based on other factors such as geography, size of customer and type of product or service purchased, and
 - (e) the period used to measure attrition *may* vary depending on circumstances. For example, for a business with monthly subscribers, one month without revenue from a particular customer would indicate a loss of that customer. In contrast, for larger industrial products, a customer might not be considered “lost” unless there have been no sales to that customer for a year or more.

- 100.6. The application of any attrition factor *should* be consistent with the way attrition was measured. Correct application of attrition factor in first projection year (and therefore all subsequent years) *must* be consistent with form of measurement.
- (a) If attrition is measured based on the number of customers at the beginning-of-period versus end-of-period (typically a year), the attrition factor *should* be applied using a “mid-period” convention for the first projection year (as it is usually assumed that customers were lost throughout the year). For example, if attrition is measured by looking at the number of customers at the beginning of the year (100) versus the number remaining at the end of the year (90), on average the company had 95 customers during that year, assuming they were lost evenly throughout the year. Although the attrition rate could be described as 10%, only half of that *should* be applied in the first year.
 - (b) If attrition is measured by analysing year-over-year revenue or customer count, the resulting attrition factor *should* generally be applied without a mid-period adjustment. For example, if attrition is measured by looking at the number of customers that generated revenue in Year 1 (100) versus the number of those same customers that had revenue in Year 2 (90), application would be different even though the attrition rate could again be described as 10%.
- 100.7. Revenue-based attrition *may* include growth in revenue from existing customers unless adjustments are made. It is generally a best practice to make adjustments to separate growth and attrition in measurement and application.
- 100.8. It is a best practice for *valuers* to input historical revenue into the model being used and check how closely it predicts actual revenue from existing customers in subsequent years. If attrition has been measured and applied appropriately, the model *should* be reasonably accurate. For example, if estimates of future attrition were developed based on historical attrition observed from 20X0 through 20X5, a *valuer should* input the 20X0 customer revenue into the model and check whether it accurately predicts the revenue achieved from existing customers in 20X1, 20X2, etc.

110. Tax Amortisation Benefit (TAB)

- 110.1. In many tax *jurisdictions*, intangible *assets* can be amortised for tax *purposes*, reducing a taxpayer’s tax burden and effectively increasing cash flows. Depending on the *purpose of a valuation* and the *valuation method* used, it *may* be appropriate to include the value of TAB in the *value* of the intangible.
- 110.2. If the market or cost approach is used to value an intangible *asset*, the price paid to create or purchase the *asset* would already reflect the ability to amortise the *asset*. However, in the income approach, a TAB needs to be explicitly calculated and included, if appropriate.
- 110.3. For some *valuation purposes*, such as financial reporting, the appropriate basis of value assumes a hypothetical sale of the subject intangible *asset*. Generally, for those *purposes*, a TAB *should* be included when the income approach is used because a typical *participant* would be able to amortise

an intangible *asset* acquired in such a hypothetical transaction. For other *valuation purposes*, the assumed transaction might be of a business or group of *assets*. For those bases of value, it *may* be appropriate to include a TAB only if the transaction would result in a step-up in basis for the intangible *assets*.

110.4. There is some diversity in practice related to the appropriate discount rate to be used in calculating a TAB. *Valuers may* use either of the following:

- (a) a discount rate appropriate for a business utilising the subject *asset*, such as a *weighted* average cost of capital. Proponents of this view believe that, since amortisation can be used to offset the taxes on any income produced by the business, a discount rate appropriate for the business as a whole *should* be used, or
- (b) a discount rate appropriate for the subject *asset* (ie, the one used in the valuation of the *asset*). Proponents of this view believe that the valuation *should* not assume the owner of the subject *asset* has operations and income separate from the subject *asset* and that the discount rate used in the TAB calculation *should* be the same as that used in the valuation of the subject *asset*.

IVS 300 Plant and Equipment

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10. Overview

- 10.1. The principles contained in the General Standards apply to valuations of plant and equipment. This standard only includes modifications, additional principles or specific examples of how the General Standards apply for valuations to which this standard applies.

20. Introduction

- 20.1. Items of plant and equipment (which *may* sometimes be categorised as a type of personal property) are tangible *assets* that are usually held by an entity for use in the manufacturing/production or supply of goods or services, for rental by others or for administrative *purposes* and that are expected to be used over a period of time.
- 20.2. For lease of machinery and equipment, the right to use an item of machinery and equipment (such as a right arising from a lease) would also follow the guidance of this standard. It *must* also be noted that the “right to use” an *asset* could have a different life span than the service life (that takes into consideration of both preventive and predictive maintenance) of the underlying machinery and equipment itself and, in such circumstances, the service life span *must* be stated.
- 20.3. *Assets* for which the highest and best use is “in use” as part of a group of *assets must* be valued using consistent assumptions. Unless the *assets* belonging to the sub-systems *may* reasonably be separated independently from its main system, then the sub-systems *may* be valued separately, having consistent assumptions within the sub-systems. This will also cascade down to sub-sub-systems and so on.
- 20.4. Intangible *assets* fall outside the classification of plant and equipment *assets*. However, an intangible *asset may* have an impact on the value of plant and equipment *assets*. For example, the value of patterns and dies is often inextricably linked to associated intellectual property rights. Operating software, technical data, production records and patents are further examples of intangible *assets* that can have an impact on the value of plant and equipment *assets*, depending on whether or not they are included in the

valuation. In such cases, the valuation process will involve consideration of the inclusion of intangible *assets* and their impact on the valuation of the plant and equipment *assets*. When there is an intangible *asset* component, the *valuer should* also follow IVS 210 Intangible Assets.

- 20.5. A valuation of plant and equipment will normally require consideration of a range of factors relating to the *asset* itself, its environment and physical, functional and economic potential. Therefore, all plant and equipment *valuers should* normally inspect the subject *assets* to ascertain the condition of the plant and also to determine if the information provided to them is usable and related to the subject *assets* being valued. Examples of factors that *may* need to be considered under each of these headings include the following:

(a) *Asset*-related:

1. the *asset's* technical specification,
2. the remaining useful, economic or effective life, considering both preventive and predictive maintenance,
3. the *asset's* condition, including maintenance history,
4. any functional, physical and technological obsolescence,
5. if the *asset* is not valued in its current location, the costs of decommissioning and removal, and any costs associated with the *asset's* existing in-place location, such as installation and re-commissioning of *assets* to its optimum status,
6. for machinery and equipment that are used for rental *purposes*, the lease renewal options and other end-of-lease possibilities,
7. any potential loss of a complementary *asset*, eg, the operational life of a machine *may* be curtailed by the length of lease on the building in which it is located,
8. additional costs associated with additional equipment, transport, installation and commissioning, etc, and
9. in cases where the historical costs are not available for the machinery and equipment that *may* reside within a plant during a construction, the *valuer may* take references from the Engineering, Procurement, Construction ("EPC") contract.

(b) Environment-related:

1. the location in relation to the source of raw material and market for the product. The suitability of a location *may* also have a limited life, eg, where raw materials are finite or where demand is transitory,
2. the impact of any environmental or other legislation that either restricts utilisation or imposes additional operating or decommissioning costs,
3. radioactive substances that *may* be in certain machinery and equipment have a severe impact if not used or disposed of

appropriately. This will have a major impact on expense consideration and the environment,

4. toxic wastes which *may* be chemical in the form of a solid, liquid or gaseous state *must* be professionally stored or disposed of. This is critical for all industrial manufacturing, and
5. licences to operate certain machines in certain countries *may* be restricted.

(c) Economic-related:

1. the actual or potential profitability of the *asset* based on comparison of operating costs with earnings or potential earnings (see IVS 200 Business and Business Interests),
2. the demand for the product manufactured by the plant with regard to both macro- and micro-economic factors could impact on demand, and
3. the potential for the *asset* to be put to a more valuable use than the current use (ie, highest and best use).

20.6. Valuations of plant and equipment *should* reflect the impact of all forms of obsolescence on value.

20.7. To comply with the requirement to identify the *asset* or liability to be valued in IVS 101 *Scope of Work*, para 20.3.(d) to the extent it impacts on value, consideration *must* be given to the degree to which the *asset* is attached to, or integrated with, other *assets*. For example:

- (a) *assets may* be permanently attached to the land and could not be removed without substantial demolition of either the *asset* or any surrounding structure or building,
- (b) an individual machine *may* be part of an integrated production line where its functionality is dependent upon other *assets*,
- (c) an *asset may* be considered to be classified as a component of the real property (eg, a Heating, Ventilation and Air Conditioning System (HVAC)).

In such cases, it will be necessary to clearly define what is to be included or excluded from the valuation. Any special assumptions relating to the availability of any complementary *assets must* also be stated (see also para 20.8).

20.8. Plant and equipment connected with the supply or provision of services to a building are often integrated within the building and, once installed, are not separable from it. These items will normally form part of the real property interest. Examples include plant and equipment with the primary function of supplying electricity, gas, heating, cooling or ventilation to a building and equipment such as elevators. If the *purpose of the valuation* requires these items to be valued separately, the scope of work *must* include a statement to the effect that the value of these items would normally be included in the real property interest and *may* not be separately realisable. When different valuation assignments are undertaken to carry out valuations of the real

property interest and plant and equipment *assets* at the same location, care is necessary to avoid either omissions or double counting.

- 20.9. Because of the diverse nature and transportability of many items of plant and equipment, additional assumptions will normally be required to describe the situation and circumstances in which the *assets* are valued. In order to comply with IVS 101 *Scope of Work*, para 20.3.(k) these *must* be considered and included in the scope of work. Examples of assumptions that *may* be appropriate in different circumstances include:
- (a) that the plant and equipment *assets* are valued as a whole, in place and as part of an operating business,
 - (b) that the plant and equipment *assets* are valued as a whole, in place but on the assumption that the business is not yet in production,
 - (c) that the plant and equipment *assets* are valued as a whole, in place but on the assumption that the business is closed,
 - (d) that the plant and equipment *assets* are valued as a whole, in place but on the assumption that it is a forced sale (See IVS 104 Bases of Value),
 - (e) that the plant and equipment *assets* are valued as individual items for removal from their current location.
- 20.10. In some circumstances, it *may* be appropriate to report on more than one set of assumptions, eg, in order to illustrate the effect of business closure or cessation of operations on the value of plant and equipment.
- 20.11. In addition to the minimum requirements in IVS 103 Reporting, a valuation report on plant and equipment *must* include appropriate references to matters addressed in the scope of work. The report *must* also include comment on the effect on the reported value of any associated tangible or intangible *assets* excluded from the actual or assumed transaction scenario, eg, operating software for a machine or a continued right to occupy the land on which the item is situated.
- 20.12. Valuations of plant and equipment are often required for different *purposes* including financial reporting, leasing, secured lending, disposal, taxation, litigation and insolvency proceedings.

30. Bases of Value

- 30.1. In accordance with IVS 104 Bases of Value, a *valuer must* select the appropriate basis(es) of value when valuing plant and equipment.
- 30.2. Using the appropriate basis(es) of value and associated premise of value (see IVS 104 Bases of Value, sections 140-170) is particularly crucial in the valuation of plant and equipment because differences in value can be pronounced, depending on whether an item of plant and equipment is valued under an “in use” premise, orderly liquidation or forced liquidation (see IVS 104 Bases of Value, para 80.1). The value of most plant and equipment is particularly sensitive to different premises of value.
- 30.3. An example of forced liquidation conditions is where the *assets* have to be removed from a property in a timeframe that precludes proper marketing

because a lease of the property is being terminated. The impact of such circumstances on value needs careful consideration. In order to advise on the value likely to be realised, it will be necessary to consider any alternatives to a sale from the current location, such as the practicality and cost of removing the items to another location for disposal within the available time limit and any diminution in value due to moving the item from its working location.

40. Valuation Approaches and Methods

- 40.1. The three principal valuation approaches described in the IVS *may* all be applied to the valuation of plant and equipment *assets* depending on the nature of the *assets*, the information available, and the facts and circumstances surrounding the valuation.

50. Market Approach

- 50.1. For classes of plant and equipment that are homogenous, eg, motor vehicles and certain types of office equipment or industrial machinery, the market approach is commonly used as there *may* be sufficient data of recent sales of similar *assets*. However, many types of plant and equipment are specialised and where direct sales evidence for such items will not be available, care *must* be exercised in offering an income or cost approach opinion of value when available market data is poor or non-existent. In such circumstances it *may* be appropriate to adopt either the income approach or the cost approach to the valuation.

60. Income Approach

- 60.1. The income approach to the valuation of plant and equipment can be used where specific cash flows can be identified for the *asset* or a group of complementary *assets*, eg, where a group of *assets* forming a process plant is operating to produce a marketable product. However, some of the cash flows *may* be attributable to intangible *assets* and difficult to separate from the cash flow contribution of the plant and equipment. Use of the income approach is not normally practical for many individual items of plant or equipment; however, it can be utilised in assessing the existence and quantum of economic obsolescence for an *asset* or *asset* group.
- 60.2. When an income approach is used to value plant and equipment, the valuation *must* consider the cash flows expected to be generated over the life of the *asset(s)* as well as the value of the *asset* at the end of its life. Care *must* be exercised when plant and equipment is valued on an income approach to ensure that elements of value relating to intangible *assets*, goodwill and other contributory *assets* is excluded (see IVS 210 Intangible Assets).

70. Cost Approach

- 70.1. The cost approach is commonly adopted for plant and equipment, particularly in the case of individual *assets* that are specialised or special-use facilities. The first step is to estimate the cost to a market *participant* of replacing the subject *asset* by reference to the lower of either reproduction or replacement cost. The replacement cost is the cost of obtaining an alternative *asset* of equivalent utility; this can either be a modern equivalent providing the same functionality or the cost of

reproducing an exact replica of the subject *asset*. After concluding on a replacement cost, the value *should* be adjusted to reflect the impact on value of physical, functional, technological and economic obsolescence on value. In any event, adjustments made to any particular replacement cost *should* be designed to produce the same cost as the modern equivalent *asset* from an output and utility point of view.

- 70.2. An entity's actual costs incurred in the acquisition or construction of an *asset* *may* be appropriate for use as the replacement cost of an *asset* under certain circumstances. However, prior to using such historical cost information, the *valuer* *should* consider the following:
- (a) Timing of the historical expenditures: An entity's actual costs *may* not be relevant, or *may* need to be adjusted for inflation/indexation to an equivalent as of the valuation date, if they were not incurred recently due to changes in market prices, inflation/deflation or other factors.
 - (b) The basis of value: Care *must* be taken when adopting a particular market *participant's* own costings or profit margins, as they *may* not represent what typical market *participants* might have paid. The *valuer* *must* also consider the possibility that the entity's costs incurred *may* not be historical in nature due to prior purchase accounting or the purchase of used plant and equipment *assets*. In any case, historical costs *must* be trended using appropriate indices.
 - (c) Specific costs included: A *valuer* *must* consider all *significant* costs that have been included and whether those costs contribute to the value of the *asset* and for some bases of value, some amount of profit margin on costs incurred *may* be appropriate.
 - (d) Non-market components: Any costs, discounts or rebates that would not be incurred by, or available to, typical market *participants* *should* be excluded.

- 70.3. Having established the replacement cost, deductions *must* be made to reflect the physical, functional, technological and economic obsolescence as applicable (see IVS 105 *Valuation Approaches and Methods*, section 80).

Cost-to-Capacity Method

- 70.4. Under the cost-to-capacity method, the replacement cost of an *asset* with an actual or required capacity can be determined by reference to the cost of a similar *asset* with a different capacity.
- 70.5. The cost-to-capacity method is generally used in one of two ways:
- (a) to estimate the replacement cost for an *asset* or *assets* with one capacity where the replacement costs of an *asset* or *assets* with a different capacity are known (such as when the capacity of two subject *assets* could be replaced by a single *asset* with a known cost), or
 - (b) to estimate the replacement cost for a modern equivalent *asset* with capacity that matches foreseeable demand where the subject *asset* has excess capacity (as a means of measuring the penalty for the lack of utility to be applied as part of an economic obsolescence adjustment).

- 70.6. This method *may* only be used as a check method unless there is an existence of an exact comparison plant of the same designed capacity that resides within the same geographical area.
- 70.7. It is noted that the relationship between cost and capacity is often not linear, so some form of exponential adjustment *may* also be required.

80. Special Considerations for Plant and Equipment

- 80.1. The following section Financing Arrangements addresses a non-exhaustive list of topics relevant to the valuation of plant and equipment.

90. Financing Arrangements

- 90.1. Generally, the value of an *asset* is independent of how it is financed. However, in some circumstances the way items of plant and equipment are financed and the stability of that financing *may* need to be considered in valuation.
- 90.2. An item of plant and equipment *may* be subject to a leasing or financing arrangement. Accordingly, the *asset* cannot be sold without the lender or lessor being paid any balance outstanding under the financing arrangement. This payment *may* or *may* not exceed the unencumbered value of the item to the extent unusual/excessive for the industry. Depending upon the *purpose of the valuation*, it *may* be appropriate to identify any encumbered *assets* and to report their values separately from the unencumbered *assets*.
- 90.3. Items of plant and equipment that are subject to operating leases are the property of third parties and are therefore not included in a valuation of the *assets* of the lessee, subject to the lease meeting certain conditions. However, such *assets may* need to be recorded as their presence *may* impact on the value of owned *assets* used in association. In any event, prior to undertaking a valuation, the *valuer should* establish (in conjunction with client and/or advisors) whether *assets* are subject to operating lease, finance lease or loan, or other secured lending. The conclusion on this regard and wider *purpose of the valuation* will then dictate the appropriate basis and valuation methodology.

IVS 400 Real Property Interests

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10. Overview

- 10.1. The principles contained in the General Standards apply to valuations of real property interests. This standard contains additional requirements for valuations of real property interests.

20. Introduction

- 20.1. Property interests are normally defined by state or the law of individual *jurisdictions* and are often regulated by national or local legislation. Before undertaking a valuation of a real property interest, a *valuer must* understand the relevant legal framework that affects the interest being valued.
- 20.2. A real property interest is a right of ownership, control, use or occupation of land and buildings. There are three main types of interest:
- (a) the superior interest in any defined area of land. The owner of this interest has an absolute right of possession and control of the land and any buildings upon it in perpetuity, subject only to any subordinate interests and any statutory or other legally enforceable constraints,
 - (b) a subordinate interest that normally gives the holder rights of exclusive possession and control of a defined area of land or buildings for a defined period, eg, under the terms of a lease contract, and/or
 - (c) a right to use land or buildings but without a right of exclusive possession or control, eg, a right to pass over land or to use it only for a specified activity.
- 20.3. Intangible *assets* fall outside the classification of real property *assets*. However, an intangible *asset may* be associated with, and have a material impact on, the value of real property *assets*. It is therefore essential to be clear in the scope of work precisely what the valuation assignment is to include or exclude. For example, the valuation of a hotel can be inextricably linked to the hotel brand. In such cases, the valuation process will involve consideration of the inclusion of intangible *assets* and their impact on the

valuation of the real property and plant and equipment *assets*. When there is an intangible *asset* component, the *valuer should* also follow IVS 210 *Intangible Assets*.

- 20.4. Although different words and terms are used to describe these types of real property interest in different *jurisdictions*, the concepts of an unlimited absolute right of ownership, an exclusive interest for a limited period or a non-exclusive right for a specified *purpose* are common to most. The immovability of land and buildings means that it is the right that a party holds that is transferred in an exchange, not the physical land and buildings. The value, therefore, attaches to the legal interest rather than to the physical land and buildings.
- 20.5. To comply with the requirement to identify the *asset* to be valued in IVS 101 *Scope of Work*, para 20.3.(d) the following matters *must* be included:
 - (a) a description of the real property interest to be valued, and
 - (b) identification of any superior or subordinate interests that affect the interest to be valued.
- 20.6. To comply with the requirements to state the extent of the investigation and the nature and source of the information to be relied upon in IVS 101 *Scope of Work*, para 20.3.(j) and IVS 102 *Investigations and Compliance*, the following matters *must* be considered:
 - (a) the evidence required to verify the real property interest and any relevant related interests,
 - (b) the extent of any inspection,
 - (c) responsibility for information on the site area and any building floor areas,
 - (d) responsibility for confirming the specification and condition of any building,
 - (e) the extent of investigation into the nature, specification and adequacy of services,
 - (f) the existence of any information on ground and foundation conditions,
 - (g) responsibility for the identification of actual or potential environmental risks,
 - (h) legal permissions or restrictions on the use of the property and any buildings, as well as any expected or potential changes to legal permissions and restrictions.
- 20.7. Typical examples of special assumptions that *may* need to be agreed and confirmed in order to comply with IVS 101 *Scope of Work*, para 20.3. (k) include:
 - (a) that a defined physical change had occurred, eg, a proposed building is valued as if complete at the valuation date,

- (b) that there had been a change in the status of the property, eg, a vacant building had been leased or a leased building had become vacant at the valuation date,
- (c) that the interest is being valued without taking into account other existing interests, and
- (d) that the property is free from contamination or other environmental risks.

20.8. Valuations of real property interests are often required for different *purposes* including secured lending, sales and purchases, taxation, litigation, compensation, insolvency proceedings and financial reporting.

30. Bases of Value

- 30.1. In accordance with IVS 104 Bases of Value, a *valuer must* select the appropriate basis(es) of value when valuing real property interests.
- 30.2. Under most bases of value, a *valuer must* consider the highest and best use of the real property, which *may* differ from its current use (see IVS 104 Bases of Value, para 30.3). This assessment is particularly important to real property interests which can be changed from one use to another or that have development potential.

40. Valuation Approaches and Methods

- 40.1. The three valuation approaches described in the IVS 105 *Valuation Approaches and Methods* can all be applicable for the valuation of a real property interest.
- 40.2. When selecting an approach and method, in addition to the requirements of this standard, a *valuer must* follow the requirements of IVS 105 *Valuation Approaches and Methods*, including para 10.3 and 10.4.

50. Market Approach

- 50.1. Property interests are generally heterogeneous (ie, with different characteristics). Even if the land and buildings have identical physical characteristics to others being exchanged in the market, the location will be different. Notwithstanding these dissimilarities, the market approach is commonly applied for the valuation of real property interests.
- 50.2. In order to compare the subject of the valuation with the price of other real property interests, *valuers should* adopt generally accepted and appropriate units of comparison that are considered by *participants*, dependent upon the type of *asset* being valued. Units of comparison that are commonly used include:
 - (a) price per square metre (or per square foot) of a building or per hectare for land,
 - (b) price per room, and
 - (c) price per unit of output, eg, crop yields.
- 50.3. A unit of comparison is only useful when it is consistently selected and applied to the subject property and the comparable properties in each analysis. To the extent possible, any unit of comparison used *should* be one commonly used by *participants* in the relevant market.

- 50.4. The reliance that can be applied to any comparable price data in the valuation process is determined by comparing various characteristics of the property and transaction from which the data was derived with the property being valued. Differences between the following *should* be considered in accordance with IVS 105 *Valuation Approaches and Methods*, para 30.8. Specific differences that *should* be considered in valuing real property interests include, but are not limited to:
- (a) the type of interest providing the price evidence and the type of interest being valued,
 - (b) the respective locations,
 - (c) the respective quality of the land or the age and specification of the buildings,
 - (d) the permitted use or zoning at each property,
 - (e) the circumstances under which the price was determined and the basis of value required,
 - (f) the effective date of the price evidence and the valuation date, and
 - (g) market conditions at the time of the relevant transactions and how they differ from conditions at the valuation date.

60. Income Approach

- 60.1. Various methods are used to indicate value under the general heading of the income approach, all of which share the common characteristic that the value is based upon an actual or estimated income that either is, or could be, generated by an owner of the interest. In the case of an investment property, that income could be in the form of rent (see paras 90.1-90.3); in an owner-occupied building, it could be an assumed rent (or rent saved) based on what it would cost the owner to lease equivalent space.
- 60.2. For some real property interests, the income-generating ability of the property is closely tied to a particular use or business/trading activity (for example, hotels, golf courses, etc). Where a building is suitable for only a particular type of trading activity, the income is often related to the actual or potential cash flows that would accrue to the owner of that building from the trading activity. The use of a property's trading potential to indicate its value is often referred to as the "profits method".
- 60.3. When the income used in the income approach represents cash flow from a business/trading activity (rather than cash flow related to rent, maintenance and other real property-specific costs), the *valuer should* also comply as appropriate with the requirements of IVS 200 Business and Business Interests and, where applicable, IVS 210 Intangible Assets.
- 60.4. For real property interests, various forms of discounted cash flow models *may* be used. These vary in detail but share the basic characteristic that the cash flow for a defined future period is adjusted to a present value using a discount rate. The sum of the present day values for the individual periods represents an estimate of the capital value. The discount rate in a discounted cash flow model will be based on the time cost of money and the risks and rewards of the income stream in question.

- 60.5. Further information on the derivation of discount rates is included in IVS 105 *Valuation Approaches and Methods*, paras 50.29-50.31. The development of a yield or discount rate *should* be influenced by the objective of the valuation. For example:
- (a) if the objective of the valuation is to establish the value to a particular owner or potential owner based on their own investment criteria, the rate used *may* reflect their required rate of return or their *weighted* average cost of capital, and
 - (b) if the objective of the valuation is to establish the market value, the discount rate *may* be derived from observation of the returns implicit in the price paid for real property interests traded in the market between *participants* or from hypothetical *participants'* required rates or return. When a discount rate is based on an analysis of market transactions, *valuers should* also follow the guidance contained in IVS 105 Valuation Approaches and Methods, paras 30.7 and 30.8.
- 60.6. An appropriate discount rate *may* also be built up from a typical “risk-free” return adjusted for the additional risks and opportunities specific to the particular real property interest.

70. Cost Approach

- 70.1. In applying the cost approach, *valuers must* follow the guidance contained in IVS 105 *Valuation Approaches and Methods*, paras 70.1-70.14.
- 70.2. This approach is generally applied to the valuation of real property interests through the depreciated replacement cost method.
- 70.3. It *may* be used as the primary approach when there is either no evidence of transaction prices for similar property or no identifiable actual or notional income stream that would accrue to the owner of the relevant interest.
- 70.4. In some cases, even when evidence of market transaction prices or an identifiable income stream is available, the cost approach *may* be used as a secondary or corroborating approach.
- 70.5. The first step requires a replacement cost to be calculated. This is normally the cost of replacing the property with a modern equivalent at the relevant valuation date. An exception is where an equivalent property would need to be a replica of the subject property in order to provide a *participant* with the same utility, in which case the replacement cost would be that of reproducing or replicating the subject building rather than replacing it with a modern equivalent. The replacement cost *must* reflect all incidental costs, as appropriate, such as the value of the land, infrastructure, design fees, finance costs and developer profit that would be incurred by a *participant* in creating an equivalent asset.
- 70.6. The cost of the modern equivalent *must* then, as appropriate, be subject to adjustment for physical, functional, technological and economic obsolescence (see IVS 105 *Valuation Approaches and Methods*, section 80). The objective of an adjustment for obsolescence is to estimate how much less valuable the subject property might, or would be, to a potential buyer than the modern equivalent. Obsolescence considers the physical condition, functionality and economic utility of the subject property compared to the modern equivalent.

80. Special Considerations for Real Property Interests

80.1. The following sections address a non-exhaustive list of topics relevant to the valuation of real property interests.

(a) Hierarchy of Interests (section 90).

(b) Rent (section 100).

90. Hierarchy of Interests

90.1. The different types of real property interests are not mutually exclusive. For example, a superior interest *may* be subject to one or more subordinate interests. The owner of the absolute interest *may* grant a lease interest in respect of part or all of his interest. Lease interests granted directly by the owner of the absolute interest are “head lease” interests. Unless prohibited by the terms of the lease contract, the holder of a head lease interest can grant a lease of part or all of that interest to a third party, which is known as a sub-lease interest. A sub-lease interest will always be shorter than, or coterminous with, the head lease out of which it is created.

90.2. These property interests will have their own characteristics, as illustrated in the following examples:

(a) Although an absolute interest provides outright ownership in perpetuity, it *may* be subject to the effect of subordinate interests. These subordinate interests could include leases, restrictions imposed by a previous owner or restrictions imposed by statute.

(b) A lease interest will be for a defined period, at the end of which the property reverts to the holder of the superior interest out of which it was created. The lease contract will normally impose obligations on the lessee, eg, the payment of rent and other expenses. It *may* also impose conditions or restrictions, such as in the way the property *may* be used or on any transfer of the interest to a third party.

(c) A right of use *may* be held in perpetuity or *may* be for a defined period. The right *may* be dependent on the holder making payments or complying with certain other conditions.

90.3. When valuing a real property interest it is therefore necessary to identify the nature of the rights accruing to the holder of that interest and reflect any constraints or encumbrances imposed by the existence of other interests in the same property. The sum of the individual values of various different interests in the same property will frequently differ from the value of the unencumbered superior interest.

100. Rent

100.1. Market rent is addressed as a basis of value in IVS 104 Bases of Value.

100.2. When valuing either a superior interest that is subject to a lease or an interest created by a lease, *valuers must* consider the contract rent and, in cases where it is different, the market rent.

- 100.3. The contract rent is the rent payable under the terms of an actual lease. It *may* be fixed for the duration of the lease or variable. The frequency and basis of calculating variations in the rent will be set out in the lease and *must* be identified and understood in order to establish the total benefits accruing to the lessor and the liability of the lessee.

IVS 410 Development Property

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10. Overview

- 10.1. The principles contained in the General Standards IVS 101 to IVS 105 apply to valuations of development property. This standard only includes modifications, additional requirements or specific examples of how the General Standards apply for valuations to which this standard applies. Valuations of development property *must* also follow IVS 400 Real Property Interests.

20. Introduction

- 20.1. In the context of this standard, development properties are defined as interests where redevelopment is required to achieve the highest and best use, or where improvements are either being contemplated or are in progress at the valuation date and include:
- (a) the construction of buildings,
 - (b) previously undeveloped land which is being provided with infrastructure,
 - (c) the redevelopment of previously developed land,
 - (d) the improvement or alteration of existing buildings or structures,
 - (e) land allocated for development in a statutory plan, and
 - (f) land allocated for a higher value uses or higher density in a statutory plan.
- 20.2. Valuations of development property *may* be required for different *purposes*. It is the *valuer's* responsibility to understand the *purpose* of a valuation. A non-exhaustive list of examples of circumstances that *may* require a development valuation is provided below:

- (a) when establishing whether proposed projects are financially feasible,
- (b) as part of general consulting and transactional support engagements for acquisition and loan security,
- (c) for tax reporting *purposes*, development valuations are frequently needed for ad valorem taxation analyses,
- (d) for litigation requiring valuation analysis in circumstances such as shareholder disputes and damage calculations,
- (e) for financial reporting *purposes*, valuation of a development property is often required in connection with accounting for business combinations, *asset* acquisitions and sales, and impairment analysis, and
- (f) for other statutory or legal events that *may* require the valuation of development property such as compulsory purchases.

- 20.3. When valuing development property, *valuers must* follow the applicable standard for that type of *asset* or liability (for example, IVS 400 Real Property Interests).
- 20.4. The residual value or land value of a development property can be very sensitive to changes in assumptions or projections concerning the income or revenue to be derived from the completed project or any of the development costs that will be incurred. This remains the case regardless of the method or methods used or however diligently the various inputs are researched in relation to the valuation date.
- 20.5. This sensitivity also applies to the impact of *significant* changes in either the costs of the project or the value on completion of the current value. If the valuation is required for a *purpose* where *significant* changes in value over the duration of a construction project *may* be of concern to the user (eg, where the valuation is for loan security or to establish a project's viability), the *valuer must* highlight the potentially disproportionate effect of possible changes in either the construction costs or end value on the profitability of the project and the value of the partially completed property. A sensitivity analysis *may* be useful for this *purpose* provided it is accompanied by a suitable explanation.

30. Bases of Value

- 30.1. In accordance with IVS 104 Bases of Value, a *valuer must* select the appropriate basis(es) of value when valuing development property.
- 30.2. The valuation of development property often includes a *significant* number of assumptions and special assumptions regarding the condition or status of the project when complete. For example, special assumptions *may* be made that the development has been completed or that the property is fully leased. As required by IVS 101 *Scope of Work*, *significant* assumptions and special assumptions used in a valuation *must* be communicated to all parties to the valuation engagement and *must* be agreed and confirmed in the scope of work. Particular care *may* also be required where reliance *may* be placed by third parties on the valuation outcome.

- 30.3. Frequently it will be either impracticable or impossible to verify every feature of a development property which could have an impact on potential future development, such as where ground conditions have yet to be investigated. When this is the case, it *may* be appropriate to make assumptions (eg, that there are no abnormal ground conditions that would result in *significantly* increased costs). If this was an assumption that a *participant* would not make, it would need to be presented as a special assumption.
- 30.4. In situations where there has been a change in the market since a project was originally conceived, a project under construction *may* no longer represent the highest and best use of the land. In such cases, the costs to complete the project originally proposed *may* be irrelevant as a buyer in the market would either demolish any partially completed structures or adapt them for an alternative project. The value of the development property under construction would need to reflect the current value of the alternative project and the costs and risks associated with completing that project.
- 30.5. For some development properties, the property is closely tied to a particular use or business/trading activity or a special assumption is made that the completed property will trade at specified and sustainable levels. In such cases, the *valuer must*, as appropriate, also comply with the requirements of IVS 200 Business and Business Interests and, where applicable, IVS 210 Intangible Assets.

40. Valuation Approaches and Methods

- 40.1. The three principal valuation approaches described in IVS 105 *Valuation Approaches and Methods* *may* all be applicable for the valuation of a real property interest. There are two main approaches in relation to the valuation the development property. These are:-
- (a) the market approach (see section 50), and
 - (b) the residual method, which is a hybrid of the market approach, the income approach and the cost approach (see sections 40-70). This is based on the completed “gross development value” and the deduction of development costs and the developer’s return to arrive at the residual value of the development property (see section 90).
- 40.2. When selecting an approach and method, in addition to the requirements of this standard, a *valuer must* follow the requirements of IVS 105 *Valuation Approaches and Methods*, including para 10.3.
- 40.3. The valuation approach to be used will depend on the required basis of value as well as specific facts and circumstances, eg, the level of recent transactions, the stage of development of the project and movements in property markets since the project started, and *should* always be that which is most appropriate to those circumstances. Therefore, the exercise of judgement in the selection of the most suitable approach is critical.

50. Market Approach

- 50.1. Some types of development property can be sufficiently homogenous and frequently exchanged in a market for there to be sufficient data from recent sales to use as a direct comparison where a valuation is required.

- 50.2. In most markets, the market approach *may* have limitations for larger or more complex development property, or smaller properties where the proposed improvements are heterogeneous. This is because the number and extent of the variables between different properties make direct comparisons of all variables inapplicable though correctly adjusted market evidence (See IVS 105 *Valuation Approaches and Methods*, section 20.5) *may* be used as the basis for a number of variables within the valuation.
- 50.3. For development property where work on the improvements has commenced but is incomplete, the application of the market approach is even more problematic. Such properties are rarely transferred between *participants* in their partially-completed state, except as either part of a transfer of the owning entity or where the seller is either insolvent or facing insolvency and therefore unable to complete the project. Even in the unlikely event of there being evidence of a transfer of another partially-completed development property close to the valuation date, the degree to which work has been completed would almost certainly differ, even if the properties were otherwise similar.
- 50.4. The market approach *may* also be appropriate for establishing the value of a completed property as one of the inputs required under the residual method, which is explained more fully in the section on the residual method (section 90).

60. Income Approach

- 60.1. Establishing the residual value of a development property *may* involve the use of a cash flow model in some markets.
- 60.2. The income approach *may* also be appropriate for establishing the value of a completed property as one of the inputs required under the residual method, which is explained more fully in the section on the residual method (see section 90).

70. Cost Approach

- 70.1. Establishing the development costs is a key component of the residual approach (see para 90.5).
- 70.2. The cost approach *may* also exclusively be used as a means of indicating the value of development property such as a proposed development of a building or other structure for which there is no active market on completion.
- 70.3. The cost approach is based on the economic principle that a buyer will pay no more for an *asset* than the amount to create an *asset* of equal utility. To apply this principle to development property, the *valuer must* consider the cost that a prospective buyer would incur in acquiring a similar *asset* with the potential to earn a similar profit from development as could be obtained from development of the subject property. However, unless there are unusual circumstances affecting the subject development property, the process of analysing a proposed development and determining the anticipated costs for a hypothetical alternative would effectively replicate either the market approach or the residual method as described above, which can be applied directly to the subject property.

- 70.4. Another difficulty in applying the cost approach to development property is in determining the profit level, which is its “utility” to a prospective buyer. Although a developer *may* have a target profit at the commencement of a project, the actual profit is normally determined by the value of the property at completion. Moreover, as the property approaches completion, some of the risks associated with development are likely to reduce, which *may* impact on the required return of a buyer. Unless a fixed price has been agreed, profit is not determined by the costs incurred in acquiring the land and undertaking the improvements.

80. Special Considerations for a Development Property

- 80.1. The following sections address a non-exhaustive list of topics relevant to the valuation of development property:
- (a) Residual Method (section 90).
 - (b) Existing Asset (section 100).
 - (c) Special Considerations for Financial Reporting (section 110).
 - (d) Special Considerations for Secured Lending (section 120).

90. Residual Method

- 90.1. The residual method is so called because it indicates the residual amount after deducting all known or anticipated costs required to complete the development from the anticipated value of the project when completed after consideration of the risks associated with completion of the project. This is known as the residual value. The residual value, derived from the residual method, *may* or *may* not equate to the market value of the development property in its current condition.
- 90.2. The residual value can be highly sensitive to relatively small changes in the forecast cash flows and the practitioner *should* provide separate sensitivity analyses for each *significant* factor.
- 90.3. Caution is required in the use of this method because of the sensitivity of the result to changes in many of the inputs, which *may* not be precisely known on the valuation date, and therefore have to be estimated with the use of assumptions.
- 90.4. The models used to apply the residual method vary considerably in complexity and sophistication, with the more complex models allowing for greater granularity of inputs, multiple development phases and sophisticated analytical tools. The most suitable model will depend on the size, duration and complexity of the proposed development.
- 90.5. In applying the residual method, a *valuer should* consider and evaluate the reasonableness and reliability of the following:
- (a) the source of information on any proposed building or structure, eg, any plans and specification that are to be relied on in the valuation, and
 - (b) any source of information on the construction and other costs that will be incurred in completing the project and which will be used in the valuation.

90.6. The following basic elements require consideration in any application of the method to estimate the market value of development property and if another basis is required, alternative inputs *may* be required.

- (a) Completed property value,
- (b) Construction costs,
- (c) Consultants fees,
- (d) Marketing costs,
- (e) Timetable,
- (f) Finance costs,
- (g) Development profit,
- (h) Discount rate.

Value of Completed Property

90.7. The first step requires an estimate of the value of the relevant interest in the real property following notional completion of the development project, which *should* be developed in accordance with IVS 105 Valuation Methods and Approaches.

90.8. Regardless of the methods adopted under either the market or income approach, the *valuer must* adopt one of the two basic underlying assumptions:

- (a) the estimated market value on completion is based on values that are current on the valuation date on the special assumption the project had already been completed in accordance with the defined plans and specification, or
- (b) the estimated value on completion is based on the special assumption that the project is completed in accordance with the defined plans and specification on the anticipated date of completion.

90.9. Market practice and availability of relevant data *should* determine which of these assumptions is more appropriate. However, it is important that there is clarity as to whether current or projected values are being used.

90.10. If estimated gross development value is used, it *should* be made clear that these are based on special assumptions that a *participant* would make based on information available on the valuation date.

90.11. It is also important that care is taken to ensure that consistent assumptions are used throughout the residual value calculation, ie, if current values are used then the costs *should* also be current and discount rates derived from analysis of current prices.

90.12. If there is a pre-sale or pre-lease agreement in place that is conditional on the project, or a relevant part, being completed, this will be reflected in the valuation of the completed property. Care *should* be taken to establish whether the price in a pre-sale agreement or the rent and other terms in a pre-lease agreement reflect those that would be agreed between *participants* on the valuation date.

- 90.13. If the terms are not reflective of the market, adjustments *may* need to be made to the valuation.
- 90.14. It would also be appropriate to establish if these agreements would be assignable to a purchaser of the relevant interest in the development property prior to the completion of the project.

Construction Costs

- 90.15. The costs of all work required at the valuation date to complete the project to the defined specification need to be identified. Where no work has started, this will include any preparatory work required prior to the main building contract, such as the costs of obtaining statutory permissions, demolition or off-site enabling work.
- 90.16. Where work has commenced, or is about to commence, there will normally be a contract or contracts in place that can provide the independent confirmation of cost. However, if there are no contracts in place, or if the actual contract costs are not typical of those that would be agreed in the market on the valuation date, then it *may* be necessary to estimate these costs reflecting the reasonable expectation of *participants* on the valuation date of the probable costs.
- 90.17. The benefit of any work carried out prior to the valuation date will be reflected in the value, but will not determine that value. Similarly, previous payments under the actual building contract for work completed prior to the valuation date are not relevant to current value.
- 90.18. In contrast, if payments under a building contract are geared to the work completed, the sums remaining to be paid for work not yet undertaken at the valuation date *may* be the best evidence of the construction costs required to complete the work.
- 90.19. However, contractual costs *may* include special requirements of a specific end user and therefore *may* not reflect the general requirements of *participants*.
- 90.20. Moreover, if there is a material risk that the contract *may* not be fulfilled, (eg, due to a dispute or insolvency of one of the parties), it *may* be more appropriate to reflect the cost of engaging a new contractor to complete the outstanding work.
- 90.21. When valuing a partly completed development property, it is not appropriate to rely solely on projected costs and income contained in any project plan or feasibility study produced at the commencement of the project.
- 90.22. Once the project has commenced, this is not a reliable tool for measuring value as the inputs will be historic. Likewise, an approach based on estimating the percentage of the project that has been completed prior to the valuation date is unlikely to be relevant in determining the current market value.

Consultants' Fees

- 90.23. These include legal and professional costs that would be reasonably incurred by a *participant* at various stages through the completion of the project.

Marketing Costs

- 90.24. If there is no identified buyer or lessee for the completed project, it will normally be appropriate to allow for the costs associated with appropriate marketing, and for any leasing commissions and consultants' fees incurred for marketing not included under para 90.23.

Timetable

- 90.25. The duration of the project from the valuation date to the expected date of physical completion of the project needs to be considered, together with the phasing of all cash outflows for construction costs, consultants' fees, etc.
- 90.26. If there is no sale agreement in place for the relevant interest in the development property following practical completion, an estimate *should* be made of the marketing period that might typically be required following completion of construction until a sale is achieved.
- 90.27. If the property is to be held for investment after completion and if there are no pre-leasing agreements, the time required to reach stabilised occupancy needs to be considered (ie, the period required to reach a realistic long-term occupancy level). For a project where there will be individual letting units, the stabilised occupancy levels *may* be less than 100 percent if market experience indicates that a number of units *may* be expected to always be vacant, and allowance *should* be considered for costs incurred by the owner during this period such as additional marketing costs, incentives, maintenance and/or unrecoverable service charges.

Finance Costs

- 90.28. These represent the cost of finance for the project from the valuation date through to the completion of the project, including any period required after physical completion to either sell the interest or achieve stabilised occupancy. As a lender *may* perceive the risks during construction to differ substantially from the risks following completion of construction, the finance cost during each period *may* also need to be considered separately. Even if an entity is intending to self-fund the project, an allowance *should* be made for interest at a rate which would be obtainable by a *participant* for borrowing to fund the completion of the project on the valuation date.

Development Profit

- 90.29. Allowance *should* be made for development profit, or the return that would be required by a buyer of the development property in the market place for taking on the risks associated with completion of the project on the valuation date. This will include the risks involved in achieving the anticipated income or capital value following physical completion of the project.
- 90.30. This target profit can be expressed as a lump sum, a percentage return on the costs incurred or a percentage of the anticipated value of the project on completion or a rate of return. Market practice for the type of property in question will normally indicate the most appropriate option. The amount of profit that would be required will reflect the level of risk that would be perceived by a prospective buyer on the valuation date and will vary according to factors such as:

- (a) the stage which the project has reached on the valuation date. A project which is nearing completion will normally be viewed as being less risky than one at an early stage, with the exception of situations where a party to the development is insolvent,
- (b) whether a buyer or lessee has been secured for the completed project, and
- (c) the size and anticipated remaining duration of the project. The longer the project, the greater the risk caused by exposure to fluctuations in future costs and receipts and changing economic conditions generally.

90.31. The following are examples of factors that *may* typically need to be considered in an assessment of the relative risks associated with the completion of a development project:

- (a) unforeseen complications that increase construction costs,
- (b) potential for contract delays caused by adverse weather or other matters outside of developer's control,
- (c) delays in obtaining statutory consents,
- (d) supplier failures,
- (e) entitlement risk and changes in entitlements over the development period,
- (f) regulatory changes, and
- (g) delays in finding a buyer or lessee for the completed project.

90.32. Whilst all of the above factors will impact the perceived risk of a project and the profit that a buyer or the development property would require, care *must* be taken to avoid double counting, either where contingencies are already reflected in the residual valuation model or risks in the discount rate used to bring future cash flows to present value.

90.33. The risk of the estimated value of the completed development project changing due to changed market conditions over the duration of the project will normally be reflected in the discount rate or capitalisation rate used to value the completed project.

90.34. The profit anticipated by the owner of an interest in development property at the commencement of a development project will vary according to the valuation of its interest in the project once construction has commenced. The valuation *should* reflect those risks remaining at the valuation date and the discount or return that a buyer of the partially completed project would require for bringing it to a successful conclusion.

Discount Rate

90.35. In order to arrive at an indication of the value of the development property on the valuation date, the residual method requires the application of a discount rate to all future cash flows in order to arrive at a net present value. This discount rate *may* be derived using a variety of methods (see IVS 105 *Valuation Approaches and Methods*, paras 50.29-50.31).

90.36. If the cash flows are based on values and costs that are current on the valuation date, the risk of these changing between the valuation date and the anticipated completion date *should* be considered and reflected in the discount rate used to determine the present value. If the cash flows are based on prospective values and costs, the risk of those projections proving to be inaccurate *should* be considered and reflected in the discount rate.

100. Existing Asset

100.1. In the valuation of development property, it is necessary to establish the suitability of the real property in question for the proposed development. Some matters *may* be within the *valuer's* knowledge and experience but some *may* require information or reports from other specialists. Matters that typically need to be considered for specific investigation when undertaking a valuation of a development property before a project commences include:

- (a) whether or not there is a market for the proposed development,
- (b) is the proposed development the highest and best use of the property in the current market,
- (c) whether there are other non-financial obligations that need to be considered (political or social criteria),
- (d) legal permissions or zoning, including any conditions or constraints on permitted development,
- (e) limitations, encumbrances or conditions imposed on the relevant interest by private contract,
- (f) rights of access to public highways or other public areas,
- (g) geotechnical conditions, including potential for contamination or other environmental risks,
- (h) the availability of, and requirements to, provide or improve necessary services, eg, water, drainage and power,
- (i) the need for any off-site infrastructure improvements and the rights required to undertake this work,
- (j) any archaeological constraints or the need for archaeological investigations,
- (k) sustainability and any *client* requirements in relation to green buildings,
- (l) economic conditions and trends and their potential impact on costs and receipts during the development period,
- (m) current and projected supply and demand for the proposed future uses,
- (n) the availability and cost of funding,
- (o) the expected time required to deal with preparatory matters prior to starting work, for the completion of the work and, if appropriate, to rent or sell the completed property, and
- (p) any other risks associated with the proposed development.

- 100.2. Where a project is in progress, additional enquires or investigations will typically be needed into the contracts in place for the design of the project, for its construction and for supervision of the construction.

110. Special Considerations for Financial Reporting

- 110.1. The accounting treatment of development property can vary depending on how it is classified by the reporting entity (eg, whether it is being held for sale, for owner occupation or as investment property). This *may* affect the valuation requirements and therefore the classification and the relevant accounting requirements need to be determined before selecting an appropriate valuation method.
- 110.2. Financial statements are normally produced on the assumption that the entity is a going concern. It is therefore normally appropriate to assume that any contracts (eg, for the construction of a development property or for its sale or leasing on completion), would pass to the buyer in the hypothetical exchange, even if those contracts *may* not be assignable in an actual exchange. An exception would be if there was evidence of an abnormal risk of default by a contracted party on the valuation date.

120. Special Considerations for Secured Lending

- 120.1. The appropriate basis of valuation for secured lending is normally market value. However, in considering the value of a development property, regard *should* be given to the probability that any contracts in place, eg, for construction or for the sale or leasing of the completed project *may*, become void or voidable in the event of one of the parties being the subject of formal insolvency proceedings. Further regard *should* be given to any contractual obligations that *may* have a material impact on market value. Therefore, it *may* be appropriate to highlight the risk to a lender caused by a prospective buyer of the property not having the benefit of existing building contracts and/or pre-leases, and pre-sales and any associated warranties and guarantees in the event of a default by the borrower.

IVS 500 Financial Instruments

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10. Overview

- 10.1. The principles contained in the General Standards apply to valuations of financial instruments. This standard only includes modifications, additional requirements or specific examples of how the General Standards apply for valuations to which this standard applies.

20. Introduction

- 20.1. A financial instrument is a contract that creates rights or obligations between specified parties to receive or pay cash or other financial consideration. Such instruments include but are not limited to, derivatives or other contingent instruments, hybrid instruments, fixed income, structured products and equity instruments. A financial instrument can also be created through the combination of other financial instruments in a portfolio to achieve a specific net financial outcome.
- 20.2. Valuations of financial instruments conducted under IVS 500 *Financial Instruments* can be performed for many different *purposes* including, but not limited to:
- (a) acquisitions, mergers and sales of businesses or parts of businesses,
 - (b) purchase and sale,
 - (c) financial reporting,
 - (d) legal or regulatory requirements (subject to any specific requirements set by the relevant authority),
 - (e) internal risk and compliance procedures,
 - (f) tax, and
 - (g) litigation.

- 20.3. A thorough understanding of the instrument being valued is required to identify and evaluate the relevant market information available for identical or comparable instruments. Such information includes prices from recent transactions in the same or a similar instrument, quotes from brokers or pricing services, credit ratings, yields, volatility, indices or any other inputs relevant to the valuation process.
- 20.4. When valuations are being undertaken by the holding entity that are intended for use by external investors, regulatory authorities or other entities, to comply with the requirement to confirm the identity and status of the *valuer* in IVS 101 *Scope of Work*, para 20.3.(a), reference *must* be made to the control environment in place, as required by IVS 105 *Valuation Approaches and Methods* and IVS 500 *Financial Instruments* paras 120.1-120.3 regarding control environment.
- 20.5. To comply with the requirement to identify the *asset* or liability to be valued as in IVS 101 *Scope of Work*, para 20.3.(d), the following matters *must* be addressed:
- (a) the class or classes of instrument to be valued,
 - (b) whether the valuation is to be of individual instruments or a portfolio, and
 - (c) the unit of account.
- 20.6. IVS 102 *Investigations and Compliance*, paras 20.2-20.4 provide that the investigations required to support the valuation *must* be adequate having regard to the *purpose* of the assignment. To support these investigations, sufficient evidence supplied by the *valuer* and/or a credible and reliable third party *must* be assembled. To comply with these requirements, the following are to be considered:
- (a) All market data used or considered as an input into the valuation process *must* be understood and, as necessary, validated.
 - (b) Any model used to estimate the value of a financial instrument shall be selected to appropriately capture the contractual terms and economics of the financial instrument.
 - (c) Where observable prices of, or market inputs from, similar financial instruments are available, those imputed inputs from comparable price(s) and/or observable inputs *should* be adjusted to reflect the contractual and economic terms of the financial instrument being valued.
 - (d) Where possible, multiple valuation approaches are preferred. If differences in value occur between the valuation approaches, the *valuer* *must* explain and document the differences in value.
- 20.7. To comply with the requirement to disclose the valuation approach(es) and reasoning in IVS 103 *Reporting*, para 20.1, consideration *must* be given to the appropriate degree of reporting detail. The requirement to disclose this information in the valuation report will differ for different categories of financial instruments. Sufficient information *should* be provided to allow users to understand the nature of each class of instrument valued and the primary factors influencing the values. Information that adds little to a users' understanding as to the nature of the *asset* or liability, or that obscures the

primary factors influencing value, *must* be avoided. In determining the level of disclosure that is appropriate, regard *must* be had to the following:

- (a) Materiality: The value of an instrument or class of instruments in relation to the total value of the holding entity's assets and liabilities or the portfolio that is valued.
- (b) Uncertainty: The value of the instrument *may* be subject to *significant* uncertainty on the valuation date due to the nature of the instrument, the model or inputs used or to market abnormalities. Disclosure of the cause and nature of any material uncertainty *should* be made.
- (c) Complexity: The greater the complexity of the instrument, the greater the appropriate level of detail to ensure that the assumptions and inputs affecting value are identified and explained.
- (d) Comparability: The instruments that are of particular interest to users *may* differ with the passage of time. The usefulness of the valuation report, or any other reference to the valuation, is enhanced if it reflects the information demands of users as market conditions change, although, to be meaningful, the information presented *should* allow comparison with previous periods.
- (e) Underlying instruments: If the cash flows of a financial instrument are generated from or secured by identifiable underlying assets or liabilities, the relevant factors that influence the underlying value *must* be provided in order to help users understand how the underlying value impacts the estimated value of the financial instrument.

30. Bases of Value

- 30.1. In accordance with IVS 104 Bases of Value, a *valuer must* select the appropriate basis(es) of value when valuing financial instruments.
- 30.2. Often, financial instrument valuations are performed using bases of value defined by entities/organisations other than the IVSC (some examples of which are mentioned in IVS 104 Bases of Value) and it is the *valuer's* responsibility to understand and follow the regulation, case law, tax law and other interpretive guidance related to those bases of value as of the valuation date.

40. Valuation Approaches and Methods

- 40.1. When selecting an approach and method, in addition to the requirements of this chapter, a *valuer must* follow the requirements of IVS 105 *Valuation Approaches and Methods*.
- 40.2. The three valuation approaches described in IVS 105 *Valuation Approaches and Methods may* be applied to the valuation of financial instruments.
- 40.3. The various valuation methods used in financial markets are based on variations of the market approach, the income approach or the cost approach as described in the IVS 105 *Valuation Approaches and Methods*. This standard describes the commonly used methods and matters that need to be considered or the inputs needed when applying these methods.

- 40.4. When using a particular valuation method or model, it is important to ensure that it is calibrated with observable market information, where available, on a regular basis to ensure that the model reflects current market conditions. As market conditions change, it *may* become necessary to change to a more suitable model(s) or to modify the existing model and recalibrate and/or make additional adjustments to the valuation inputs. Those adjustments *should* be made to ensure consistency with the required valuation basis, which in turn is determined by the *purpose* for which the valuation is required; see the *IVS Framework*.

50. Market Approach

- 50.1. A price obtained from trading on a liquid exchange on, or very close to, the time or date of valuation is normally the best indication of the market value of a holding of the identical instrument. In cases where there have not been recent relevant transactions, the evidence of quoted or consensus prices, or private transactions *may* also be relevant.
- 50.2. It *may* be necessary to make adjustments to the price information if the observed instrument is dissimilar to that being valued or if the information is not recent enough to be relevant. For example, if an observable price is available for similar instruments with one or more different characteristics to the instrument being valued, then the implied inputs from the comparable observable price are to be adjusted to reflect the specific terms of the financial instrument being valued.
- 50.3. When relying on a price from a pricing service, the *valuer must* understand how the price was derived.

60. Income Approach

- 60.1. The value of financial instruments *may* be determined using a discounted cash flow method. The terms of an instrument determine, or allow estimation of, the undiscounted cash flows. The terms of a financial instrument typically set out:
- (a) the timing of the cash flows, ie, when the entity expects to realise the cash flows related to the instrument,
 - (b) the calculation of the cash flows, eg, for a debt instrument, the interest rate that applies, or for a derivative instrument, how the cash flows are calculated in relation to the underlying instrument or index (or indices),
 - (c) the timing and conditions for any options in the contract, eg, put or call, prepayment, extension or conversion options, and
 - (d) protection of the rights of the parties to the instrument, eg, terms relating to credit risk in debt instruments or the priority over, or subordination to, other instruments held.
- 60.2. In establishing the appropriate discount rate, it is necessary to assess the return that would be required on the instrument to compensate for the time value of money and potential additional risks from, but not limited to the following:
- (a) the terms and conditions of the instrument, eg, subordination,

- (b) the credit risk, ie, uncertainty about the ability of the counterparty to make payments when due,
- (c) the liquidity and marketability of the instrument,
- (d) the risk of changes to the regulatory or legal environment, and
- (e) the tax status of the instrument.

60.3. Where future cash flows are not based on fixed contracted amounts, estimates of the expected cash flows will need to be made in order to determine the necessary inputs. The determination of the discount rate *must* reflect the risks of, and be consistent with, the cash flows. For example, if the expected cash flows are measured net of credit losses then the discount rate *must* be reduced by the credit risk component. Depending upon the *purpose of the valuation*, the inputs and assumptions made into the cash flow model will need to reflect either those that would be made by *participants*, or those that would be based on the holder's current expectations or targets. For example, if the *purpose of the valuation* is to determine market value, or fair value as defined in IFRS, the assumptions *should* reflect those of *participants*. If the *purpose* is to measure performance of an *asset* against management determined benchmarks, eg, a target internal rate of return, then alternative assumptions *may* be appropriate.

70. Cost Approach

70.1. In applying the cost approach, *valuers must* follow the guidance contained in IVS 105 *Valuation Approaches and Methods*, paras 70.1-70.14.

80. Special Considerations for Financial Instruments

80.1. The following sections address a non-exhaustive list of topics relevant to the valuation of financial instruments:

- (a) Valuation Inputs (section 90).
- (b) Credit Risk (section 100).
- (c) Liquidity and Market Activity (section 110).
- (d) Control Environment (section 120).

90. Valuation Inputs

90.1. As per IVS 105 *Valuation Approaches and Methods*, para 10.7, any data set used as a valuation input, understanding the sources and how inputs are adjusted by the provider, if any, is essential to understanding the reliance that *should* be given to the use of the valuation input.

90.2. Valuation inputs *may* come from a variety of sources. Commonly used valuation input sources are broker quotations, consensus pricing services, the prices of comparable instruments from third parties and market data pricing services. Implied inputs can often be derived from such observable prices such as volatility and yields.

90.3. When assessing the validity of broker quotations, as evidence of how *participants* would price an *asset*, the *valuer should* consider the following:

- (a) Brokers generally make markets and provide bids in respect of more popular instruments and *may* not extend coverage to less liquid instruments. Because liquidity often reduces with time, quotations *may* be harder to find for older instruments.
- (b) A broker is concerned with trading, not supporting valuation, and they have little incentive to research an indicative quotation as thoroughly as they would an executable quotation. A *valuer* is required to understand whether the broker quote is a binding, executable quote or a non-binding, theoretical quote. In the case of a non-binding quote, the *valuer* is required to gather additional information to understand if the quote *should* be adjusted or omitted from the valuation.
- (c) There is an inherent conflict of interest where the broker is the counterparty to an instrument.
- (d) Brokers have an incentive to encourage trading.

90.4. Consensus pricing services operate by collecting price or valuation input information about an instrument from several participating subscribers. They reflect a pool of quotations from different sources, sometimes with adjustment to compensate for any sampling bias. This overcomes the conflict of interest problems associated with single brokers. However, as with a broker quotation, it *may* not be possible to find a suitable input for all instruments in all markets. Additionally, despite its name, a consensus price *may* not necessarily constitute a true market “consensus”, but rather is more of a statistical estimate of recent market transactions or quoted prices. Therefore, the *valuer* needs to understand how the consensus pricing was estimated and if such estimates are reasonable, given the instrument being valued. Information and inputs relevant to the valuation of an illiquid instrument can often be gleaned through comparable transactions (see section 110 for further details).

100. Credit Risk Adjustments

- 100.1. Understanding the credit risk is often an important aspect of valuing a financial instrument and most importantly the issuer. Some of the common factors that need to be considered in establishing and measuring credit risk include the following:
- (a) Own credit and counterparty risk: Assessing the financial strength of the issuer or any credit support providers will involve consideration of not only historical and projected financial performance of the relevant entity or entities but also consideration of performance and prospects for the industry sector in which the business operates. In addition to issuer credit, the *valuer must* also consider the credit exposure of any counterparties to the *asset* or liability being valued. In the case of a clearing house settlement process, many *jurisdictions* now require certain derivatives to be transacted through a central counterparty which can mitigate risk, however residual counterparty risk needs to be considered.
 - (b) The *valuer* also needs to be able to differentiate between the credit risk of the instrument and the credit risk of the issuer and/or counterparty. Generally, the credit risk of the issuer or counterparty does not consider specific collateral related to the instrument.

- (c) Subordination: Establishing the priority of an instrument is critical in assessing the default risk. Other instruments *may* have priority over an issuer's *assets* or the cash flows that support the instrument.
 - (d) Leverage: The amount of debt used to fund the *assets* from which an instrument's return is derived can affect the volatility of returns to the issuer and credit risk.
 - (e) Netting agreements: Where derivative instruments are held between counterparties, credit risk *may* be reduced by a netting or offset agreement that limits the obligations to the net value of the transactions, ie, if one party becomes insolvent, the other party has the right to offset sums owed to the insolvent party against sums due under other instruments.
 - (f) Default protection: Many instruments contain some form of protection to reduce the risk of non-payment to the holder. Protection might take the form of a guarantee by a third party, an insurance contract, a credit default swap or more *assets* to support the instrument than are needed to make the payments. Credit exposure is also reduced if subordinated instruments take the first losses on the underlying *assets* and therefore reduce the risk to more senior instruments. When protection is in the form of a guarantee, an insurance contract or a credit default swap, it is necessary to identify the party providing the protection and assess that party's creditworthiness. Considering the credit worthiness of a third party involves not only the current position but also the possible effect of any other guarantees or insurance contracts the entity has written. If the provider of a guarantee has also guaranteed other correlated debt securities, the risk of its non-performance will likely increase.
- 100.2. For parties for which limited information is available, if secondary trading in a financial instrument exists, there *may* be sufficient market data to provide evidence of the appropriate risk adjustment. If not, it might be necessary to look to credit indices, information available for entities with similar risk characteristics, or estimate a credit rating for the party using its own financial information. The varying sensitivities of different liabilities to credit risk, such as collateral and/or maturity differences, *should* be taken into account in evaluating which source of credit data provides the most relevant information. The risk adjustment or credit spread applied is based on the amount a *participant* would require for the particular instrument being valued.
- 100.3. The own credit risk associated with a liability is important to its value as the credit risk of the issuer is relevant to the value in any transfer of that liability. Where it is necessary to assume a transfer of the liability regardless of any actual constraints on the ability of the counterparties to do so, eg, in order to comply with financial reporting requirements, there are various potential sources for reflecting own credit risk in the valuation of liabilities. These include the yield curve for the entity's own bonds or other debt issued, credit default swap spreads, or by reference to the value of the corresponding *asset*. However, in many cases the issuer of a liability will not have the ability to transfer it and can only settle the liability with the counterparty.

- 100.4. Collateral: The *assets* to which the holder of an instrument has recourse in the event of default need to be considered. In particular, the *valuer* needs to be understand whether recourse is to all the *assets* of the issuer or only to specified *asset(s)*. The greater the value and liquidity of the *asset(s)* to which an entity has recourse in the event of default, the lower the overall risk of the instrument due to increased recovery. In order not to double count, the *valuer* also needs to consider if the collateral is already accounted for in another area of the balance sheet.
- 100.5. When adjusting for own credit risk of the instrument, it is also important to consider the nature of the collateral available for the liabilities being valued. Collateral that is legally separated from the issuer normally reduces the credit exposure. If liabilities are subject to a frequent collateralisation process, there might not be a material own credit risk adjustment because the counterparty is mostly protected from loss in the event of default.

110. Liquidity and Market Activity

- 110.1. The liquidity of financial instruments range from those that are standardised and regularly transacted in high volumes to those that are agreed between counterparties that are incapable of assignment to a third party. This range means that consideration of the liquidity of an instrument or the current level of market activity is important in determining the most appropriate valuation approach.
- 110.2. Liquidity and market activity are distinct. The liquidity of an *asset* is a measure of how easily and quickly it can be transferred in return for cash or a cash equivalent. Market activity is a measure of the volume of trading at any given time, and is a relative rather than an absolute measure. Low market activity for an instrument does not necessarily imply the instrument is illiquid.
- 110.3. Although separate concepts, illiquidity or low levels of market activity pose similar valuation challenges through a lack of relevant market data, ie, data that is either current at the valuation date or that relates to a sufficiently similar *asset* to be reliable. The lower the liquidity or market activity, the greater the reliance that will be needed on valuation approaches that use techniques to adjust or *weight* the inputs based on the evidence of other comparable transactions to reflect either market changes or differing characteristics of the *asset*.

120. Valuation Control and Objectivity

- 120.1. The control environment consists of the internal governance and control procedures that are in place with the objective of increasing the confidence of those who *may* rely on the valuation in the valuation process and conclusion. Where an external *valuer* is placing reliance upon an internally performed valuation, the external *valuer must* consider the adequacy and independence of the valuation control environment.
- 120.2. In comparison with other *asset* classes, financial instruments are more commonly valued internally by the same entity that creates and trades them. Internal valuations bring into question the independence of the *valuer* and hence this creates risk to the perceived objectivity of valuations. Please reference 40.1 and 40.2 of the IVS *Framework* regarding valuation performed by internal *valuers* and the need for procedures to be in place

to ensure the objectivity of the valuation and steps that *should* be taken to ensure that an adequate control environment exists to minimise threats to the independence of the valuation. Many entities which deal with the valuation of financial instruments are registered and regulated by statutory financial regulators. Most financial regulators require banks or other regulated entities that deal with financial instruments to have independent price verification procedures. These operate separately from trading desks to produce valuations required for financial reporting or the calculation of regulatory capital guidance on the specific valuation controls required by different regulatory regimes. This is outside the scope of this standard. However, as a general principle, valuations produced by one department of an entity that are to be included in financial statements or otherwise relied on by third parties *should* be subject to scrutiny and approval by an independent department of the entity. Ultimate authority for such valuations *should* be separate from, and fully independent of, the risk-taking functions. The practical means of achieving a separation of the function will vary according to the nature of the entity, the type of instrument being valued and the materiality of the value of the particular class of instrument to the overall objective. The appropriate protocols and controls *should* be determined by careful consideration of the threats to objectivity that would be perceived by a third party relying on the valuation.

- 120.3. When accessing your valuation controls, the following include items you *should* consider in the valuation process:
- (a) establishing a governance group responsible for valuation policies and procedures and for oversight of the entity's valuation process, including some members external to the entity,
 - (b) systems for regulatory compliance if applicable,
 - (c) a protocol for the frequency and methods for calibration and testing of valuation models,
 - (d) criteria for verification of certain valuations by different internal or external experts,
 - (e) periodic independent validation of the valuation model(s),
 - (f) identifying thresholds or events that trigger more thorough investigation or secondary approval requirements, and
 - (g) identifying procedures for establishing *significant* inputs that are not directly observable in the market, eg, by establishing pricing or audit committees.

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PART II

PHILIPPINE CONTEXT FOCUSING ON VALUATION FOR TAXATION AND OTHER PURPOSES

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- 40. Valuations of Private Lands for Acquisition by the Government
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List of Acronyms

AIPO	Accredited and Integrated Professional Organization
BIR	Bureau of Internal Revenue
CA	Compulsory Acquisition
CAMA	Computer Assisted Mass Appraisal
CARP-LAD	Comprehensive Agrarian Reform Program-Land Acquisition and Distribution
CPD	Continuing Professional Development
DAR	Department of Agrarian Reform
DARAB	DAR Adjudication Board
DENR	Department of Environment and Natural Resources
IVS	International Valuation Standards
LAR	Local Assessment Regulations
LGC	Local Government Code
MAG	Mass Appraisal Guidebook
MRPAO	Manual on Real Property Appraisal and Assessment Operations
NIRC	National Internal Revenue Code
PRBRES	Professional Regulatory Board of Real Estate Service
PRC	Professional Regulation Commission
PVS	Philippine Valuation Standards
RA	Republic Act
RCN	Reproduction/Replacement Cost New
RESA	Real Estate Service Act
RROW	Road Right-of-Way
SMFV	Schedule of Fair Market Value
SMV	Schedule of Market Value
TRAIN	Tax Reform for Acceleration and Inclusion
VOS	Voluntary Offer to Sell
ZV	Zonal Value

Introduction

Part II contains contextual discussions to clarify and explain any departure from the IVS 2017, considering applicable legal and statutory requirements in the Philippines. In particular, it takes into account existing laws, rules and regulations pertaining to national and local valuation for taxation and other purposes.

A departure, as defined in IVS 2017, “is a circumstance where specific legislative, regulatory or other authoritative requirements must be followed that differs from some of the requirements within IVS. Departures are mandatory in that a valuer must comply with legislative, regulatory and other authoritative requirements appropriate to the purpose and jurisdiction of the valuation to be in compliance with IVS. A valuer may still state that the valuation was performed in accordance with IVS when there are departures in these circumstances.” (para 60.1, IVS Framework)

“The requirement to depart from IVS pursuant to legislative, regulatory or other authoritative requirements takes precedence over all other IVS requirements.” (para 60.2, IVS Framework)

This section likewise contains clarificatory statements or other terminologies widely used in the Philippines, which may be an elaboration of terminologies in the IVS.

Glossary

10. Overview of Glossary

- 10.1. This glossary is an addendum that defines several terms used in the IVS.
- 10.2. It provides additional terminologies related to selected IVS-defined terms as they are widely used in the Philippines.

20. Addendum to IVS-Defined Terms

20.1. Client

This refers to the party requesting a valuation or appraisal. In the public sector, the word “client” also refers to the property owners, national and local government units, including any interested person or entity in the general public requiring appraisal or valuation services.

20.2. Jurisdiction

This refers to the sectors (e.g. finance, agriculture, trade, etc.) where existing laws, rules and regulations clearly define which government agencies, institutions, instrumentalities, or government-owned and controlled corporation (GOCC) are mandated to regulate them. In spatial terms, the tiers of local government jurisdictions in the Philippines are provinces, cities, municipalities and barangays. It also refers to regional jurisdictions and revenue districts.

20.3. Machinery. *see in Plant the definition of the Machinery.*

20.4. Market Data Approach

It is also known as the Comparative or Comparable Sales Approach. This is an appraisal procedure in which the market value estimate is predicated upon prices paid in actual market transactions and current listings. It is a process of analyzing sales of similar sold properties (sold at or around the relevant date) in order to derive an indication of the most probable sale prices of the property being appraised. The reliability of this technique is dependent upon: a) the availability of comparable sales data; b) the verification of the sales date; c) the degree of compatibility or extent of adjustment necessary for time differences; and d) the absence of non-typical conditions affecting the sales price. [*p. x, Mass Appraisal Guidebook (MAG) issued under DO No. 10-2010*]

20.5. Must

The term “shall” is used to indicate statutory compliance in the Philippines.

20.6. Plant

In the Philippines, the term “Plant” is not in general use. ‘Machinery and Equipment’ is the collective term adopted by Appraisers, while ‘Plant and Equipment’ is adopted by Accountants. Reference to “Plant and Equipment” in the IVS is therefore extended to “Plant, Machinery and Equipment”. [*p.15, Philippine Valuation Standards (PVS) 2009*]

Machinery is defined to embrace machines, equipment, mechanical contrivances, instruments, appliances, or apparatus which may or may not be attached, permanently or temporarily, to the real property. It includes the physical facilities for production, the installations and appurtenant service facilities, those which are mobile, self-powered or self-propelled, and those not permanently attached to the real property which are actually, directly, and exclusively used to meet the needs of the particular industry, business or activity and which by their very nature and purpose are designed for, or necessary to its manufacturing, mining, logging, commercial, industrial or agricultural purposes. *[Section 199 (o), RA No. 7160]*

20.7. Purpose

The word “objective” is used to indicate the reason for conducting a valuation.

20.8. Valuation Reviewer

Valuation reviewers are licensed appraisers or public officers who perform valuation review based on the agency mandates.

20.9. Valuer

The term “valuer” may also refer to “appraiser” and “assessor”, which are widely used in the Philippines. Specifically, Section 3 of the RA No. 9646 or the Real Estate Service Act (RESA) of 2009, the following terms were defined:

- (a) “Appraiser,” also known as valuer, refers to a person who conducts valuation/appraisal; specifically, one who possesses the necessary qualifications, license, ability and experience to execute or direct the valuation/appraisal of real property.
- (b) “Assessor” refers to an official in the local government unit, who performs appraisal and assessment of real properties, including plants, equipment, and machineries, essentially for taxation purposes. This definition also includes assistant assessors.

Framework

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10. Introduction

- 10.1. In relation to the IVS Framework, other synonymous terminologies used in the Philippines, as well as the statutory requirements, are herein defined and discussed to ensure objectivity and compliance.

20. Valuer

- 20.1 Under RESA, professional real estate appraisers, including local government assessors, require Real Estate Appraiser License. This is granted by the Professional Regulation Commission (PRC) through a licensure examination (Section 12 of RA No. 9646), or by registration without examination during the early part of RESA implementation (Section 20, RA No. 9646).

For government officers performing valuations based on their agency mandates, Section 28 of RESA applies, to wit:

“Section 28. *Exemptions from the Acts Constituting the Practice of Real Estate Service.* - The provisions of this Act and its rules and regulations shall not apply to the following: xxx...

(e) Public officers in the performance of their official duties and functions, except government assessors and appraisers.”

30. Objectivity

- 30.1 The Code of Ethics and Responsibilities for real estate service practitioners is provided under Section 35 of RA No. 9646, which shall be prescribed and issued by the Accredited and Integrated Professional Organization (AIPO) to be adopted and promulgated by the Professional Regulatory Board of Real Estate Service (PRBRES).
- 30.2 Government appraisers and assessors are covered by RA No. 6713, or the Code of Conduct and Ethical Standards of Public Officials and Employees.
- 30.3. These codes of conduct shall serve as guide to valuation practitioners in the attainment of transparent and impartial analysis, judgments, opinions and conclusions.

40. Competence

- 40.1 The PRC, through the PRBRES, shall formulate the operational guidelines in the implementation of RA No. 10912, or the Continuing Professional

Development (CPD) Act of 2016. The CPD Programs would ensure that real estate service professionals shall be equipped with the proper skills and knowledge in the face of emerging trends and technologies in the property market. These would be evaluated by the CPD Council as they accredit CPD providers. Prescribed CPD credit units are required for the renewal of professional licenses.

50. Departures

- 50.1 A departure from the IVS as a result of existing legislative, regulatory or other authoritative requirements in the Philippines is permitted in valuations performed in accordance with IVS.

General Standards

PVS 101 Scope of Work

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10. Introduction

- 10.1. The standards are intended to apply to national and local valuation for taxation and other purposes in the Philippines.

20. General Requirements

- 20.1. A valuer must communicate the scope of work to its client prior to completion of the assignment, including the following:
- (a) **Identity of the valuer.** The valuer may be the assessor, appraiser or appraisal committee or other individuals performing valuation for taxation and other purposes.
 - (b) **Identity of the client(s).** Under IVS 101, Section 20, General Requirements, para. 20.1 (b), confirmation of those for whom the valuation assignment is being produced is important when determining the form and content of the report to ensure that it contains information relevant to their needs. In the Philippines, property owners/administrators, local government units, national agencies, and other interested parties are the clients.
 - (c) **Identity of other intended users.** Other users of valuation reports are public and the private sector depending upon the purpose of the appraisal.
 - (d) **Asset(s)/Properties being valued.** The identity of the real estate being valued shall include land, building, machineries and other improvements.
 - (e) **The valuation currency.** The Philippine peso is the standard valuation currency, unless otherwise stated.
- 20.6 **Purpose of the valuation.** For national and local taxation, and other purposes in the Philippines, subject to applicable governing laws, rules and regulations.

PVS 102 Investigations and Compliance

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10. Investigations

- 10.1. For the Local Government Units, the Manual on Real Property Appraisal and Assessment Operations (MRPAAO), issued under Local Assessment Regulations No. 1-04, prescribes the process, procedures and requirements for the appraisal of land, building, and other improvements, and machineries, which includes gathering of data and other evidences, and inspection of property, to ensure that the valuation is properly supported.
- 10.2. For other national agencies, such as the Bureau of Internal Revenue (BIR), investigation and collection of information should be in accordance with their own internal policies and regulations.

20. Valuation Record

- 20.1. The local assessor shall maintain the prescribed forms under the MRPAAO. Other government agencies and stakeholders shall maintain and keep their valuation records in accordance with their established policies and procedures.

PVS 103 Reporting

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10. Valuation Reports

10.1. Valuation reports can be categorized into:

- a) Narrative reports, either long or short, prepared by licensed appraisers or local assessors and the provincial/city appraisal committee;
- b) Schedule of real property values such as Schedule of Fair Market Value (SMFV) and Zonal Value (ZV);
- c) Form reports prescribed by appropriate government agencies; and
- d) Oral reports or depositions during testimonies before adjudicatory or collegial bodies.

20. Valuation Review Reports

20.1. Valuation review reports should be conducted by competent and licensed appraiser exercising impartial judgment, or in accordance with Section 28 of RESA. The valuation review report should contain the scope of review, documents reviewed in terms of form and content, findings and comments, including recommendations, and the date of review.

PVS 104 Bases of Value

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10. Defined Basis of Value – Fair Market Value

- 10.1. Fair market value is defined as “the price at which a property may be sold by a seller who is not compelled to sell and bought by a buyer who is not compelled to buy”. *[Section 199 (l), RA No. 7160]*

20. Premise of Value – Actual/Predominant Use

- 20.1. Real Property shall be classified, valued and assessed on the basis of its actual use regardless of where located, whoever owns it, and whoever uses it. *(Section 217, RA No. 7160)*

“Actual Use” refers to the purpose for which the property is principally or predominantly utilized by the person in possession thereof. *[Section 199 (b), RA No. 7160]*

PVS 105 Valuation Approaches and Methods

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10. Introduction

The three (3) widely used valuation approaches to value are: (1) Market Data Approach; (2) Income Capitalization Approach; and (3) Cost Approach. These approaches are also used by the LGUs in the preparation of their SFMV pursuant to Department of Finance (DOF) Order No. 10-2010.

20. Market Approach

- 20.1. It is also known as Market Data Approach which is synonymous to Comparative or Comparable Sales Approach. This is an appraisal procedure in which the market value estimate is predicated upon prices paid in actual market transactions and current listings. It is a process of analyzing sales of similar sold properties (sold at or around the relevant date) in order to derive an indication of the most probable sale prices of the property being appraised. The reliability of this technique is dependent upon: a) the availability of comparable sales data; b) the verification of the sales date; c) the degree of compatibility or extent of adjustment necessary for time differences; and d) the absence of non-typical conditions affecting the sales price. [*p. x, Mass Appraisal Guidebook (MAG)*]

In some cases, Abstraction Method, as part of Market Data Approach, is used to estimate the value of the land by estimating first the value of the improvement pursuant to Section 210 of the Local Government Code (LGC). The value of the improvement is deducted from the sales price to derive the land value. [*Section 21 (A) (2), Local Assessment Regulations (LAR) No. 1-92*]

In the LGU, valid sales data are gathered from:

- (a) Registrar of Deeds and Notaries Public (*Section 278, RA No. 7160*);
- (b) Sworn statements of property owner (*Section 202, RA No. 7160*);
- (c) Insurance Companies (*Section 279, RA No. 7160*);
- (d) Building Officials Issuing Permit or Certificate of Registration of Machinery (*Section 210, RA No. 7160*); and
- (e) Other reliable sources of information on sales data.

- 20.2 To determine the relationship between the amount of consideration contained in a sales transaction and the current value of the subject property, certain factors in the sales transaction should be thoroughly analyzed. Only sales transactions which meet, more or less, the following criteria shall be considered for sales analysis:

- i. The date of the transaction must be reasonably near the general assessment date. Sales transactions for the current year or preceding year, if adequate, would also serve as a good basis for studies on trends of market values. This should not extend for more than three (3) years from general re-assessment date.
- ii. The type of conveyance representing a normal transaction is one which envisions willing, able and well-informed buyers and sellers. Quitclaims and transfers between relatives, inter-related corporations and the like, should not be considered as these are not “arm’s-length” transactions.
- iii. The amount of consideration reflects a strong presumption of the fair market value of the property involved.

30. Income Approach

- 30.1. It is also known as Income Capitalization Approach. It is a direct approach to estimate the value of property. It is based on the theory that the value of an income-producing property is no more than the return derived from it. It requires an analysis of the income produced by the property in order to estimate the sum which might be invested in the purchase of the property. A detailed financial study must be made of the property. Gross annual income is either determined from actual figures or is estimated. Annual expense figures are obtained from the owner. The income, operating expenses and fixed charges of the subject property are analyzed and the expenses derived thereof are deducted from gross income. The resultant net income capitalized at a rate which the investor of the property can expect as a reasonable return or interest prevailing in the locality. The capitalized value of the income represents the present value of the property. *[Section 21 (B), LAR No. 1-92]*

40. Cost Approach

- 40.1. Cost Approach is also known as the Reproduction/Replacement Cost New (RCN) Approach, which is a factual approach used exclusively in appraising man-made improvements such as buildings and other structures. This approach depends on guides and standards, based on such data materials and labor costs. *[Section 21 (C), LAR No. 1-92]*
- 40.2. The “reproduction or replacement cost approach” makes use of a value estimate of reproducing a new replica property within the same or closely similar materials on the basis of current construction materials and labor costs. Unit base construction cost is developed on a per square meter or per cubic meter basis for typical buildings or structures. The unit cost is multiplied by the ground area or volume, as the case may be, of the subject structure to derive its total reproduction or replacement cost, allowance for depreciation is deducted to arrive at depreciated cost of subject property. *[Section 21 (C), LAR No. 1-92]*

Asset Standards

PVS 300 Plant, Machinery and Equipment

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10. Introduction

- 10.1. Section 199(o) of the Local Government of Code of 1991 (RA No. 7160) defines the term “Machinery” to refer to plant and equipment.

20. Valuation of Machinery

- 20.1 The appraisal and assessment of machinery for real property tax purposes is provided under Section 224 of the RA No. 7160 which states:
- (a) The fair market value of brand-new machinery shall be the acquisition cost. In all other cases, the fair market value shall be determined by dividing the remaining economic life of the machinery by its estimated economic life and multiplied by the replacement or reproduction cost.
 - (b) If the machinery is imported the acquisition cost includes freight, insurance and other charges, brokerage, arrastre and handling, duties and taxes, plus cost of in-land transportation, handling, and installation charges at the present site.
- 20.2 Acquisition cost for newly acquired machinery not yet depreciated and appraised within the year of its purchase, refers to the actual cost of the machinery to its present owner plus the cost of transportation, handling, and installation at the present site. *[Section 199 (a), RA No. 7160]*
- 20.3 The valuation of Plant, Machinery, Equipment and Facilities of Renewable Energy pursuant to Chapter VII (c) of RA No. 9513, otherwise known as Renewable Energy Act of 2008, shall be the original cost less accumulated normal depreciation or net book value.

PVS 400 Real Property Interests

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10. Introduction

- 10.1. Real property refers to all the rights, interests and benefits related to the ownership of real estate. These rights are lumped into a “bundle of rights.” [p. xii, *Mass Appraisal Guidebook (MAG)*]
- 10.2. Real Property shall be classified, valued and assessed on the basis of its actual use regardless of where located, whoever owns it, and whoever uses it. (*Section 217, RA No. 7160*)
- 10.3. “Actual Use” refers to the purpose for which the property is principally or predominantly utilized by the person in possession thereof. [*Section 199 (b), RA No. 7160*]

20. Legal Bases for the Valuation of Real Property

Pursuant to Section 201 of the RA No. 7160, the DOF shall promulgate the necessary rules and regulations for the classification, appraisal and assessment of real property in the form of Department Order, LAR and other form of issuances.

30. Modes of Acquiring Ownership of Real Property

The Civil Code of the Philippines (RA No. 386) enumerates the mode of acquiring ownership of real property under Article 712, Book III, thereof:

- (a) Occupation;
- (b) Intellectual creation;
- (c) By law;
- (d) By donation;
- (e) By testate and intestate succession;
- (f) In consequence of certain contracts;
- (g) By tradition; and
- (h) By means of prescription.

40. Classes of Real Property for Assessment Purposes

- 40.1. For purposes of assessment, real property shall be classified as the following:
 - (a) Residential,
 - (b) Agricultural,
 - (c) Commercial,
 - (d) Industrial,
 - (e) Mineral,
 - (f) Timberland, or
 - (g) Special.

PVS 410 Development Property

Contents	Paragraphs
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10. Introduction

- 10.1. Valuation of Development Property for taxation purposes must follow the PVS 400 Real Property Interests on Asset Standards. However, valuation for other purposes must refer to IVS 410 Development Property on Asset Standards.

Valuation for Taxation Purposes

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10. Introduction

- 10.1. Valuation, also known as appraisal, refers to the systematic, analytic and logical analysis of the recording of property facts, circumstances, investments and other relevant data resulting in a supportable estimate or professional opinion of the value of the property as of a specific date and for a specific purpose.
- 10.2 The valuation of real properties shall be used as basis for property taxation.

20. Real Property Valuation

20.1. Valuation Practice

RA No. 9646 declares the policy of the State to recognize the “*vital role of real estate service practitioners in the social, political, economic development and progress of the country by promoting the real estate market, stimulating economic activity and enhancing government income from real property - based transactions. Hence, it shall nurture through proper and effective regulation and supervision a corps of technically competent, responsible and respected professional real estate service practitioners whose standards of practice and service shall be globally competitive and will promote the growth of the real estate industry.*”

Real Estate Service Practitioners, pursuant to RA No. 9646, consist of the following:

- (a) Real estate consultant - a duly registered and licensed -natural person who, for a professional fee, compensation or other valuable consideration, offers or renders professional advice and judgment on:
 - (i) the acquisition, enhancement, preservation, utilization or disposition of lands or improvements thereon; and (ii) the conception, planning, management and development of real estate projects
- (b) Real estate appraiser - a duly registered and licensed natural person who, for a professional fee, compensation or other valuable consideration, performs or renders, or offers to perform services in estimating and arriving at an opinion of or acts as an expert on real estate values, such services of which shall be finally rendered by the preparation of the report in acceptable written form.
- (c) Real estate assessor - a duly registered and licensed natural person who works in a local government unit and performs appraisal and assessment of real properties, including plants, equipment, and machineries, essentially for taxation purposes.

- (d) Real estate broker - a duly registered and licensed natural person who, for a professional fee, commission or other valuable consideration, acts as an agent of a party in a real estate transaction to offer, advertise, solicit, list, promote, mediate, negotiate or effect the meeting of the minds on the sale, purchase, exchange, mortgage, lease or joint venture, or other similar transactions on real estate or any interest therein.
- (e) Real estate salesperson - a duly accredited natural person who performs service for, and in behalf of a real estate broker who is registered and licensed by the Professional Regulatory Board of Real Estate Service for or in expectation of a share in the commission, professional fee, compensation or other valuable consideration.

The real estate service practitioners are under the supervision and control of the Philippine Regulation Commission (PRC) through the Philippine Regulatory Board of Real Estate Service (PBRES).

20.2. **Authority of the Department of Finance (DOF)**

The Department of Finance promulgates the necessary rules and regulations for the classification, appraisal, and assessment of real property pursuant to the provisions of the Local Government Code of 1991. (*Section 201, RA No. 7160*)

20.3. **Preparation of Schedule of Market Value**

The preparation of the Schedule of Market Value (SMV) is a mass appraisal process which facilitates the appraisal of multiple properties at a given date by a systematic and uniform application of appraisal methods and techniques. The blend of cost and sales/direct comparison approaches to value may be used as technique for preparing the SMV.

There are four (4) major stages of the SMV development, namely: 1) Preparatory stage, 2) Data Collection Stage, 3) Data Analysis Stage, and (4) Testing of SMV Stage. The electronic system for developing the Schedule of Market Values is also called Computer-Assisted Mass Appraisal (CAMA).

Section 219 of the RA No. 7160 prescribes the conduct of General Revision of Assessments and Property Classification where the provincial, city or municipal assessor shall undertake a general revision of real property assessment within two (2) years after the effectivity of this Code and every three (3) years thereafter.

30. **Zonal Valuation**

The Commissioner of the BIR has the authority to prescribe real property values pursuant to Section 4 of the Tax Reform for Acceleration and Inclusion (TRAIN), or RA No. 10963, which amended Section 6(E) of the National Internal Revenue Code (NIRC) which states:

Section 4. Section 6 of the NIRC, as amended, is hereby further amended to read as follows:

“Sec. 6. Power of the Commissioner to Make Assessments and Prescribe Additional Requirements for Tax Administration and Enforcement. - xxx...

(E) *Authority of the Commissioner to Prescribe Real Property Values* – The Commissioner is hereby authorized to divide the Philippines into different zones or areas and shall, upon mandatory consultation with competent appraisers both from the private and public sectors, and with prior notice to affected taxpayers, determine the fair market value of real properties located in each zone or area, subject to automatic adjustment once every three (3) years through rules and regulations issued by the Secretary of Finance based on current Philippine valuation standards: *Provided*, That no adjustment in zonal valuation shall be valid unless published in a newspaper of general circulation in the province, city or municipality concerned, or in the absence thereof, shall be posted in the provincial capitol, city or municipal hall and in two (2) other conspicuous public places therein: *Provided, further*, That the basis of any valuation, including the records of consultations done, shall be public records open to the inquiry of any taxpayer. For purposes of computing any internal revenue tax, the value of the property shall be, whichever is the higher of:

- (1) the fair market value as determined by the Commissioner; or
- (2) the fair market value as shown in the schedule of values of the Provincial and City Assessors.”

References for Valuation of Property for Other Purposes

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10. Introduction

- 10.1. This section serves as a reference for valuation of property for other governmental purposes. For other purposes not herein listed, reference must be made to the applicable Standards under Part I hereof.

20. Creation of Appraisal Committee

Executive Order No. 132, s. 1937, as amended, provides the composition of appraisal committee for province and city; and municipality within the Metro Manila Area as follows:

1. Provincial/City Appraisal Committee

Provincial/City Assessor	-	Chairperson
Provincial/City Engineer	-	Member
Provincial/City Treasurer	-	Member

If the province has no position of Provincial Engineer, the Public Works Engineer or Highways District Engineer shall sit as member.

2. Metro Manila Area

a. City Appraisal Committee

MMDA Chairperson	-	Chairperson
City Assessor	-	Member
City Treasurer	-	Member
City Engineer	-	Member
DPWH District Engineer	-	Member

b. Municipal Appraisal Committee

MMDA Chairperson	-	Chairperson
DPWH District Engineer	-	Member
Municipal Assessor	-	Member
Municipal Engineer	-	Member
Municipal Treasurer	-	Member

National government agencies and other instrumentalities also have their own internal appraisers, and may create their own appraisal committee depending on the nature of their agency, and the specific mandated functions and authorities.

30. Acquisition of Private Property

National and local governments, GOCCs, and instrumentalities of government may acquire private properties for government programs and projects.

30.1. Agricultural Land Valuation and Landowner Compensation

The specific provisions of Section 7 of RA No. 9700, as implemented by the Department of Agrarian Reform (DAR) under Administrative Order (AO) No. 2, series of 2009, are presented as follows:

- a) the amount determined in accordance with the criteria provided for in Section 7 of the said law and existing guidelines on land valuation; or
- b) the value based on the order of the Department of Agrarian Reform Adjudication Board (DARAB) or the regular court, which has become final and executory.

The specific guidelines governing the valuation of lands under Voluntary Offer to Sell (VOS) or Compulsory Acquisition (CA), pursuant to RA No. 6657, as amended by RA No. 9700, are provided in Comprehensive Agrarian Reform Program - Land Acquisition and Distribution (CARP-LAD) process.

30.2. Valuation of Lands for Socialized Housing

The specific provisions of Section 13 of RA No. 7279 are presented as follows:

"Sec. 13. *Valuation of Lands for Socialized Housing.* — Equitable land valuation guidelines for socialized housing shall be set by the Department of Finance on the basis of the market value reflected in the Zonal valuation, or in its absence, on the latest real property tax declaration.

For site already occupied by qualified Program beneficiaries, the Department of Finance shall factor into the valuation the blighted status of the lands as certified by the local government unit or the National Housing Authority."

For this purpose, the DOF issued Local Finance Circular No. 3-92, as amended by Local Finance Circular No. 1-97, which provides the determination of the market values of "blighted lands," which shall be computed primarily on the basis of the land market value determined on the basis of the zonal values shown in the approved zonal valuation of real properties for the area or locality applicable to the corresponding period, shall be used as the basis for computing the fair market value of the subject land.

40. Valuations of Private Lands for Acquisition by the Government

- 40.1 Valuation of private land for acquisition of the national government for road right-of-way (RROW) shall be governed by the provisions of Sections 5, 6 and 7 of RA No. 10752, an Act Facilitating the Acquisition of Right-of-Way Site or Location for National Government Infrastructure Projects, and its implementing rules and regulations.
- 40.2 Valuation of private lands for acquisition of local governments shall be through the power of Eminent Domain as provided for in Section 19 of RA No. 7160. Local governments may adopt the prescribed procedures in Sections 5, 6, and 7 of RA No. 10752. Otherwise, they shall be guided by Executive Order

No. 132, series of 1937, as amended, which provides the procedures to be followed in the acquisition of private property for public use and creating appraisal committees.

50. Valuation for Disposition of Properties

50.1. Appraisal of Public Lands and Other Patrimonial Property

The specific provisions of Section 3 of Department of Environment and Natural Resources (DENR) Administrative Order (AO) No. 98-20 are presented as follows:

Manner of Conducting the Appraisal – In conducting the appraisal or re-appraisal, the fair market value of the property shall, as much as possible, be ascertained by considering the following factors, to wit:

- a) Extent, classification, location, actual use and development trends of the area;
- b) Assessed value and BIR zonal valuation;
- c) Sales and holding prices of lands of similar character located in the area;
- d) Highest and best use or potential of the property;
- e) The purpose for which the property is to be disposed of; and
- f) Other relevant factors or circumstances.

Subject to the limitations that may be imposed under the pertinent laws, proclamations or any other presidential issuances, in no case shall the appraised or re-appraised value of the property classified as residential or agricultural or primarily used for institutional or recreational purposes, be less than the current assessed value thereof. If the property is classified as commercial or industrial, the appraised or re-appraised value shall be not less than the average of the assessed and zonal values thereof.

If the property has not yet been declared for taxation purposes or its assessed value is not available, the assessed value of another property located in the area which is of similar character with that of the property being appraised shall be used.

If the zonal value is not available, the fair market value or the current selling price of properties of similar character in the area as determined by the Assessor's Office concerned, shall be utilized. The foregoing, notwithstanding, part or parts not exceeding thirty percent (30%) of its total area, of industrial or commercial land covered by lease application or contract, which are devoted to and actually utilized for growing of trees, shall be appraised or re-appraised at zero value.

60. Valuation for Other Purposes

- 60.1. Valuation for other purposes by other government agencies shall be guided by the standards, subject their respective governing laws, rules and regulations and procedures.

PART III

GUIDANCE NOTES

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Introduction

Part III contains the Guidance Notes that cover issues in the application of the Standards which frequently arise in valuation practice and from those who use valuation services. The herein Guidance Notes provide guidance on valuation issues and how Standards are to be applied in more specific situations. The Guidance Notes, as in the PVS 2009, complement and expand certain discussions on the Standards and Applications, with which they have equal importance and relevance.

In PVS 2009, it included fifteen (15) guidance notes. In the succeeding editions of IVS 2007, however, the guidance notes were removed for various reasons. Either the elements were carried forward and merged with other relevant topics/sections, or the IVSC produced revised Technical Information Papers (TIP) on the topics. After deliberations and consultations with stakeholders, the Guidance Notes in the earlier PVS are still deemed valid with regard to valuation practice in the Philippines, and identified and updated eight (8) Guidance Notes to form part of the PVS 2nd Edition, 2018.

Compliance with the Guidance Notes, as with the Standards, is therefore enjoined for Valuers, in preparing valuation assignments, as far as practicable.

PVS GN 100 Mass Appraisal for Property Taxation¹

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10. Introduction

- 10.1. The objective of this Guidance Note is to provide a framework for the performance of *Mass Appraisal* assignments for *Ad Valorem Property Taxation* throughout the Philippines. The Guidance Note provides assistance in understanding recognized Mass Appraisal methods, the design and implementation of property taxation systems and the relationship of Mass Appraisal to International Valuation Standards.
- 10.2. The *Mass Appraisal Process* may be utilized as a methodology for *Ad Valorem Property Taxation*, or statistical and economic studies under government administrative programs. The appraisal outputs facilitate revenue raising, revenue equalization, and the distribution of financial benefits or grants to government authorities. In this Guidance Note, reference to Mass Appraisal implies Mass Appraisal for the above purposes.
- 10.3. For a property taxation system to be effective, the following elements must be in place:
- a legal system and legal infrastructure that define, support and protect property rights;
 - a recording and inventory system for all parcels of land, which represents the basis of taxation;
 - sufficient market data from which valuations may be determined;
 - sufficient resources and trained personnel to implement the system;
 - continued maintenance of the inventory and databases to ensure more refined data, more accurate valuations, and more equitable taxation; and
 - a process for sampling and testing developed models to ensure consistency in methodology and application.
- 10.4. The Mass Appraisal process includes:
- identifying properties to be appraised;
 - defining the market area in terms of consistent behavior on the part of property owners and would-be purchasers;
 - identifying characteristics of supply and demand that affect the creation of value in the defined market area;

¹ Formerly Guidance Notes GN 13 of PVS 1st Edition 2009

- d) developing a model structure that reflects the relationship among the characteristics affecting value in the market area;
 - e) calibrating the model structure to determine, among other attributes, the contribution of the individual property features affecting value;
 - f) applying the conclusions reflected in the model to the characteristics of the property(ies) being appraised;
 - g) validating the adopted mass appraisal process, model, measurements or other readings including the performance measures, on an ongoing basis and/or at discrete stages throughout the process; and
 - h) reviewing and reconciling the Mass Appraisal results.
- 10.5. The valuation basis for Mass Appraisal is *Market Value* as defined in *IVS 104, Section 30*, subject to any modification of the concept as specified under relevant instructions or legislation. If such instructions or legislation stipulate a valuation basis other than *Market Value* as defined above, Valuers should apply appropriate valuation methods to accomplish the objectives of IVSC Standards under these circumstances. See *IVS 104, Section 120, Other Basis of Value – Fair Value (Legal/Statutory) in different jurisdictions. In the Philippines, for purposes of ad valorem real property taxation, Section 199(I) of Republic Act No. 7160, also known as the Local Government Code of 1991, defines fair market value as “the price at which a property may be sold by a seller who is not compelled to sell and bought by a buyer who is not compelled to buy”.*
- 10.6. Mass Appraisal can be prepared with or without computer assistance. While computerized methodology has made the Mass Appraisal process more efficient and more widespread, it has not altered that process. Data banks and computerized applications are used in data storage, mapping, data analysis, and testing of the results.
- 10.7. The development of Mass Appraisal systems for Property Taxation should follow recognized scientific standards in statistical applications.
- 10.8. Republic Act No. 9646, also known as the Real Estate Service Act, particularly Section 35 thereof, provides the adoption and promulgation of the Code of Ethics and Responsibilities for real estate service practitioners by the Professional Regulatory Board of Real Estate Service (PRBRES) as prescribed and issued by the accredited and integrated professional organization of real estate service practitioners.
- Requirements under the *Code of Ethics and Responsibilities* apply to:
- a) the mass valuation process itself; and
 - b) the use of computers and computer-generated models in the mass appraisal process.
- 10.9. According to the National Internal Revenue Code (NIRC), capital gains tax or CGT is a tax that is imposed on earnings the seller has gained from the sale of capital assets. It is charged at a flat tax rate of 6% of the gross selling price.

Section 5 of the Revenue Regulations No. 17-2003 prescribes that the final capital gains tax on the sale or other onerous disposition of real property considered as capital asset which must be taken/withheld from the seller by the buyer and remitted within thirty (30) days from the date of notarization of the transfer document to the collecting agent of the RDO having jurisdiction over the place where the property is located.

When there is a sale of real estate, automatically people think that they have to pay Capital Gains Tax (CGT). This is not necessarily the case. CGT is a tax on the gain from the sale of capital assets. Regular corporate income tax (RCIT) [for corporations] and regular income tax [for individuals] apply to the sale of ordinary assets while CGT applies to the sale of capital assets.

Thus, there is a need to determine first whether the asset being sold is a capital or an ordinary asset so as to know the proper tax rate to be used and the BIR form to be used, among others. When the real property which is a capital asset to the seller is sold, the gross selling price or fair market value (FMV) [zonal value], whichever is higher, will be subject to 6% CGT.

Zonal values are the values of real properties which can more or less approximate the present fair market values of real properties as basis for computing the Property Tax (capital gains tax, documentary stamp tax, estate tax when the property is sold or transferred), based on the rules and regulations of the BIR and the Department of Finance consistent with the established standards.

20. Scope

- 20.1. The professional responsibility of Valuers is, in most instances, prescribed by statute or regulations affecting Mass Appraisal assignments. It is the professional duty of the Valuer to be familiar with, adhere to, and administer the provisions of the law established in the Ad Valorem property taxing jurisdiction.
- 20.2. The various outputs from Mass Appraisal programs have financial implications in government administration. For purposes of revenue raising, revenue equalization, or the distribution of benefits or grants, any departure from an accurate basis of assessment will result in inequities. Local statutes prescribe the basis and definitions of values to be returned (i.e., the assessments and/or indices developed in Mass Appraisal assignments), the administrative procedures for the collection and delivery of valuation data, the time-frames between undertaking Mass Appraisals, and the processes for appeal of assessments or indices.
- 20.3. The scope of the completed assignment shall be consistent with:
 - a) the expectations of participants in the market for the same or similar valuation services; and
 - b) the requirements of IVSC Standards, Guidance Notes and Applications for the same or a similar assignment.

30. Definitions

- 30.1. *Ad Valorem Property Taxation.* A revenue-raising procedure, based on the assessed value of property related to a scale of charges defined by statute within a specified time-frame.

- 30.2. *Calibration.* The process of analyzing sets of property and market data to determine the specific parameters operating upon a model.
- 30.3. *Mass Appraisal.* The practice of appraising multiple properties as of a given date by a systematic and uniform application of appraisal methods and techniques that allow for statistical review and analysis of results.
- 30.4. *Mass Appraisal Process.* The procedures applied in mass appraisal assignments for arriving at assessments and/or indices. This process includes the eight steps, identified in para. 10.4 above.

40. Relationship to Accounting Standards

- 40.1. Mass Appraisal does not fall under the governance of national or international accounting standards.
- 40.2. Valuers should be aware that revaluation procedures for financial reporting purposes are unrelated to Mass Appraisal procedures for Ad Valorem Property Taxation.
- 40.3. Legislative requirements and standards of appraisal level and uniformity in valuations for Ad Valorem Property Taxation are likely to produce variations in property values from those determined for financial reporting purposes.

50. Guidance

50.1. Data Collection and System Recording

- 50.1.1. A robust data collection system must be available to the Valuer. The recording of data has evolved from the use of manual methods to the creation of sophisticated data banks that facilitate computer-assisted appraisal, often incorporating geographic information systems (GIS). Property data may be quantitative (e.g., land areas, dimensions, building specifications) and/or qualitative (assessment of the physical condition, character, or market desirability of the improvements).
- 50.1.1.1. Appraisal data banks are built around land tenure records, e.g., title deeds, transfer documents, and sales information, in national, federal, state or local government jurisdictions that define property ownership or interests in land.
- 50.1.2. Characteristics of the market that are relevant to the purpose and intended use of the Mass Appraisal shall be recorded in the system including:
 - (a) location of the defined market area;
 - (b) physical, legal, and economic attributes of the properties;
 - (c) time-frame of market activity; and
 - (d) property interests reflected in the market.

50.2. The Development and Maintenance of Assessment Lists (Assessment Roll)

- 50.2.1. Assessment Lists/Roll will contain information on property ownership, value definitions, details of the assessment, date of the assessment, and date on which the assessment comes into force.

- 50.2.2. Assessment Lists/Roll must allow for periodic adjustments or alterations to ensure the currency and consistency of assessed values.

50.3. **Mass Appraisal Value Definitions**

- 50.3.1. Where mass appraisal is undertaken for the purpose of Ad Valorem Property Taxation, value definitions are generally mandated by local statute. Specific valuation methodologies may be required under different value definitions.

50.4. **Standards of Appraisal Level and Uniformity**

- 50.4.1. In the interests of assessment equity, standards of appraisal level (the proximity between assessments and actual prices) and uniformity (the statistical measure of valuation consistency) must be observed in the application of mass appraisal systems.

50.5. **Disclosure in Mass Appraisal Assignment Reports**

- 50.5.1. Valuers undertaking Mass Appraisal assignments are subject to the provisions of *IVS 103, Reporting*. The Valuer shall disclose the following essential data that is specific to Mass Appraisal reporting:

- (a) the client and other intended users;
- (b) the purpose and intended use of the appraisal;
- (c) the scope of work necessary to complete the assignment, including any special limiting conditions;
- (d) any extraordinary assumptions and hypothetical conditions needed to carry out the assignment, provided these are reasonable and result in a credible analysis;
- (e) the relevant basis of valuation if, under reasonable terms and conditions, the value opinion to be developed is other than *Market Value*;
- (f) the characteristics of the properties that are relevant to the purpose and intended use of the Mass Appraisal;
- (g) a reference to each individual property in the Assessment List or grouping, indicating where information is stored in the property record relating to its identity;
- (h) the characteristics of the market that are relevant to the purpose and intended use of the Mass Appraisal (see para. 5.1.2).

50.6. **Departure**

- 50.6.1. Departure from the instructions in this Guidance Note should only result from required compliance with statutory provisions, administrative instructions, or the agreed or amended terms of appraisal contracts.
- 50.6.2. Further discussion on Departure provisions is set out in the IVS Framework, Section 60, Departures.

PVS GN 200 Valuation of Agricultural Properties²

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10. Introduction

- 10.1. In many regions of the world, lands devoted to the production of agricultural commodities are the major economic asset and, frequently, the sole economic base of a region.
- 10.2. Lands devoted to agricultural use are thus a principal subject of valuation services for a multitude of reasons including private and public transfer of ownership, taxation, determination of collateral for financing, and economic, land-use, and investment studies. Reliable valuations of agricultural lands are essential to ensure the availability of capital necessary to support the continuity of the economic base, to promote the productive use of the land, to maintain the confidence of capital markets, and to meet the needs for general financial reporting.
- 10.3. Providing a reliable and accurate valuation service for agricultural properties requires that the Valuer have a sound knowledge and understanding of the physical and economic elements that affect the productive capacity of agricultural lands and the value of the commodities produced thereon.
- 10.4. The physical and economic characteristics of agricultural lands differ from those of non-agricultural or urban environments in degree of importance.
 - 10.4.1. Soils in an urban environment must be suitable for bearing the improvements that stand upon them. In agricultural properties, the soil is the principal agent in production, varying in its capacity to support a given amount of a particular commodity or class of commodities.
 - 10.4.2. In urban environments, the economic use of a property and/or the amenities it provides may remain unchanged over a period of years and may even be guaranteed by contractual arrangements. While for some agricultural properties, the same use may extend over a long duration (e.g., forests harvested after 25 years), for others, the economic benefits can vary from year to year, depending on the commodities the property is capable of producing.
 - 10.4.3. The income stream associated with agricultural property will vary from year to year, depending on the type of agriculture for which it is used, the commodities produced, and the cyclical nature of the commodity markets.

²Formerly Guidance Notes GN 10 of PVS 1st Edition 2009

20. Scope

- 20.1. This GN encompasses those characteristics of value associated with agricultural properties, and the basic requirements of the Valuation Standards and Applications as they apply to the valuation of agricultural properties.

30. Definitions

The agricultural uses of properties may be classified in several broad groups definitions of which follow:

- 30.1. **Crop(ping) Farms.** Agricultural properties used for growing commodities that are typically planted and harvested within a twelve-month cycle. Properties used for annual crop production may grow more than one type of annual crop over the same period and may or may not make use of irrigation to produce the crops. Some commodities are annual crops that may be left in the ground beyond a twelve-month cycle, per contract provisions or in circumstances where market conditions are unfavorable. These crops will last for more than one year after harvest but are considered less than permanent. Also see irrigated land (para. 30.4), perennial plantings (para. 30.6).
- 30.2. **Dairy Farms.** Agricultural properties used for the production of milk from cows, carabaos, goats, etc. or for other dairy products. These properties usually have extensive structural improvements (barns, milking parlors, silos) and equipment (feed bins, milking machines). Feed may be produced on the property, imported, or supplied by both sources.
- 30.3. **Forestry/Timberland.** Agricultural property used for the growing of non-orchard trees that are periodically harvested over extended growing periods (10 to 20 or more years). Considered to be agricultural properties because they produce a crop, i.e., wood, even though that crop requires a long-term growing period. Also see perennial plantings. *There are also non-orchard trees periodically harvested over growing periods of less than 10 years (e.g. Gemelina trees).*
- 30.4. **Irrigated Land.** Lands used to produce crops of forage for livestock and which require the application of water other than that from natural rainfall, are called irrigated crop(ping) farms or irrigated grazing land. Properties that lack a water source other than natural rainfall are referred to as dry land agricultural properties.
- 30.5. **Livestock Ranches/Stations.** Agricultural properties used to raise and feed animals such as cattle, sheep, pigs, goats, horses, or combinations thereof. The actual use of these properties can take many forms. The animals may be bred, raised, and sold within the operation of the property. Young animals may be acquired from outside the property and then raised within the property. The animals may be raised for consumptive use or for breeding stock. Feed for the animals may be produced on the property, imported, or supplied by both sources. Properties used for the production and feeding of livestock have significant capital investment in the structural improvements (pens, livestock shelters, sheds, division fencing) and the livestock, which may or may not be depreciable depending on the laws and regulations of the local jurisdiction.

- 30.6. **Perennial Plantings.** Crops grown from plantings that have a life extending beyond one year or one-crop cycle. Examples are vineyards and orchards. These types of properties can have significant capital investment in the plantings, which represent a depreciable asset. Also see forestry/timberland (para. 30.3).
- 30.7. **Specialized Livestock Facilities.** See dairy farms, livestock ranches/stations.
- 30.8. **Specialized, or Special Purpose Properties.** Agricultural properties that do not typically produce a crop but are used for the handling, processing, or storage of crops following harvest. These properties frequently have a small land base that is extensively developed with structural improvements (grain elevators) and equipment (lifting machinery). Properties may also be classified as special purpose by the nature of the commodity produced. Examples are truck farms, poultry farms, farms that produce certified crop seeds or fresh cut flowers, and racehorse breeding or training stables.
- 30.9. **Agricultural Activity.** Management by an entity of the biological transformation of biological assets for sale, into agricultural produce, or into additional biological assets. See International Accounting Standard 41 [IAS 41], Agriculture, para.5). *Agricultural Land as defined in RA 7160, Local Government Code of 1991, Sec. 199, para. (d): —is land devoted principally to the planting of trees, raising of crops, livestock and poultry, dairying, salt making, inland fishing and similar aqua-cultural activities, and other agricultural activities, and is not classified as mineral, timber, residential, commercial or industrial land.*
- 30.10. **Biological Asset.** A living animal or plant. (IAS 41, para. 5).
- 30.11. **Integrated Unit.** An agricultural entity that has common ownership of all or part of the processes involving the production and marketing of its products and/or commodities.

Integrated Forest Management Agreement (IFMA) as defined in DENR Administrative Order (AO) No. 99-53: —is a production sharing contract entered into by and between the DENR and a qualified applicant wherein the DENR grants to the latter the exclusive right to develop, manage, protect and utilize a specified area of forestland and forest resources therein for a period of 25 years and may be renewed for another 25-year period, consistent with the principle of sustainable development and in accordance with an approved Comprehensive Development and Management Plan (CDMP) and under which both parties share in its produce.

40. Relationship to Accounting Standards

- 40.1. International Accounting Standards 16 (Property, Plant and Equipment), 40 (Investment Property), and 41 (Agriculture) apply to the valuation of agricultural property. An entity follows IAS 16 or IAS 40, depending on which standard is appropriate in the circumstances. IAS 16 requires that land be measured either at its cost less any accumulated depreciation and accumulated impairment losses or at a revalued amount. IAS 40 requires land that is investment property to be measured at its fair value, or cost less any accumulated depreciation and accumulated impairment losses. IAS 41, which establishes no new principles for land related to agricultural activity, requires that biological assets physically attached to land (e.g., trees in a plantation

forest) be measured at their fair value less estimated point-of-sale costs, separately from the land.

- 40.2. IAS 41 acknowledges that there may be no separate market for biological assets attached to the land but that an active market may exist for the combined assets, i.e., the biological assets, raw land, and land improvements, as a package. An entity may, therefore, use information regarding the combined assets to determine fair value for the biological assets. The fair value of raw land and land improvements may be deducted from the fair value of the combined assets to arrive at the fair value of the biological assets. (See IAS 41, para. 25.) IAS 41 also gives guidance on how to determine fair value for a biological asset or agricultural produce where an active market exists as well as in the absence of an active market.

- 40.3. Agricultural property assets can be classified as:

- Land;
- Structural improvements;
- Plant, machinery and equipment (attached to the land);
- Plant, machinery and equipment (not attached to the land);
- Biological assets (attached to the land); and
- Biological assets (not attached to the land).

The Fair Value/Market Value of biological assets is the contributory, or added, value they give to the land. IAS 41 requires that biological assets be measured upon initial recognition and at each balance sheet date.

- 40.4. IAS 16 recommends frequent revaluations, stating that every three to five years may be sufficient. IAS 40 requires revaluation on an annual basis.

50. Guidance

- 50.1. Diverse forms of commodity production and methods of operation are characteristic of agricultural properties. These properties may also represent various combinations of land, buildings, equipment, and crop plantings. Generally Accepted Valuation Principles (GAVP) are as applicable to agricultural properties as they are to the valuation of other forms of real property.

- 50.1.1. The Valuer must have competence in valuing the various assets that comprise the property. (See IVS Framework, section 50, Competence.)

- 50.2. Market Value must be recognized as the fundamental basis of valuation (IVS 104).

- 50.2.1 The Valuer shall arrive at the Market Value for the agricultural property, ensuring that the valuation is market-derived.

- 50.2.2. For financial reporting, the Valuer shall apportion the Market Value in accordance with the requirements of the IAS.

- 50.3. Where other bases of valuation are used, they must be distinguished from the Market Value basis.

50.3.1. When estimating values other than Market Value as required for financial reporting, depreciation schedules, or tax purposes, the Valuer must ensure that the distinction is clearly defined and noted.

50.4. **Non-Realty Elements**

50.4.1. When the valuation is made of an agricultural property that may include non-realty elements such as livestock, stored crops, and equipment, the Valuer needs to understand when a crop or other commodity is real property and when it may become personal property. Timber for example, is part of the real property while growing but becomes personal property when it is removed from the land.

50.5. The Valuer must understand the unique nature of agricultural productive factors, commodity markets, production practices, and cycles in the market region.

50.5.1. In the valuation of agricultural properties, the physical and environmental aspects of the property assume special importance. These include features such as climate, soil types and their productive capability, the availability or absence of water for irrigation, and the feeding/carrying capacity for livestock. External factors to be considered include the availability and adequacy of support facilities required for storage, processing, and transportation. The relative importance of these factors will vary depending upon the type of agriculture for which the property is suited or used. The Valuer needs to consider both internal and external factors in making a determination of which class of agricultural use the property is best suited for.

50.5.2. In keeping with the definition of Market Value, a highest and best use analysis of the property should always be conducted in order 1) to warrant that an agricultural use is to be continued, especially when it appears that another land use, e.g., subdivision development occasioned by encroaching urban/suburban expansion, might be more appropriate, and 2) to determine whether the specific agricultural use is to be continued.

50.5.3. Where the Valuer is specifically instructed to ignore uses other than the current agricultural use, the resulting valuation will not necessarily indicate the Market Value of the property, and this should be fully disclosed.

50.6. The estimate of stabilized income to the agricultural property must be based on the crop patterns and cycles in the market area.

50.6.1. The cash flow to agricultural properties is a function of both the production cycle followed on the property and cyclical forces such as commodity markets. The Valuer should understand the impact of these cycles on cash flows. The valuation of the property must be based on the stabilized pattern of income that is consistent with the production cycles commonly practiced in the region in which the property is located.

50.7. The Valuer of agricultural property that has more than one physical component of class of agricultural use must clearly state whether the value of each component or use is its value contribution as part of the whole property or its value as a separate, free-standing component.

50.7.1. The various components of a whole property may have value as separate parts which is greater or lesser than their value as part of the whole. The

Valuer must determine whether each component is to be valued individually or as part of the whole property.

- 50.7.2. Agricultural properties may be managed to simultaneously produce more than one class of commodity based on different physical conditions within the property or on management decisions. In the valuation of agricultural properties on which crops of more than one agricultural classification are cultivated and harvested at different times, the value of each agricultural classification must be based on its contribution to total property value and not its stand-alone value.
- 50.7.3. The agricultural use of the property may require extensive building improvements, e.g., barns, silos, dairy machinery. Such improvements, while requisite to the proper operation of the property, are frequently secondary to the principal land asset. Their value must be based on their contribution to the total value of the property regardless of their cost or other measure.

Typically, such improvements have a value in use, i.e., their contributory value to the enterprise/entity. On those occasions where an allocation of value between the assets may be required, such an allocation is not to be taken as an indication of the individual value of the improvements as separate assets.

- 50.8. The requirements for valuation reports are addressed in the IVS Framework and IVS 103, Valuation Reporting.

PVS GN 300 Reviewing Valuations³

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10. Introduction

10.1. A valuation review is a review of a Valuer's work undertaken by another Valuer exercising impartial judgment.

10.2. Because of the need to ensure the accuracy, appropriateness, and quality of Valuation Reports, valuation reviews have become an integral part of professional practice. In a valuation review, the correctness, consistency, reasonableness, and completeness of the valuation are considered.

10.2.1. A valuation review may call for input from experts with specialist knowledge of construction costs, property income, legal and tax matters, or environmental problems.

10.2.2. A valuation review provides a credibility check on the valuation under review, and tests its strength by focusing upon

10.2.2.1. The apparent adequacy and relevance of the data used and enquiries made;

10.2.2.2. The appropriateness of the methods and techniques employed;

10.2.2.3. Whether the analysis, opinions, and conclusions are appropriate and reasonable; and

10.2.2.4. Whether the overall product presented meets or exceeds Generally Accepted Valuation Principles (GAVP).

10.3. Valuations reviews are performed for a variety of reasons, including:

10.3.1. Due diligence required of financial reporting and asset management;

10.3.2. Expert testimony in legal proceeding and circumstances;

10.3.3. A basis for business decisions; and

10.3.4. Determination of whether a report complies with regulatory requirements, where

10.3.4.1. Valuations are used as part of the mortgage lending process, especially mortgages insured or regulated by the government; and

³Formerly Guidance Notes GN 11 of PVS 1st Edition 2009

- 10.3.4.2. It is necessary to test whether Valuers have met regulatory standards and requirements within their jurisdiction.

20. Scope

- 20.1. The requirements in this GN apply to the development and reporting of valuation reviews.
- 20.2. Compliance with this GN is incumbent upon any Valuer who, in a supervisory or managerial capacity, signs a valuation review, thereby accepting responsibility for the contents of that review.

30. Definitions

- 30.1. **Administrative (Compliance) Review.** A valuation review performed by a client or user of valuation service as an exercise in due diligence when the valuation is to be used for purposes of decision-making such as underwriting, purchasing, or selling the property. A Valuer may, on occasion, perform an administrative review to assist a client with these functions. An administrative review is also undertaken to ensure that a valuation meets or exceeds the compliance requirements or guidelines of the specific market and, at a minimum.
- 30.2. **Desk Review.** A valuation review that is limited to the data presented in the report, which may or may not be independently confirmed. Generally performed using a checklist of items. The reviewer checks for the accuracy of calculation, the reasonableness of data, the appropriateness of methodology, and compliance with client guidelines, regulatory requirements, and professional standards. Also see field review.
- 30.3. **Field Review.** A valuation review that includes inspection of the exterior and sometimes the interior of the subject property and possibly inspection of the comparable properties to confirm the data provided in the report. Generally performed using a checklist that covers the items examined in a desk review and may also include confirmation of market data, research to gather additional data, and verification of the software used in preparing the report. Also see desk review.
- 30.4. **Technical Review.** A valuation review performed by a Valuer to form an opinion as to whether the analyses, opinions, and conclusions in the report under review are appropriate, reasonable, and supportable.
- 30.5. **Valuation Review.** A valuation assignment that covers a range of types and purposes. The principal characteristic of all valuation reviews have in common is that one Valuer exercises impartial judgment in considering the work of another Valuer. A valuation review may support the same value conclusion in the valuation under review or it may result in disagreement with that value conclusion. Valuation reviews provide a credibility check on the valuation as well as check on the strength of the work of the Valuer who developed it, as regards the Valuer's knowledge, experience, and independence. In some countries a valuation review may also be an update done by a Valuer of the same valuation firm that carried out the original valuation.

Valuation organizations around the world distinguish between various types of review, e.g., administrative (compliance) reviews, technical reviews, desk reviews, field reviews, reviews to ensure that a valuation has been carried

out in accordance with professional standards (where the bases of valuation used in the valuation under review are accepted), reviews that muster general market information to support or contest the value conclusion, and reviews that examine the specific data in the valuation under review with comparable data from a sample group.

40. Relationship to Accounting Standards

- 40.1. Where the valuation review is in connection with financial reporting purposes, IVS 102 Investigation and Compliance Section 40 Compliance with Other Standards should be considered.

50. Guidance

- 50.1. In developing a valuation review, the Review Valuer shall:
- 50.1.1. Identify the client and intended users of the Valuation Review, the intended use of the Review Valuer's opinions and conclusions, and the purpose of the assignment;
 - 50.1.2. Identify the subject property, the date of the valuation review, the property and ownership interest valued in the report under review, the date of the report under review, the effective date of the opinion in the report under review, and the Valuer(s) who completed the report under review;
 - 50.1.3. Identify the scope of the review process to be performed;
 - 50.1.4. Identify all assumptions and limiting conditions in the valuation review;
 - 50.1.5. Develop an opinion as to the completeness of the report under review within the scope of work applicable to the assignment;
 - 50.1.6. Develop an opinion as to the apparent adequacy and relevance of the data and any adjustments;
 - 50.1.7. Develop an opinion as to the appropriateness of the methods and techniques used and develop the reason for agreement or any disagreement with the report under review; and
 - 50.1.8. Develop an opinion as to whether the analyses, opinions, and conclusions in the work under review are appropriate, reasonable, and supportable.
- 50.2. In reporting the result of a valuation review, the Review Valuer shall:
- 50.2.1. State the identity of the client and intended users, the intended use of the assignment results, and the purpose of the assignment;
 - 50.2.2. State the information that must be identified in accordance with paragraphs 50.1.1-50.1.4 above;
 - 50.2.3. State the nature, extent, and detail of the review process undertaken;
 - 50.2.4. State the opinions, reasons and conclusions required in paragraphs 50.1.5-50.1.8 above;
 - 50.2.5. Indicate whether all known pertinent information is included; and
 - 50.2.6. Include a signed Compliance Statement in the review report.

- 50.3. The Review Valuer shall not consider events affecting the property or market that occurred subsequent to a valuation, but only information that was readily available in the market at the time of the valuation.
- 50.4. Reasons for agreement or disagreement with the conclusions of a valuation report should be fully explained by the Review Valuer.
 - 50.4.1. Where the Review Valuer agrees with the conclusions of a valuation report, reasons for such agreement should be fully explained and disclosed.
 - 50.4.2. Where the Review Valuer does not agree with the conclusions of a valuation report, the reasons for such disagreement should be fully explained and disclosed.
 - 50.4.3. Where the Review Valuer is not in possession of all the facts and information on which the Valuer relied, the Review Valuer must disclose the limitations of his or her conclusions.
- 50.5. Where the scope of the work undertaken is sufficient to constitute a new valuation, such valuation must conform to the requirement of the International Valuation Standards and IVSC Code of Conduct.

PVS GN 400 Valuation of Properties in the Extractive Industries⁴

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10. Introduction

- 10.1. The purpose of this Guidance Note (GN) is to provide clarification and guidance on the valuation of assets or property interests (rights) held by entities involved in the Extractive Industries. It distinguishes among the various property interests that must be recognized, and discusses concepts that should be understood by financial reporting and regulatory authorities, courts, financiers, investors, participants in natural resource transactions, and other users of valuation services for property involved in the Extractive Industries.
- 10.2. Reliable valuations of Extractive Industries assets, including interests (rights) in natural resource properties, are essential to ensure the availability of capital necessary to support the continuity of the Extractive Industries component of the world's economic base, to promote the productive use of Mineral and Petroleum natural resources, and to maintain the confidence of capital markets.
- 10.3. Extractive Industries comprise the Minerals Industry and Petroleum Industry, but do not include activities focused on the extraction of water from the earth.
- 10.4. The Minerals and Petroleum Industries are characterized by the extraction from the earth of natural resources, which may pass through a series of ownership, processing and measurement stages. It is important to Valuers and the users of valuation services that distinctions are made among real property, personal property, and business interests involved in these stages. Financial reporting requires the recognition of various asset classifications into which these interests may fall. Additionally, clear and precise understanding of these distinctions is necessary for valuations to be performed and used in the public interest, regardless of the application.
- 10.5. Valuations in the Extractive Industries often must rely heavily on information provided by (a) Technical Expert(s) or other accredited specialist(s) specific to the industry.
- 10.6. A typical characteristic of the Extractive Industries that sets them apart from other industries or economic sectors is the depletion or wasting of natural resources, which can be replaced in their original state by natural

⁴Formerly Guidance Notes GN 14 of PVS 1st Edition 2009

actions following extraction only in special cases. Special cases of natural replacement may occur for water transported minerals and geothermal fluid. The means of production is extraction from the earth of natural resources that form part of the Real Estate.

- 10.6.1. The ultimate quantity and quality of material of economic interest that might be extracted from an Extractive Industry natural resource property is often not known at the Effective Date of Valuation.
- 10.7. Examples of depleting or wasting natural resources include, but are not limited to:
 - 10.7.1. Metallic Mineral deposits containing metals such as copper, aluminum, gold, iron, manganese, nickel, cobalt, zinc, lead, silver, tin, tungsten, uranium, and platinum group metals;
 - 10.7.2. Non-metallic Mineral deposits such as coal, potash, phosphates, sulphur, magnesium, limestone, salt, mineral sands, diamonds and other gemstones;
 - 10.7.3. Construction materials such as sand, gravel, crushed stone, and dimension stone;
 - 10.7.4. Petroleum deposits including oil, natural gas, natural gas liquids, other gases, heavy oil, and oil sands.
- 10.8. There are contrasts between the production and transportation phases of the Minerals and Petroleum Industries that must be understood:
 - 10.8.1. Items 10.7.1, 10.7.2 and 10.7.3 above include products of the Mineral Industry, which extracts valuable mineralization, generally by mining in a surface mine (open pit, open-cast, open-cut, or strip mine; a quarry used to produce construction material is also considered a surface mine), or an underground mine. Some extraction is undertaken through wells, for example, sulphur extraction, and in situ leaching (solution mining) of various salts and uranium minerals. Some extraction is also done by dredging the floors of bodies of water, such as for gravel, mineral sands, diamonds, and alluvial gold. Extraction of mineral products from water, such as halite (common salt) and magnesium, is also part of the Mineral Industry.
 - 10.8.2. The Minerals Industry generally has a planned extraction phase, though this phase is often extended through Mineral Reserve additions. Once extraction is completed, no more known economically recoverable asset remains in place at that time.
 - 10.8.3. The raw materials cited in para.10.7.4 above are produced by the Petroleum Industry, which extracts valuable product generally through wells drilled into the earth's crust. Some extraction is also undertaken using mining methods, for example, open pit mining of oil sand and oil shale. The extraction of a solid asset is more labor intensive than the extraction of fluid asset. A single person may operate oil and gas extraction by pumps or valves, with the occasional need for well maintenance or well work-over crews.
 - 10.8.4. The Petroleum Industry frequently has more than one economical extraction phase for crude oil. At the conclusion of the initial (primary) extraction phase, much of the initial Petroleum Reserve of crude oil may remain. Secondary and/or enhanced recovery methods are often applied to recover more oil and natural gas. Generally, a large percent of the initial oil in place remains in

place at the conclusion of production operations.

- 10.8.5. Another significant difference between the Minerals and Petroleum Industries relates to land surface requirements for processing plant and infrastructure. Relatively little surface area is required for oil or gas well operations. A mining operation often requires a larger land area for stockpiles and disposal of waste material, as well as an open pit if applicable.
- 10.8.6. Crude oil, natural gas, and refined Petroleum products are more often than not transported to market or port by pipeline. In contrast, a mined product is generally transported to market or port by rail or truck, resulting in differing start-up costs and environmental impacts.
- 10.9. The Minerals and Petroleum Industries are both major industries throughout the world. Their products are essential in all modern economies by provision of raw and refined materials for other downstream industries, such as energy generation, construction, manufacturing, transport and communications.
- 10.10. Exploration of Minerals and Petroleum properties is a high-risk activity. Considerable work and study must be undertaken to determine the technical and economic viability of production. The larger majority of Mineral and Petroleum properties do not reach the production stage.
- 10.11. The projected net earnings derived or potentially derived from an Extractive Industry natural resource property is its main source of value. The net earnings may vary from year to year, depending on the type of natural resource commodity, the cyclical nature of the commodity markets and prices, and variations in production rate and costs.
- 10.12. Mineral and Petroleum natural resource properties are valued primarily based on the presence of Mineral or Petroleum Reserves, and Mineral or Petroleum Resources, or the potential for discovery of Resources. The quantity and quality of such Reserves/Resources may vary over time due to changing economic and technical advances, as well as exploration success. Nevertheless, they are ultimately finite and will deplete over time.
- 10.13. The fixed assets and specialized plant, machinery and equipment (see Glossary for definitions) used in the extraction and processing of raw products of the Extractive Industries, may retain relatively little or no value when separated from production at the site.
- 10.14. Exploration Properties have asset value derived from their potential for the existence and discovery of economically viable Mineral or Petroleum deposits contained within. Exploration Property interests are bought and sold in the market. Many of these transactions involve partial interest arrangements, such as farm-in, option or joint venture arrangements.
 - 10.14.1. The value of an Exploration Property is largely dependent upon surface and subsurface geological and related information, and its interpretation. Little may be known about the characteristics of a deposit that may be contained within the property until the deposit is discovered and explored.
 - 10.14.2. Extractive Industries deposits are often located in remote areas and are generally substantially or completely buried below the land surface, and sometimes below the floor of bodies of water or under the sea.

- 10.15. The residual value of the real property interest, plant, machinery and equipment as well as environmental reclamation requirements (as liabilities and property improvements), are pertinent factors in the valuation process for Extractive Industries properties.

20. Scope

- 20.1. This Guidance Note provides specific guidance for the valuation of assets and interests of the Extractive Industries. It provides supplemental guidance for application of the IVS General Standards and IVS Asset Standards.

IVS 200 Business and Business Interests;
 IVS 210 Intangible Assets;
 IVS 300 Plant and Equipment;
 IVS 400 Real Property Interests; and
 IVS 500 Financial Instruments.

- 20.2. The ownership of, or rights to, an industrial water supply and water storage system, can form an important component in the valuation of Properties in the Extractive Industries. Water rights may attach to land or may be obtained elsewhere. Adequate rights and facilities for transportation and storage of off-site water may be required for a reliable water supply. Valuation of the contribution of such rights poses special problems that must be addressed by the Valuer. However, this GN does not provide specific guidance for valuation of water ownership, rights, transportation and storage.
- 20.3. Where mark-to-market financial reporting procedures apply or are contemplated. Valuers should observe the provisions of *IVS 103 Reporting*, in conjunction with this GN. In some countries, securities exchanges and administrations may have specific reporting requirements for the Minerals and Petroleum Industries that override IVSs provisions.
- 20.4. While providing supplemental guidance for the conduct and reporting of valuations of Extractive Industries property and interests in accordance with para. 20.1 above, the provisions of this GN do not replace provisions elsewhere in the current edition of the International Valuation Standards.

30. Definitions

- 30.1. *Extractive Industries.* Those industries involved in the finding, extracting and associated processing of natural resources located on, in or near the earth's crust. They are composed of the Minerals Industry and the Petroleum Industry. They do not include the industry sector focused on extraction of water from the earth, but they do include extraction of geothermal fluid for its energy content.
- 30.2. *Exploration Property or Area.* A Mineral or Petroleum real property interest that is being actively explored for Mineral deposits or Petroleum accumulations, but for which economic viability has not been demonstrated.
- 30.3. *Feasibility Study in the Extractive Industries.* A comprehensive study of a Mineral deposit or Petroleum accumulation, in which all geological, engineering, operating, economic, marketing, environmental, regulatory and other relevant factors are considered in sufficient detail. The study could reasonably serve as the basis for a final decision by a proponent or financial institution to proceed with, or finance, the development of the prospective property for Mineral or Petroleum production. See also *Prefeasibility Study*.

30.4. *Mineral*. Any naturally occurring material useful to, and/or having a value placed on it by humankind, and found in or on the earth's crust. For the purposes of this GN, Minerals include metallic or non-metallic minerals, industrial minerals, aggregates, precious stones and fuel minerals; but Minerals do not include Petroleum, which is defined separately. *Minerals* as defined in RA 7942, Philippine Mining Act of 1995, Section 3: paragraph (aa): —refers to all naturally occurring inorganic substance in solid, gas, liquid, or any intermediate state excluding energy materials such as coal, petroleum, natural gas, radioactive materials, and geothermal energy.

30.5. *Mineral Reserve*. As defined by the Combined [Mineral] Reserves International Reporting Standard Committee (CRIRSCO): —the economically mineable part of a Measured and/or Indicated Mineral Resource. It includes diluting materials and allowances for losses, which may occur when the material is mined. Appropriate assessments that may include Feasibility Studies, have been carried out, and include consideration of, and modification by, realistically assumed mining, metallurgical, economic, marketing, legal, environmental, social and governmental factors. These assessments demonstrate at the time of reporting that extraction is justified. Mineral Reserves are subdivided in order of increasing confidence into Probable Mineral Reserves and Proved Mineral Reserves.

The United Nations Framework Classification (UNFC) similarly defines a Mineral Reserve and its subdivisions, applying the UNFC coding system. Entities electing to adopt the UNFC or other definitions of Mineral Reserve for public financial reporting purposes must reconcile the Mineral Reserves to the CRIRSCO Proved and Probable Mineral Reserve categories for valuation purposes.

30.6. *Mineral Resources*. As defined by CRIRSCO: —a concentration of occurrence of material of intrinsic economic interest in or on the earth's crust (a deposit) in such form and quantity that there are reasonable prospects for eventual economic extraction. The location, quantity, grade, geological characteristics and continuity of a Mineral Resource are known, estimated or interpreted from specific geological evidence and knowledge. Mineral Resources are subdivided, in order of increasing geological confidence, into Inferred, Indicated and Measured categories. Portions of a deposit that do not have reasonable prospects for eventual economic extraction must not be included in a Mineral Resource.

The United Nations Framework Classification (UNFC) similarly defines a Mineral Resource and its subdivision, applying the UNFC coding system. For the purposes of this GN, mineralization classified into the UNFC's G4 (Reconnaissance StudyII) category, is excluded from a Mineral Resource. Entities electing to adopt the UNFC or other definitions of Mineral Resources for public financial reporting purposes must reconcile the Mineral Resources to the CRIRSCO Inferred, Indicated and Measured Mineral Resource categories for valuation purposes.

30.7. *Minerals Industry*. Entities involved in exploration for Minerals, and the mining, processing and marketing of Minerals. This GN is not designed to cover assets downstream from the metals refineries or minerals processing plants, such as assets involved in the distribution of refined metals to metals fabricators, or mineral products to retailers or the final market.

- 30.8. *Petroleum*. Any naturally occurring hydrocarbon, whether in a gaseous, liquid or solid state. Raw Petroleum products are primarily crude oil and natural gas.
- 30.9. *Petroleum Industry*. Entities involved in exploration for Petroleum, and the extraction processing, refining and marketing of crude Petroleum and associated gases. This GN is not designed to cover assets downstream from the petroleum refineries and natural gas processing plants, such as assets involved in the distribution of refined petroleum products to retailers.
- 30.10. *Petroleum Reserves*. As defined by the Society of Petroleum Engineers (SPE) and the World Petroleum Congress (WPC): "those quantities of Petroleum, which are anticipated to be commercially recovered from known accumulations from a given date forward. All (Petroleum) Reserve estimates involve some degree of uncertainty. The uncertainty depends chiefly on the amount of reliable geologic and engineering data available at the time of the estimate and the interpretation of these data. The relative degree of uncertainty may be conveyed by placing reserves into one of two principal classifications either Proved or Unproved. Unproved Reserves are less certain to be recovered than Proved Reserves and may be further sub-classified as Probable and Possible Reserves to denote progressively increasing uncertainty in their recoverability." Proved Reserves can be categorized as Developed or Undeveloped.

The United Nations Framework Classification (UNFC) similarly defines Petroleum Reserves and their subdivisions, applying the UNFC coding system.

- 30.11. *Petroleum Resources*. For the purpose of this GN, petroleum resources comprise only Petroleum Reserves and Contingent Resources. Contingent Resources as defined by the Society of Petroleum Engineers (SPE)/World Petroleum Congress (WPC), in conjunction with the American Association of Petroleum Geologists (AAPG), are —those quantities of petroleum, which are estimated on a given date, to be potentially recoverable from known accumulations, but which are not currently considered to be commercially recoverable.

The United Nations Framework Classification (UNFC) similarly defines Petroleum Reserves and their subdivisions, applying the UNFC coding system. For the purpose of this GN, petroleum accumulations classified into the UNFC's G4 ("Potential Geological Conditions") category are excluded from Petroleum Resources.

- 30.12. *Prefeasibility Study in the Extractive Industries*. A study of a Mineral or Petroleum deposit, in which all geological, engineering, operating, economic, environmental and other relevant factors, are considered in sufficient detail to serve as the reasonable basis for a decision to proceed to a Feasibility Study.
- 30.13. *Royalty or "Royalty Interest" in the Extractive Industries*. The landowner's or lessor's shares of production, in money or product, free of charge for expenses of production. An *"Overriding Royalty"* is a share of mineral or petroleum produced, free of the expense of production, paid to someone other than the lessor, over and above any lessor's Royalty.

- 30.14. *Technical Expert in the Extractive Industries* (called *Technical Expert* elsewhere in this GN). A person, who is responsible for all or part of the *Technical Assessment* that supports an Extractive Industry Valuation. A Technical Expert must have appropriate experience relevant to the subject matter, and in countries where required by statute or regulation, must be a member or license-holder in good standing of a professional organization that has the authority to sanction members or licensees. An accredited specialist may not take responsibility for all or part of a *Technical Assessment* without also being a Technical Expert.
- 30.15. *Technical Assessment in the Extractive Industries*. A technical document, prepared by (a) Technical Expert(s) that supports the Extractive Industry Valuation and is appended to, or forms part of, a Valuation Report.

40. Relationship to Accounting Standards

- 40.1. Amongst national GAAPs there are currently several approaches to measurement of "Upstream Activities", which include exploration for, discovery of, and acquisition or development of, Mineral or Petroleum Resources up to when the Reserves are first capable of being sold or used. The extreme high-risk element in expenditure on these activities has led to two main approaches to Historical Cost accounting for the Extractive Industries, those being:
- a) all "exploration and evaluation costs" to be expected to be recouped, or the activities have not yet established whether the costs are economically recoverable (i.e., adaptations of a "successful efforts" approach). In applications of this approach, there are variations as to which types of cost are permitted to be capitalized and the treatment of costs prior to the determination of "success", or otherwise, of the exploration and evaluation activities; and
 - b) all expenditures incurred in finding and developing Mineral and Petroleum Reserves to be capitalized and treated as a part of the cost of whatever Reserves may have been found (i.e., a full cost approach).
- 40.2. In December 2004, the International Accounting Standards Board (IASB) released IFRS 6 Exploration for and Evaluation of Mineral Resources. Under the provisions of the Standard, entities are permitted to recognize their exploration and evaluation expenditures as exploration and evaluation assets^{II}. The Standard requires such assets to be measured at cost at initial recognition. After initial recognition, an entity may choose to apply a cost or revaluation model (as outlined in either IAS 16 Property, Plant and Equipment, or IAS 38 Intangible Assets) to measure their "exploration and evaluation assets".
- 40.2.1. The concept of exploration and evaluation assets, and the costs that make up those assets, apply equally to the cost and revaluation models described above.
- 40.3. IFRS 6 states at paragraph 9 that: An entity shall determine a policy for which expenditures are recognized as exploration and evaluation assets and apply the policy consistently. In making this determination, an entity considers the degree to which the expenditure can be associated with finding specific mineral resources. The following are examples of

expenditures that might be included in the initial measurement of exploration and evaluation assets (the list is not exhaustive):

- acquisition of rights to explore;
- topographical, geological, geochemical, and geophysical studies;
- exploratory drilling;
- trenching;
- sampling; and
- activities in relation to evaluating technical feasibility and commercial viability of extracting a mineral resource.

- 40.3.1. IFRS 6, paragraph 5(a), excludes “activities that precede the exploration for an evaluation of mineral resources” from the scope of the Standard.
- 40.3.2. IFRS 6, paragraph 10, additionally states that: “Once the technical feasibility and commercial viability of extracting a mineral resource are demonstrable, expenditures related to the development of that mineral resource shall not be recognized as exploration and evaluation assets. The Framework and IAS 38 Intangible Assets provide guidance on the recognition of assets arising from development (or the development phase of an internal project).”
- 40.3.3. When facts and circumstances stated in paragraph 20 of the Standard suggest that the carrying amount of exploration and evaluation assets may exceed their recoverable amount, entities are required to measure and disclose any resulting impairment loss. The level at which such assets are assessed for impairment may comprise one or more cash-generating-units, which is a higher level of aggregation than that otherwise allowed under IAS 36.
- 40.3.4. In the context of the IFRS 6, a minerals resource includes minerals, oil, natural gas and similar non-regenerative resources (see the Defined Terms in Appendix A to IFRS 6) and also in paragraphs 30.6 and 30.11 above.
- 40.4. IASB Standards that require value determinations to be provided under the provisions of this GN include:
- IAS 36 Impairment of Assets – for determining the recoverable amount of an asset (including assets that incorporate reserves and resources) in order to ascertain whether the asset is impaired. This process requires determination of “fair value less costs to sell” and/or “value in use” as defined in the Standard.
 - IFRS 3 Business Combinations – for determining the carrying amount of assets that were acquired in the acquisition of a business (including assets that incorporate reserves and resources); and
 - IAS 16 Property, Plant and Equipment – for the revaluation (if chosen) of property, plant and equipment that relates to extractive operations.
- 40.5. This Guidance Note recognizes that the Historical Cost of finding and developing Mineral and Petroleum Reserves is usually not indicative of the realizable value of such Reserves once they have become established.

50. Guidance**50.1. Valuation Concepts**

- 50.1.1. The provisions of this GN are designed to assure application of the valuation concepts to Extractive Industries Valuations, in accordance with the valuation fundamentals expressed in the IVS Framework and IVS General Standards.
- 50.1.2. The standard of value is Market Value defined in IVS 104, Section 30. If some other type of value is to be determined in accordance with IVS 104, a clear definition of that value should be provided by the Valuer and highlighted in the Valuation Report as prescribed in IVS 103, and a clear and conspicuous explanation provided.
- 50.1.3. The property type(s) involved in valuation of Minerals and Petroleum Industry property must be correctly identified in order to correctly select the applicable IVSC Standards and GNs. Naturally occurring in situ Minerals and Petroleum are a part of physical land and Real Estate. The ownership of such in situ Minerals and Petroleum, an interest in such natural resources, and the right to explore and extract such natural resources, are Real Property, except where otherwise defined by statute. Minerals and Petroleum are Personal Property during transportation and processing. The operation of a mine, quarry or petroleum well is a business activity, as is the transportation and processing of Minerals and Petroleum. Such business activity is generally conducted by an Extractive Industries business enterprise that owns real property and personal property assets, and the activity contributes to the Going Concern Value of the enterprise.
- 50.1.4. A key aspect of the valuation of an Extractive Industry natural resource property is that the property interests and related rights being valued must be properly identified.
- 50.1.5. A Market Valuation of an Extractive Industry property as Real Property must be based on the Highest and Best Use (HABU) of the property. This requires consideration of non-Minerals or non-Petroleum uses for the property, if such uses are possible. Consideration must also be given to a change in exploration, development or operating strategy, or potential for leasing the property, in order to maximize its economic benefit.
- 50.1.6. In determining the HABU, the Valuer should determine the most probable use that is physically possible, appropriately justified, legally permissible, financially feasible, and which results in the highest value of the property being valued.
- 50.1.7. In conducting a Market Valuation, the three Valuation Approaches -Market Approach, Income Approach and Cost Approach are generally available for consideration under IVS 200 Business and Business Interests and IVS 210 Intangible Assets.
- 50.1.8. Where one or more of the above Valuation Approaches has been applied in preference to others, the reason must be stated.
- 50.1.9. As applied to Mineral and Petroleum natural resource property interests, the appropriate Valuation Methods employed depend upon the stage of exploration or development of the property. For convenience, such Mineral

and Petroleum properties can be categorized as four main types, though the categorization is sometimes the subject of the opinion of a Valuer or Technical Expert.

- Exploration properties;
- Resource properties;
- Development properties; and
- Production properties.

50.1.10. Exploration Properties are defined at para. 30.2.

50.1.11. Resource properties contain a Mineral Resource or Petroleum Resource but have not been demonstrated by a Prefeasibility Study or a Feasibility Study to be economically viable.

50.1.12. Development properties, in general, have been demonstrated by a Feasibility Study to be economically viable but are not yet in production.

50.1.13. Production properties contain a Mineral or Petroleum producing operation active at the time of Valuation.

50.1.14. The different stages of exploration and development carry different levels of risk. The risk pertains to the likelihood of eventual or continued Mineral or Petroleum production. As an Exploration Property is advanced to a Resource property, to a Development property, and to a Production property, more technical information is collected, enabling technical analysis, including Prefeasibility and Feasibility Studies, to be carried out, and thereby reducing the risk factor, as the amount of capital investment at risk rapidly increases.

50.1.15. The results from the Valuation Approaches and Methods employed must be weighed and reconciled into a concluding opinion of value. The reasons for giving a higher weighting to one Approach or Method over another must be stated.

50.2. **Competence and Impartiality**

50.2.1. Valuations prepared under this Guidance Note shall comply with all provisions of the IVSC Code of Conduct.

50.2.2. To develop a Valuation of an Extractive Industry asset or interest, the Valuer must have competence relevant to the subject asset or interest, or retain the services of (an) appropriately skilled Technical Expert(s).

50.2.3. Providing a reliable and accurate valuation typically requires the Valuer to have specialized training, or assistance from (a) Technical Expert(s) or other accredited specialist(s), in geology, Resource and Reserve estimation, engineering, and economic and environmental aspects relevant to the subject natural resource type and geographic setting. The defined term Technical Expert includes "Competent Person", "Independent Valuer", and similar requirements that may apply in some countries, if the intended use of the Valuation Report is related to public financial reporting or other regulatory purpose.

50.2.4. The Valuer is responsible for the decision to rely on a Technical Assessment, data, or opinion provided by other experts or specialists. This includes responsibility for conducting reasonable verification that those persons are appropriately qualified and competent and that their work is credible.

50.3. **Special Considerations of Extractive Industries Valuations**

- 50.3.1. Each Mineral deposit, Petroleum accumulation and Exploration Property is unique. Therefore, direct comparison of Mineral or Petroleum natural resource property transaction is often difficult or inappropriate. However, sales analysis is an important valuation tool. Sales adjustment or ratio analysis can frequently be applied for indirect sales comparison purposes. Sales analysis and other market analysis can often yield market factors such as a market discount rate, a risk factor or uncertainty factor that may be used in the Income Approach.
- 50.3.2. For a Valuation Report to provide an estimate of Market Value, the valuation analysis must be based on market evidence and current expectations and perceptions of market participants for the property valued, and such market evidence must be consistently applied in the Valuer's analysis.
- 50.3.3. The method most commonly used by businesses for investment decision-making within the Extractive Industries is net present value analysis/ discounted cash flow analysis (NPV analysis/DCF analysis). The Valuer is cautioned that this and other methods, such as those based on option theory, will yield other than Market Value estimates of Investment Value or Value in Use, unless great care is taken to assure that a Market Value estimate is obtained. For the Valuer to report a Market Value estimate resulting from such an analysis, all inputs and assumptions must reflect available market-based evidence and current expectations and perceptions of market participants, in accordance with GN 9. Any departure from the requirements and analysis protocol of GN 9 must be specified.
- 50.3.4. The Market Value of Extractive Industries' natural resource properties and businesses are usually more or less than the value of the sum of their parts or component values. For example, the Market Value of a real estate tract owned in fee simple, that contains a Mineral deposit, is rarely the sum of the independent values of the Minerals, land surface, and plant, machinery and equipment. Similar situations may often occur in the Petroleum Industry.
- 50.3.5. For a producing Mineral or Petroleum Industry natural resource property, there may be separate ownership rights over component parts utilized by the enterprise, such as the Reserve, Royalties, and plant, machinery and equipment. It is important for a Valuer of the enterprise to correctly recognize these. There may also be a requirement to provide valuations of the separate ownership interests.
- 50.3.6. Material data relied on in developing the value estimate should be verified for accuracy whenever reasonable to do so. This may include selective review of drill hole information and samples and related analytical data for a subject natural resource property, and confirmation of published information pertaining to transactions of similar properties.
- 50.3.7. If there is more than one estimate of the quantity and quality of Resources and Reserves for a subject natural resource property, the Valuer shall

decide which estimates it is appropriate to disclose and discuss, and which estimate to use as the basis in the Valuation process, and shall state the reasons. A critique of alternative estimates may be submitted with the Valuation Report.

50.3.8. The Valuer shall take account of, and make reference to other matters that have a material impact on the Valuation. Dependent on the property type and rights being valued, these may include:

- the status of tenements, rights and other interests;
- all Mineral or Petroleum deposits within the boundaries of the tenements or rights;
- access to markets and the quality and quantity of product that can be sold;
- services and infrastructures, and any toll arrangements, fees or liabilities related thereto;
- environmental assessments and rehabilitation liabilities;
- any Native Title or Ancestral Domain Title aspects;
- capital and operating costs;
- timing and completion of capital projects;
- residual value estimates;
- material agreements and statutory/ legal requirements;
- taxation and Royalties;
- liabilities and financial exposures;
- site rehabilitation, reclamation and closure costs; and
- any other aspect that has a material bearing on the Valuation.

50.4 **Disclosure in Extractive Industries Valuation Reports**

50.4.1. The Valuation Report shall properly identify the property type(s), specific property interest(s) and related rights being valued as specified in IVS 103.

50.4.2. The Valuation Report shall disclose the name, professional qualifications and relevant industry experience of the Valuer, and other Technical Expert(s) who's Technical Assessment has been relied upon to support the Valuation.

50.4.3. The Valuation Report shall be supported by disclosure of relevant Extractive Industries Codes, Standards or Rules of Practice applicable to the Valuation and supporting Technical Assessment. All estimates of a Mineral or Petroleum Resource or Reserve disclosed in the valuation report or supporting Technical Assessment shall abide by the definitions provided in Section 3 above, and the classification systems referenced in those definitions, unless jurisdictional or other reasonable cause is disclosed.

- 50.4.4. Maps, geological sections, diagrams, and photographs shall be included in the Valuation Report, if appropriate and possible, to aid the communication of information. Relevant technical information supporting the Valuation of a subject natural resource property (ies), including estimates of Resources and Reserves being valued, shall be disclosed and discussed in a Technical Assessment.
- 50.4.5. The Valuation Report shall disclose whether or not the entity employing/retaining the Valuer, or the owner of the subject asset or its operating management, has provided the Valuer with a statement that all available data and information requested by the Valuer or otherwise relevant to the Valuation have been supplied to the Valuer.

PVS GN 500 Valuation of Historic Property⁵

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10. Introduction

- 10.1. Historic properties are assets that embody a cultural, historic, and/ or architectural heritage.
- 10.2. Historic properties may have legal or statutory protection because of their cultural and economic importance. Many governments have enacted measures to safeguard specific historic properties or to protect whole areas of special architectural or historic interest.
- 10.3. Private organizations play a significant role in promoting historic preservation and education about historic properties. In some cases, historic properties also bring economic benefits through increased tourism in the communities where they are located.
- 10.4. The valuation of historic properties requires consideration of a variety of factors that are associated with the importance of these properties, including the legal and statutory protections to which they are subject; the various restraints upon their use, alteration and disposal; and possible financial grants or rate/tax exemption to the owners of such properties in some jurisdictions.
- 10.5. The costs to restore and maintain historic properties may be considerable and these costs, in turn, affect the value of properties.
- 10.6. The assessment of the highest and best use of historic properties will depend on the specific restrictions that apply to them. In some situations, the use of historic properties is limited to restoration for non-commercial use while in others, adaptation to some other use, including commercial use, is permissible.

20. Scope

- 20.1. This Guidance Note covers real property that has cultural and historic significance, specifically focusing upon historic buildings and sites. It does not address either natural heritage assets or heritage assets that are personal property, e.g., works of art.
- 20.2. This Guidance Note applies to the valuation of historic properties in both public and private sectors. Historic properties owned either by public sector or private sector entities are distinguished by similar cultural characteristics, and share common problems in regard to restoration and/or adaptation. But the specific legal and statutory protections that apply to private and public historic properties and other restraints upon their use, alteration and disposal may differ.

⁵Formerly Guidance Notes GN 15 of PVS 1st Edition 2009

- 20.3. Historic property is a broad term, encompassing many property types. Some historic properties have been restored to their original condition; some have been partially restored (e.g., the building façade); and others have not been restored. Historic property also includes properties partially adapted to current standards (e.g., the interior space), and properties that have been extensively modernized. All historic properties (buildings and/or sites) have some degree of historic character. This Guidance Note addresses properties having historic character to some degree or other.

30. Definitions

International Valuation Standards

- 30.1. Historic House Owner Associations. Not-for-profit membership associations that promote the preservation of historic properties and provide their owner-members with advice on matters such as the management, repair, maintenance, taxation and insurance of historic properties.
- 30.2. Historic Property. Real property publicly recognized or officially designated by a government-chartered body as having cultural or historic importance because of its association with an historic event or period, with an architectural style, or with the nation's heritage. Four characteristics are commonly associated with historic properties: 1) their historic, architectural and/or cultural importance; 2) the statutory or legal protection to which they may be subject; 3) restraints and limitations placed upon their use, alteration and disposal; and 4) the frequent obligation in some jurisdictions that they be accessible to the public. The terms, historic property and heritage asset, often overlap but are not in all cases equivalent. See also Heritage Asset, Publicly Designated Historic Properties.
- 30.3. Listing of (Heritage) Buildings or Historic Properties Register. A recording of officially designated historic properties. Not all historic properties are necessarily listed in registers. Many properties publicly recognized as having cultural and historic importance also qualify as historic properties.
- 30.4. Preservation Incentives. Incentives to the owners of historic properties, primarily fiscal in nature, to promote the restoration and maintenance of such properties. Examples may include exemptions on inheritance taxes for conservation properties gifted to heritage trusts, government subsidies, and exemptions in municipal rates/taxes, investment tax credits to owners of historic properties, transferable development rights and deductions taken on conservation easements donated to nonprofit organizations.
- 30.5. Publicly Designated Historic Properties. Those properties, the historic status of which is officially recognized by government-chartered bodies to identify historic properties and to promote historic preservation. Such bodies may be established by national/federal, state/provincial, or county/municipal governments. Local non-governmental historical societies may also designate historic properties and maintain private historic property lists or registers that confer many of the same benefits while remaining exempt from government restrictions. See also listing of (Heritage) Buildings or Historic Register.

International Public Sector Accounting Standards

- 30.5. Heritage Asset. An asset having some cultural, environmental, or historical significance. Heritage assets may include historical buildings and monuments,

archeological sites, conservation areas and natural reserves, and works of art. Heritage assets often display the following characteristics (although these characteristics are not necessarily limited to heritage assets):

- a) Their economic benefit in cultural, educational and historic terms is unlikely to be fully reflected in a financial value based purely on market price;
- b) Legal and/or statutory obligations may impose prohibitions or severe restrictions on disposal by sale;
- c) They are often irreplaceable and their economic benefits may increase over time even if their physical condition deteriorates; and
- d) It may be difficult to estimate their useful lives, which in some cases could be hundreds of years.

The above definition is consistent with the description of heritage and conservation assets in IPSAS 17.9 (n.b., IPSAS 17 does not currently contain a formal definition of heritage assets).

UNESCO Glossary of World Heritage Terms

30.7. Cultural Heritage. Three groups of assets are recognized:

1. Monuments: architectural works, works of monumental sculpture and painting, elements or structures of an archaeological nature, inscriptions, cave dwellings and combinations of features, which are of outstanding universal value from the point of view of history, art or science;
2. Groups of buildings: groups of separate or connected buildings which, because of their architecture, their homogeneity or their place in the landscape, are of outstanding universal value from the point of view of history, art or science; and
3. Sites: works of man or the combined works of nature and man, and areas including archaeological sites, which are of outstanding universal value from the historical, aesthetic, ethnological or anthropological point of view.

World Heritage Convention, Article I, UNESCO, 1972

30.8. Cultural Property. Property inscribed in the World Heritage List after having met at least one of the cultural heritage criteria and the test of authenticity. (World Heritage Convention, Article II, UNESCO, 1972)

40. Relationship to Accounting Standards

40.1. International Public Sector Accounting Standard 17 (Property, Plant and Equipment) does not require an entity to recognize heritage assets that would otherwise meet the definition of, and recognition criteria for, property, plant and equipment. If an entity does recognize heritage assets, it must apply the disclosure requirements of IPSAS 17 and may, but is not required to, apply the measurement requirements of IPSAS 17 (paras. 2 and 8). The IPSAS Discussion Paper on Heritage Assets (January 2006) requires the valuation of heritage assets where this is practicable. However, where valuation is impracticable, an entity would be required to make relevant disclosures, including reasons why valuation is not practicable.

- 40.2. This Guidance Note considers historic structures as a category of heritage and conservation assets.

50. Guidance

- 50.1. The valuation of historic properties involves special considerations dealing with the nature of older construction methods and materials, the current efficiency and performance of such properties in terms of modern equivalent assets, the appropriateness of methods used to repair, restore, refurbish, or rehabilitate the properties, and the character and extent of legal and statutory protections affecting the properties.
- 50.2. The cost and income capitalization approaches may be employed in the valuation of historic properties. The selection of the approach or approaches to be used depends on the availability of data required to apply that or those approaches
- 50.2.1. In applying the sales comparison approach, the historic nature of the property may change the order of priority normally given to attributes of comparable properties. It is especially important that the Valuer find comparable properties with historic features similar to those of the subject. Criteria for the selection of comparable properties include similarity in location (i.e., in zoning, permissible use, legal protection, and concentration of historic properties), architectural style, property size, and the specific cultural or historic associations of the subject property. Variety of adjustments may have to be made to the comparable sales. These involve differences in location, costs of restoration or rehabilitation, or specific encumbrances. Adjustments are made in the following situations:
- 50.2.1.1. When costs must be incurred to restore or rehabilitate the subject property, but not the comparable sales; and
- 50.2.1.2. Where the specific encumbrances upon the subject, e.g., restrictive covenants or preservation easements, differ from those upon the comparable properties.
- 50.2.2. Historic properties having a commercial use are often valued by means of the income capitalization approach. Where the distinctive architecture and ambiance of an historic property contribute to its drawing power under an income-producing use and that income-producing use is considered to be the highest and best use of the historic property, the valuation will address the following:
- 50.2.2.1. All work proposed to restore, adapt or rehabilitate the historic property must meet existing zoning requirements and covenant obligations;
- 50.2.2.2. Where listed building consents or a zoning variance and/or building code exemptions are required, the projected timeframe to obtain such authorization needs to be taken into consideration; and
- 50.2.2.3. The income capitalization approach should consider the cost effectiveness of an income-producing historic property in terms of the rental and/or commercial income the property is able to generate. In particular, it should address the additional costs involved in maintaining the property, especially those costs incurred due to functional obsolescence or reserves set aside for any required sinking fund.

- 50.2.3. When applying the cost approach to historic property, the Valuer needs to consider whether the historic features of a building would be of intrinsic value in the market for that property. Some historic buildings will be of value simply because of their symbolic status, for example a famous art gallery where the building is as, or more, important than the function it fulfills. In other words, the service potential of such a building is inseparable from its historic features. The modern equivalent of such properties would need to reflect either the cost of reproducing a replica, or if this is not possible because the original materials or techniques are no longer available, the cost of the modern building with a similarly distinctive and high specification.

In many cases the historic features will add no value, or be viewed as an encumbrance by a purchaser in the market, for example a hospital operating in an historic building. In such cases the modern equivalent would reflect the cost of a new building constructed to a conventional modern specification.

In all cases the adjustments for physical deterioration and functional obsolescence will need to reflect factors such as the higher cost of maintenance associated with historic property and the loss of flexibility for adapting the building to the changing needs of an occupier.

The land or site, upon which an historic property stands, may be subject to constraints upon its use. In turn, any such constraints will affect land and overall property value.

- 50.3. Historic, or heritage assets, for which there is no reliable or relevant sales evidence, which have no potential for generating income, and which would or could not be replaced may be incapable of reliable valuation. An example could be a partially ruined building with no income generating potential. Where liable assessment of value is not possible, the Valuer must disclose the reasons for this conclusion in the report.
- 50.4. Legal measures to safeguard historic properties may limit or restrict the use, intensity of use or alteration of an historic property. Examples include restrictive covenants that run with the land regardless of the owner; preservation easements that prohibit certain physical changes, usually based on the condition of the property at the time the easement was acquired or immediately after proposed restoration of the property; and conservation easements that limit the future use of a property so as to protect open space, natural features, or wildlife habitat.
- 50.4.1. Restrictive covenants and preservation easements, whether existing or proposed, may have a major influence on the highest and best use of an historic property, and thereby have a significant effect on property value. Preservation easements can be donated, purchased, or obtained by compulsory acquisition/ eminent domain.
- 50.5. The valuation conclusion shall be reported in accordance with IVS 103 Reporting.

PVS GN 600 Consideration of Hazardous and Toxic Substances in Valuation⁶

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10. Introduction

- 10.1. The objective of this Guidance Note (GN) is to assist Valuers in preparing valuations when specific hazardous or toxic substances may influence property values.
- 10.2. Hazardous and toxic substances are included among a number of possible environmental factors that, when appropriate, are specifically considered by Valuers. This GN is limited to consideration of hazardous and toxic substances because other environmental factors that may be encountered in valuations typically have less involvement with scientific and associated technical issues, including related law.
- 10.3. This GN addresses general concepts, principles, and considerations that guide Valuers in preparing valuations when hazardous or toxic materials that may influence property values are present. It also discusses concepts that must be understood by accountants, regulatory authorities, and other users of valuation services.
- 10.4. Valuers rarely have special qualifications in legal, scientific, or other technical areas that involve evaluating risks associated with hazardous or toxic substances. When considering the market effects of such risks in property valuations, Valuers commonly rely upon other experts' advice. As specified in the IVS Framework Section 50 Competence significant reliance upon other experts' advice must be disclosed and explained in the context of the property addressed in the Valuation Report.
- 10.5. Fundamental to the application of this GN are the Valuer's adherence to market-based valuations, objectivity, and full disclosure of relevant matters. Similarly, Valuers are obliged to write Valuation Reports that may be reasonably understood by clients and others. The obligation for clarity and full disclosure is particularly important when scientific, technical, and legal issues are involved.
- 10.6. This GN also provides for proper treatment and disclosure of hazardous and toxic substance issues when valuing specialized properties and in other situations, which preclude the application of Market Value concepts.

⁶Formerly Guidance Notes GN 7 of PVS 1st Edition 2009

20. Scope

- 20.1. This GN applies to all valuations of property including plant, machinery and equipment. Special provisions of the GN should be observed when hazardous or toxic substances that may influence the property's Market Value or other defined value are known or reasonably believed to be present.

30. Definitions

- 30.1. Hazardous or toxic substances within the context of this GN involve specific materials that, by their presence or proximity, may have adverse effect on property value because of their potential to cause harm to life-forms. Such materials may be incorporated into improvements to or on the site, or they may be found in or on the land. They may also be offsite, but nearby. In some instances, they may be airborne.
- 30.1.1. In a more general use beyond this GN, environmental factors may be characterized as influences external to the property being valued which may have positive effect, negative effect, or no effect at all on the property's value. Hazardous or toxic substances may be found either on or off the site of the property valued.
- 30.2. Hazardous substance within the context of a valuation is any material within, around, or near the property being valued that has sufficient form, quantity, and bio-availability to create a negative impact on the property's Market Value.
- 30.3. Toxic describes the status of a material, whether gas, liquid, or solid, that in its form, quantity, and location at the date of valuation has capacity to cause harm to life-forms. Toxicity refers to the degree or extent of such capacity.

40. Relationship to Accounting Standards

- 40.1. While the expressions Market Value and Fair Value may not always be synonymous (see IVS 104 Bases of Value, each type of value reflects market behavior under conditions contained within the respective definitions. To the extent that property values reported under either type of value may be affected by hazardous or toxic substances, proper disclosure and the application of proper valuation procedures to the circumstances are necessary in making and reporting valuations.
- 40.2. This GN is applicable to all circumstances involving public disclosure of property values, whether reported individually or in the aggregate, when hazardous or toxic substances may have adverse effect on such values. In addition to the possible effect of such considerations on the properties valued, it is possible that there are other accompanying issues such as curative or restoration costs, maintenance or monitoring costs, third-party or regulatory liabilities, and the like. Thus, proper disclosure and handling are essential in valuations used for preparation of financial statements and related accounts.
- 40.3. In the ordinary course of conducting an asset valuation, the Valuer will be instructed by the Directors of the entity as provided in IVS 101 Scope of Work. Any special instructions to the Valuer concerning the handling of hazardous or toxic substance issues that may have negative impact upon property value are, under IVS 101 Scope of Work, important disclosures to be discussed by the Valuer in the Valuation Report. Such disclosures shall

be accompanied by the Valuer's explanation of how the issues are handled in the Valuation Process; any assumptions that are made; and the effect, if any, such considerations have upon the value reported.

- 40.4. Although the value effects of hazardous or toxic substances are derived from the market in a Market Value assignment, such effects may not be as readily discerned when valuing property for which a Depreciated Replacement Cost method is appropriate. To comply with IVS 101 Scope of Work and IVS 105 Valuation Approaches and Methods, when applying the DRC method, Valuers should apply the principles of this GN to the extent possible and should fully disclose the extent of their analysis and the basis for their conclusions.
- 40.5. The accounting definition of impairment loss is the amount by which the carrying amount of an asset or a cash-generating unit exceeds its recoverable amount (IAS 36, para. 6). The negative impact of hazardous or toxic materials that are present in a property may contribute to its impairment. The impairment loss incurred by a property where such substances are present may include the adverse effect of those substances upon property value. (See para. 50.4 below.)

50. Guidance

- 50.1. In dealing with a client or prospective client in matters pertaining to the valuation of property when known or reasonably discoverable environmental conditions that may have adverse influence on the property values are present, the Valuer should disclose to the client the extent of his or her knowledge, experience, and competency to deal with the situation.
- 50.1.1. If the environmental factors are known or are suspected to exist at the time the Valuer and prospective client are discussing the potential engagement, the Valuer should satisfy himself or herself that the client understands the Valuer's competency and disclosure obligations and that undertaking the engagement will in no way compromise these obligations.
- 50.1.2. If the environmental factors are discovered after commencing the engagement, the Valuer should make known to the client the knowledge, experience, and competency disclosures specified by this Guidance, and should then comply with all other IVSs disclosure requirements.
- 50.2. Recognizing that many environmental situations will require advice on physical, legal, scientific, and other technical issues, if the engagement is otherwise acceptable to both the client and the Valuer, the Valuer should take the necessary steps to complete the assignment competently. These steps may include appropriate personal study; association with another Valuer who has the requisite knowledge, experience, and competency; or obtaining the professional assistance of others who possess the requisite knowledge, experience, and competency.
- 50.3. Disclosure of the existence of any hazardous or toxic substance that may have adverse effect upon a property's value is included among the general reporting requirements of IVS 103 Reporting. Also required is the Valuer's disclosure regarding how that factor has been dealt with in the engagement. If the engagement calls for valuation of the property as though no such environmental factor existed, full disclosure must be made of the limiting assumption, and reporting must comply with the statements above. (See

paras. 50.1 and 50.2.) The requirements for valuation reports are addressed in the IVS Framework and IVS 103 Reporting.

- 50.4. Where impairment is present in a property, the Valuer should estimate the value of the property as if the impairment had been removed. Where possible, the Valuer should identify the cost of remediation; but if this is not possible, the Valuer should disclose the fact that the property is impaired.
- 50.5. If a property is valued as if unaffected by hazardous substances, and such substances are known or suspected to exist, the Valuation Report should contain a qualification that clearly limits the scope of the valuation, an appropriate statement of purpose, properly qualified conclusions, and a restriction against use of the valuation conclusion without accompanying disclosure of the qualification and its explanation.
- 50.6. The names and nature of expert assistance of others who contributed specific information concerning effects of environmental factors on the property valued should be acknowledged in the Valuation Report.
- 50.7. When there are no known environmental factors that may have adverse effect on property value, the Valuer should, as a matter of routine practice, include within the Valuation Report a contingent and limiting condition or other disclaimer affirming that the valuation was made on the assumption that no such factors were known to exist at the date of valuation, but if such factors did exist they could well have an adverse effect on value.
- 50.8. There is growing worldwide concern about the effects of hazardous and toxic substances upon lives and property. Many potential hazards have been recently identified, and others are likely to be added over time as new and additional discoveries are made and subsequent controls are invoked by governments or are required by the market.
- 50.9. Some hazardous or toxic substances can have material effect on property values. However, as Valuers normally deal with Market Values, it is the market's reaction to these substances that is at issue in Market Value engagements. Over time, substances once believed to have no adverse effect on property value may be determined to have such an effect. Conversely, materials once believed to have substantial effect may be found to have little or no property value effect, or to have such effect only under certain market conditions.
- 50.10. The handling of physical, legal, scientific, and technical issues involved with hazardous or toxic substances is frequently beyond the skill of the Valuer. However, the Valuer's role in consideration of such factors will be facilitated if
 - 50.10.1. Situations involving environmentally sensitive substances are recognized and dealt with in accordance with this Guidance, whether they are encountered prior to undertaking an engagement or subsequent to its commencement;
 - 50.10.2. Proper reliance is placed upon the professional advice and assistance of others when special skills, knowledge, training, and experience are required, and any such assistance is acknowledged and explained in the Valuation Report;
 - 50.10.3. Proper treatment is given to the influence of hazardous or toxic substances in the Valuation Process, or alternatively, if the Valuation Process is to exclude consideration of such matters, proper disclosure is made of any assumptions

made, the purpose of the analysis, and the likely effect of the assumptions on the defined value; and

- 50.10.4. Reasonable effort is made to ensure that reports and the value estimates they contain are not misleading and can be reasonably used only for the purpose for which they are intended.
- 50.11. Valuers are expected to correctly apply those recognized methods and techniques that are necessary to comply with this Guidance. When valuing property subject to some hazardous or toxic substance that adversely influences property value, the Valuer should apply those processes necessary to adequately reflect any such value losses, taking care to neither over- or understate the value effects. In a Market Value engagement, it is the Valuer's responsibility to reflect the market effect of the particular condition or circumstance.
- 50.12. Valuers are cautioned that there can be considerable controversy among legal, scientific, and other technical experts upon whose advice the Valuer may need to rely. Particular differences may be found in the methods experts use to determine the extent of clean-up, maintenance, or monitoring that may be associated with hazardous or toxic substances and the costs required to accomplish such clean-up, maintenance, or monitoring.
- 50.12.1. Engagements may require valuation of the affected property under an assumption that any value effect of the hazardous or toxic substances is excluded from the reported value. Such engagements are acceptable, provided that the resulting valuation is not misleading, that the client is informed of and agrees to this limiting assumption, and that the Valuation Report clearly sets forth the limitation and the reasons therefore.
- 50.12.2. Because of technical issues frequently involved in environmental matters, it is common for Valuers, directly or through the client, to seek the counsel of and rely upon the opinions of those who hold appropriate qualifications to evaluate problems involving hazardous or toxic substances. Valuers should disclose the level and nature of reliance placed upon such opinions.
- 50.13. Valuers and users of valuation services should recognize that the effect of a particular hazardous or toxic substance may vary widely with differences in properties, locations, and markets. Adverse value effects may range from none to those that are more than costs of cure and remediation. The latter may occur, for example, where highest and best use of the property affected is changed by the condition and where marketability or other usefulness of the property is altered. In any case, it is the role of the Valuer to research and reflect the effects of the environmental factor on a particular property in its market.
- 50.14. In the typical valuation engagement, the Valuer will not have the skills to make legal, scientific, or technical findings regarding hazardous or toxic substances, or other environmental factors that may have adverse effect on value. It is important to the credibility and usefulness of the reported value that if any such conditions exist, they be properly considered and reported.
- 50.14.1. When the Valuer knows that a hazardous or toxic substance is present in the property specified in a valuation engagement, the Valuer should follow all requirements of this Guidance. Normally, any technical experts upon whose advice the Valuer may rely will be engaged by the client or by others.

- 50.14.2. When the Valuer has some reason to believe that a potentially adverse hazardous or toxic substance may be present, the Valuer should immediately make the client aware of the concern and request that the client take steps to resolve pertinent questions. By handling this concern on a private, confidential basis, the Valuer keeps information confidential that in itself could affect the property involved.
- 50.15. Guidelines for Valuers' responsibilities to observe, locate, and identify hazardous or toxic substances or circumstances may vary from time to time within and among jurisdictions. In general, determination of the nature, extent and physical effects of environmental conditions is beyond the scope of service of Valuers.
- 50.16. In dealing with hazardous or toxic substances, the Valuer should research all related issues in a confidential manner so as not to raise undue speculation concerning the property.
- 50.17. It is common for individuals unfamiliar with hazardous or toxic substances issues to assume that if there is a physical effect of such substances, there must be an adverse economic reaction. Market experience shows there can be, and frequently are, important differences between general public perceptions and actual market effects of the presence of such substances. The Valuer's role is to avoid such generally held but possibly erroneous assumptions and to carefully consider all significant factors, perform competent market research, and reflect relevant market attitudes towards the situation in Valuation Reports.

PVS GN 700 Valuation of Trade Related Property⁷

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10. Introduction

- 10.1. Trade Related Properties (TRPs) are individual properties, such as hotels, fuel stations, and restaurants that usually change hands in the marketplace while remaining operational. These assets include not only land and buildings, but also fixtures and fittings (furniture, fixtures and equipment) and a business component made up of intangible assets, including transferable goodwill.
- 10.2. This Guidance Note provides direction on the valuation of TRPs as operating assets as well as the allocation of TRP value into its main components. Component values are usually required for depreciation and tax purposes. This Guidance Note should also be read in conjunction with the Guidance Notes cited in para. 20.1 below.
- 10.3. Some concepts involved in the valuation of a business not classed as a TRP (see IVS 200 Business and Business Interests) must be distinguished from those involved in the valuation of Trade Related Property. (Also see paras. 50.6 and 50.7.2 below.)

20. Scope

- 20.1. This Guidance Note focuses on TRP valuation. For further insight into the application of valuation principles, the following IVS Asset Standards should be consulted:
- 20.1.1. IVS 200 Business and Business Interests;
- 20.1.2. IVS 210 Intangible Assets;
- 20.1.3. IVS 300 Plant and Equipment;
- 20.1.4. IVS 400 Real Property Interests;
- 20.1.5. IVS 410 Development Property; and
- 20.1.6. IVS 500 Financial Instruments.

30. Definitions

- 30.1. Capitalization. At a given date the conversion into the equivalent capital value of net income or a series of net receipts, actual or estimated, over a period.

⁷Formerly Guidance Notes GN 12 of PVS 1st Edition 2009

30.2. Discounted Cash Flow. A financial modeling technique based on explicit assumptions regarding the prospective cash flow to a property or business. The most widely used applications of DCF analysis are the Internal Rate of Return (IRR) and Net Present Value (NPV).

30.3. **Goodwill**

30.3.1. Future economic benefits arising from assets that are not capable of being individually identified and separately recognized. (IFRS 3, Appendix A).

30.3.2. Personal Goodwill. The value of profit generated over and above market expectations, which would be extinguished upon sale of the trade related property, together with those financial factors related specifically to the current operator of the business, such as taxation, depreciation policy, borrowing costs and the capital invested in the business.

30.3.3. Transferrable Goodwill. That intangible asset that arises as a result of property-specific name and reputation, customer patronage, location, products, and similar factors, which generate economic benefits. It is inherent to the trade related property, and will transfer to a new owner on sale.

30.4. Reasonably Efficient Operator, or Average Competent Management. A market based concept whereby a potential purchaser, and thus the Valuer, estimates the maintainable level of trade and future profitability that can be achieved by a competent operator of a business conducted on the premises, acting in an efficient manner. The concept involves the trading potential rather than the actual level of trade under the existing ownership so it excludes personal goodwill.

30.5. Trade Related Property. Certain classes of real property, which are designed for a specific type of business and that are normally bought and sold in the market, having regard to their trading potential.

40. **Relationship to Accounting Standards**

40.1. Under International Financial Reporting Standards (IFRSs), like other types of real property, a TRP may be carried on an entity's balance sheet at either cost or at fair value (see IVS 104 Bases of Value). It may be necessary to allocate the value of a TRP between its different components for depreciation purposes.

50. **Guidance**

50.1. This Guidance Note describes that category of property referred to as TRPs and explains how TRPs are valued in accordance with IVS 104 Bases of Value.

50.2. When performing a TRP valuation, the Valuer may also find relevant asset standard in the six asset standards cited in para. 20.1 above. If the valuation is for inclusion in a Financial Statement, the Valuer should refer to IVS 104 Bases of Value Section 90.

50.3. TRPs are considered as individual trading concerns and typically are valued on the basis of their potential Earnings Before Interest, Taxes,

Depreciation and Amortization (EBITDA), as adjusted to reflect the trading of a reasonably efficient operator and often on the basis of either DCF methodology or by use of a capitalization rate applied to the EBITDA.

- 50.4. Valuations of TRPs are usually based on assumptions that there will be a continuation of trading by a Reasonably Efficient Operator, with the benefit of existing licenses, trade inventory, fixtures, fittings and equipment, and with adequate working capital. The value of the property including transferrable goodwill is derived from an estimated maintainable level of trade. If the valuation is required on any other assumption, the Valuer should make such assumption explicit through disclosure. While the actual trading performance may be the starting point for the assessment of the fair maintainable level of trade, adjustments should be made for atypical revenues or costs so as to reflect the trade of a reasonably efficient operator.
- 50.5. Profit generated, in excess of market expectations that may be attributed to the manager is not included. The manager's particular tax position, depreciation policy, borrowing costs and capital invested in the business are not considered for the purpose of establishing a common basis to compare different properties under different managers.
- 50.6. Although the concepts and techniques are similar to those often used in the valuation of a large-scale business, to the extent that the valuation of a TRP does not usually consider tax, depreciation, borrowing costs and capital invested in the business, the valuation is based on different inputs from those of a valuation of a sizable business.
- 50.7. The valuation conclusion may need to be broken down between the different asset components for the purposes of financial reporting, for property taxation or, when required, for property lending purposes.
 - 50.7.1. The components of TRP entity value are typically:
 - 50.7.1.1. land;
 - 50.7.1.2. building(s);
 - 50.7.1.3. fixtures and fittings (furniture, fixtures and equipment), including software;
 - 50.7.1.4. inventory, which may or may not be included (this should be disclosed);
 - 50.7.1.5. intangible assets, including transferrable goodwill; and
 - 50.7.1.6. any licenses and permits required to trade.
 - 50.7.2. Items such as working capital and debt are considered in valuing equity for business, but equity is not valued for TRPs. TRPs may, however, comprise part of a business.
 - 50.7.3. An estimation of the individual values of the components can only represent an apportionment, unless direct market evidence is available for one or more of these components to isolate component value from the overall TRP value.
- 50.8. TRPs are by their nature, specialized assets that are usually designed for a specific use. Changes in market circumstances, whether structural to

the industry or due to the local competition or another reason, can have a material impact on value.

- 50.9. It is necessary to distinguish between the asset value of a Trade Related Property and the ownership value of the business. In order to undertake a valuation of a TRP, a Valuer will require sufficient knowledge of the specific market sector so as to be able to judge the trading potential achievable by a Reasonably Efficient Operator, as well as knowledge of the value of the individual component elements.

PVS GN 800 Valuation for Compulsory Acquisition⁸

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10. Introduction

- 10.1. Compulsory acquisition is the power of government to acquire private rights in land without the willing consent of its owner or occupant in order to benefit society. This power is often necessary for social and economic development and the protection of the natural environment.
- 10.2. Republic Act No. 10752 entitled “An Act Facilitating the Acquisition of Right-of-Way, Site or Location for National Government Infrastructure Projects” or “The Right-of-Way Act” was enacted on 07 March 2016. Right-of-Way or ROW means a part or the entirety of a property, site or location, with defined physical boundaries, used or required by a national government project.
- 10.3. This Guidance Note provides direction on the valuation of right-of-way as interpreted in the Implementing Rules and Regulations of Republic Act. No.10752.
- 10.4. Notwithstanding the implementing rules and regulations on the engagement/ procurement of independent property appraisers (IPA), the importance of the qualifications of the IPA, should be consistent with the IVS Framework and the valuations consistent with IVS General Standards.

20. Scope

- 20.1. As provided in Section 4 of the Act, the following are the regular modes of ROW acquisition:

- a) Donation;
- b) Negotiated Sale; and
- c) Expropriation.

The other modes of ROW acquisition are the following:

- a) Acquisition of Properties under Commonwealth Act (CA) No. 141;
- b) Exchange or Barter;
- c) Easement of Right of Way;
- d) Acquisition of Subsurface Right-of-Way; and
- e) Other modes authorized by law.

⁸ Additional Guidance Notes GN based on the Discussion with the Stakeholders

20.2. This Guidance Note focuses on the valuation of the affected right-of-way, sites or location. As provided in Section 5 of the Act, the Implementing Agency may acquire through negotiated sale the required ROW project, by offering to the property owner as compensation price, the sum of the:

- a) current market value of the land;
- b) replacement cost of structures and improvements therein; and
- c) current market value of crops and trees therein.

For further insight into the application of valuation principles, the following IVS Asset Standards should be consulted:

20.2.1. IVS 300, Plant and Equipment;

20.2.2. IVS 400 Real Property Interests;

20.2.3. IVS 410 Development Property; and

20.2.4. PVS GN 200 Valuation of Agricultural Properties.

20.3. With regard to Section 6.1b of the Implementing Rules and Regulations of the Act, the Replacement Cost of a structure or improvement affected by ROW shall be based on the current market prices of materials, equipment, labor, contractors profit and overhead and all other attendant costs associated with the acquisition and installation of a similar asset in place of the affected asset.

If the affected structure has been damaged, then the Replacement Cost should be based on the pre-damaged condition of that structure. The Replacement Cost of the structure may vary from the market value of the existing structure since the structure that would actually it may have a different cost at current market prices. The replacement structure has to perform the same functions and meet the performance specifications as the original structure.

30. Definitions

30.1. Implementing Agency or IA – refers to any department, bureau, office, commission, authority or agency of the national government, including any government-owned and controlled corporation or state college or university, authorized by law or its respective charter to undertake national government projects.

30.2. National Government Projects. As defined in Section 3 of the Act, refers to all national government infrastructure projects and their public service facilities, engineering works and service contracts, including projects undertaken by government-owned and controlled corporation, all projects covered by Republic Act No. 6957, as amended by RA No. 7718, otherwise known as the “Build-Operate-and-Transfer-Law” and other related laws including those involving private sector participation and all necessary activities or projects that are intended for public use or purpose, such as site acquisition, supply and/or installation of equipment and materials, implementation, construction, completion, operation, maintenance, improvement, repair and rehabilitation, regardless of the source of funding. These projects shall include, but not limited to the following:

- a. Highways, including expressways, roads, bridges, interchanges, overpasses, tunnels, viaducts and related facilities;
- b. Railways and mass transit facilities;
- c. Port infrastructure, like piers, wharves, quays, storage handling and ferry services;
- d. Airports and air navigation facilities;
- e. Power generation, transmission and distribution facilities;
- f. Radio/television broadcasting and telecommunications infrastructure;
- g. Information technology infrastructure;
- h. Irrigation, flood control and drainage systems;
- i. Water and debris retention structures and dams;
- j. Water supply, sanitation, sewerage and waste management facilities;
- k. Land reclamation, dredging and development;
- l. Industrial and tourism estates;
- m. Government school buildings, hospitals, clinics and other buildings and housing projects;
- n. Public markets and slaughterhouses; and
- o. Other similar or related infrastructure works and services of the national government.

40. Relationship to Accounting Standards

- 40.1. Under International Financial Reporting Standards (IFRSs), like other types of real property, a real property subject of compulsory acquisition may be carried on an entity's balance sheet at either cost or at fair value. It may be necessary to allocate the value of the property between its different components for depreciation purposes.

50. Guidance

- 50.1. This Guidance Note describes that category of property referred to as the subject of compulsory acquisition are valued in accordance with the asset standards cited in Sections 20.2.1 through 20.2.4 above.

Acknowledgment

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Consulted Stakeholders

National Government Agencies

Bureau of Internal Revenue
Board of Investments
Bangko Sentral ng Pilipinas
Commission on Audit
Department of Agrarian Reform
Department of Public Works and Highways
Department of Environment and Natural Resources
Forest Management Bureau
Home Guaranty Corporation
Insurance Commission
Land Registration Authority
Land Management Bureau
National Housing Authority
National Economic and Development Authority
Professional Regulatory Board of Real Estate Service
Securities and Exchange Commission
Social Housing Finance Corporation

Academe

UP Open University (Faculty of Management and Development Studies)

Assessors of Local Government Units

Province of Agusan del Sur
Province of Bohol
Province of Davao del Norte
Province of Davao Del Sur
Province of Davao Occidental
Province of Laguna
Province of Nueva Vizcaya
Province of Rizal
Province of Sorsogon

Province of Surigao del Sur
 Baguio City
 Butuan City
 Davao City
 Digos City
 Pasay City
 San Juan City
 Manila City
 Valenzuela City
 Municipality of Buguias, Benguet
 Municipality of Camalaniugan, Cagayan
 Municipality of Laguindingan, Misamis Oriental
 Municipality of Malay, Aklan
 Municipality of Taytay, Rizal

Private Valuers, Developers and Other Stakeholders

ACNN Property Appraisal
 AVISO Valuation and Advisory
 Bank Appraisers Association of the Philippines
 Cuervo Appraisers Inc.
 Institute of Philippine Real Estate Appraisers
 J&BB Realty
 Leechiu & Associates, Inc.
 Philippine Association of Assessing Officers
 Philippine Association of Local Treasurers and Assessors
 Philippine Association of Municipal Assessors
 Philippine Association of Realty Appraisers
 Philippine Constructors Association, Inc. – Monark Equipment
 Philippine Franchise Association
 Punongbayan & Araullo
 Pinnacle Real Estate Consulting Services, Inc.
 Robinsons Land Corporation, ALTUS, Inc.
 SM Investments Corporation
 SM Prime Holdings

Banks

Development Bank of the Philippines
 Landbank of the Philippines
 Philippine Postal Savings Bank
 Philippine Veterans Bank

Notes on the Second Printing of PVS 2nd Edition, 2018

Page viii	Aligned Valuation for Taxation Purposes and References for Valuation of Property for Other Purposes to Asset Standards (the two items are not part of the Asset Standards)
Page 118	Aligned Valuation for Taxation Purposes and References for Valuation of Property for Other Purposes to Asset Standards (the two items are not part of the Asset Standards)
Page 136	Removed Asset Standards from the edge (not part of Asset Standards)
Page 137	Replaced the header Asset Standards with Valuation for Taxation Purposes , and removed Asset Standards from the edge (not part of Asset Standards)
Page 138	Removed Asset Standards from the edge (not part of Asset Standards)
Page 139	Replaced the header Asset Standards with References for Valuation of Property for other Purposes , and removed Asset Standards from the edge (not part of Asset Standards)
Page 140	Removed Asset Standards from the edge (not part of Asset Standards)
Page 141	Replaced the header Asset Standards with References for Valuation of Property for other Purposes , and removed Asset Standards from the edge (not part of Asset Standards)
Page 175	Realigned International Valuation Standards and International Public Sector Accounting Standards and removed the word Definition therefrom (subtitle for Definition is already given)
Page 176	Deleted the words “ Definitions from the ” to retain UNESCO Glossary of World Heritage Terms and realigned World Heritage Convention, Article I, UNESCO, 1972



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